

Large Area μ TPC Micromegas for the X17 Search at n_TOF

Micromegas built at CEA Saclay for an X17
search at n_TOF's EAR2 facility at CERN

Outline

- 1 The X17 particle and the n_TOF channel**

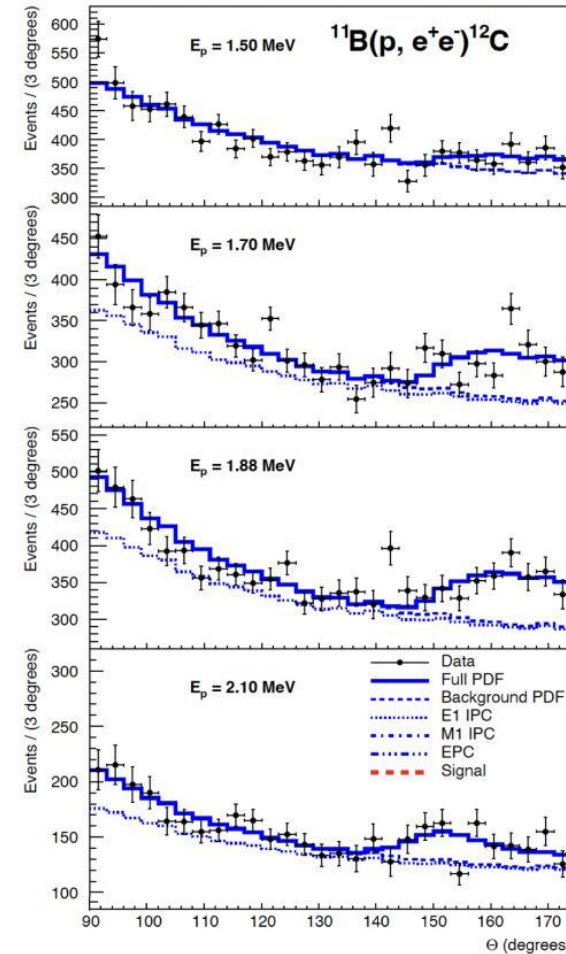
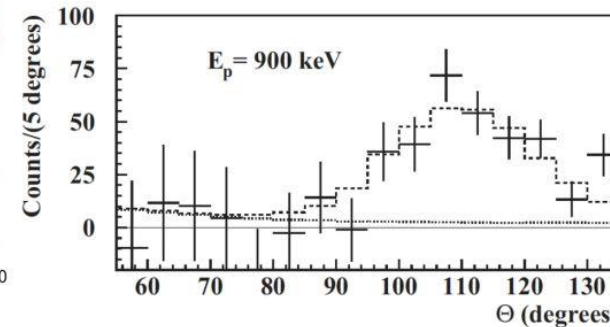
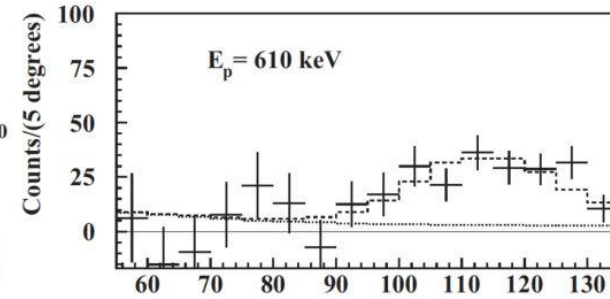
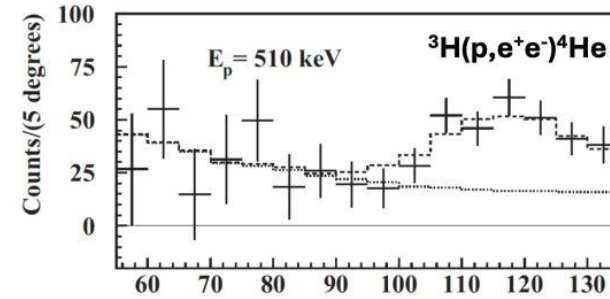
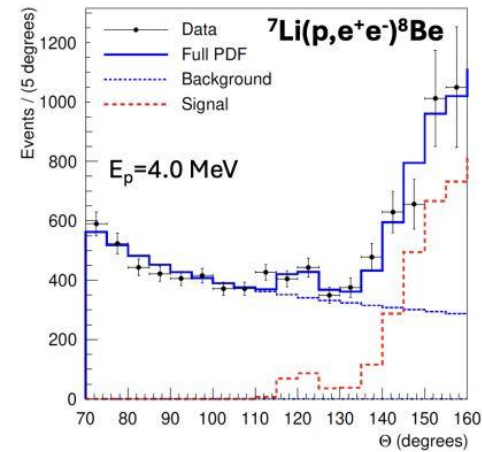
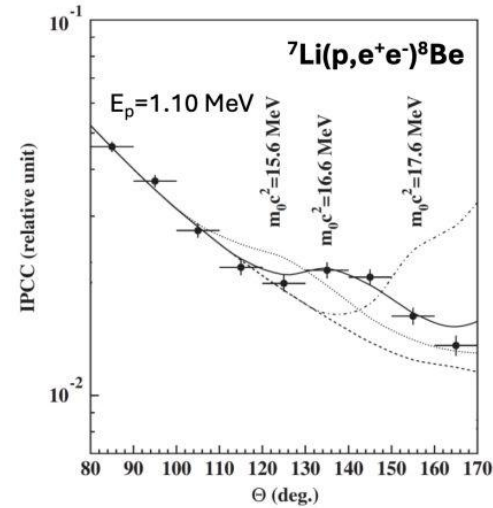
- 2 Micromegas μ TPC detectors**

- 3 X17 search at n_TOF —
experimental setup**

- 4 Experiment Summary and Early Results**

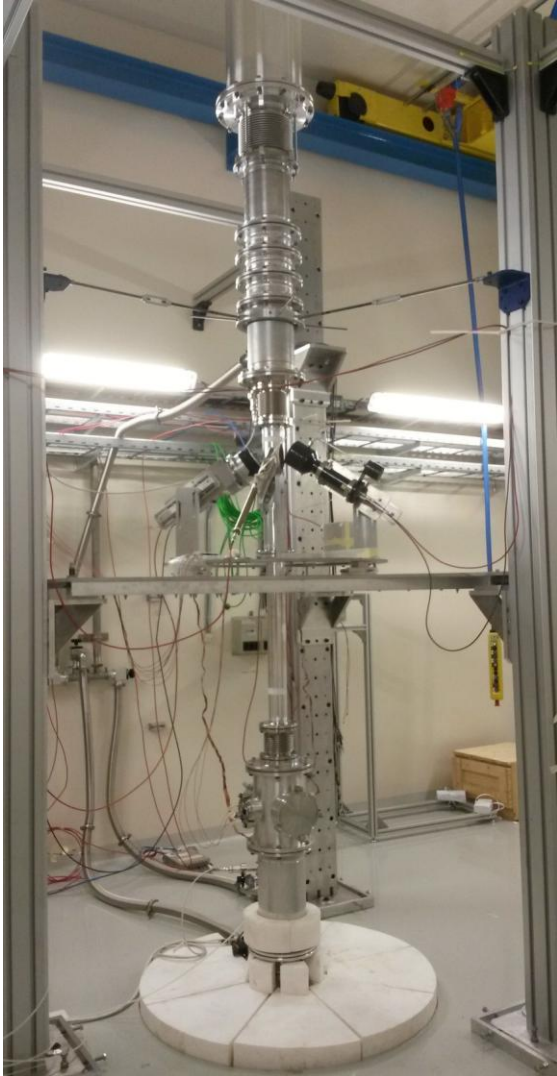
A possible new boson at 17 MeV?

- ▶ **ATOMKI, 2016:** an excess of e^+e^- pairs at large opening angle, in three nuclear transitions
- ▶ Consistent with a new particle: **X17**, ~17 MeV, decaying to e^+e^-
- ▶ Can we make an independent measurement?
- ▶ The same $^4\text{He}^*$ state, generated with neutrons + ^3He ?

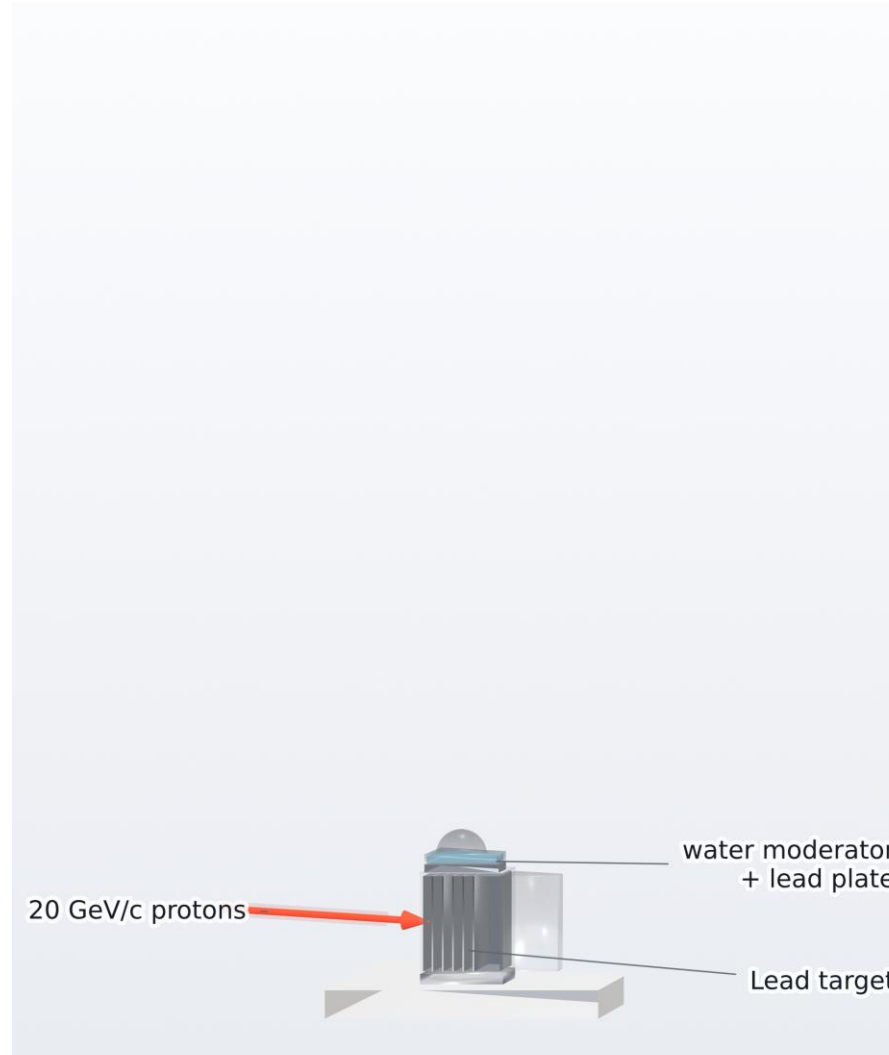


A.J. Krasznahorkay et al., *PRL* **116**, 042501 (2016) · *PRC* **104**, 044003 (2021) · *PRC* **106**, L061601 (2022)

n_TOF EAR2: a vertical neutron beam line at CERN



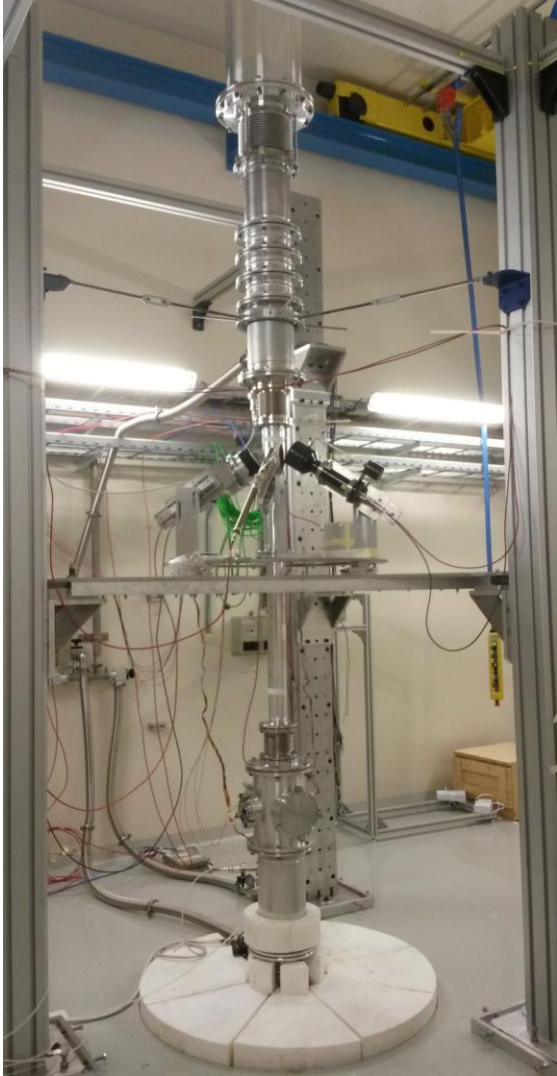
The EAR2 hall — © CERN



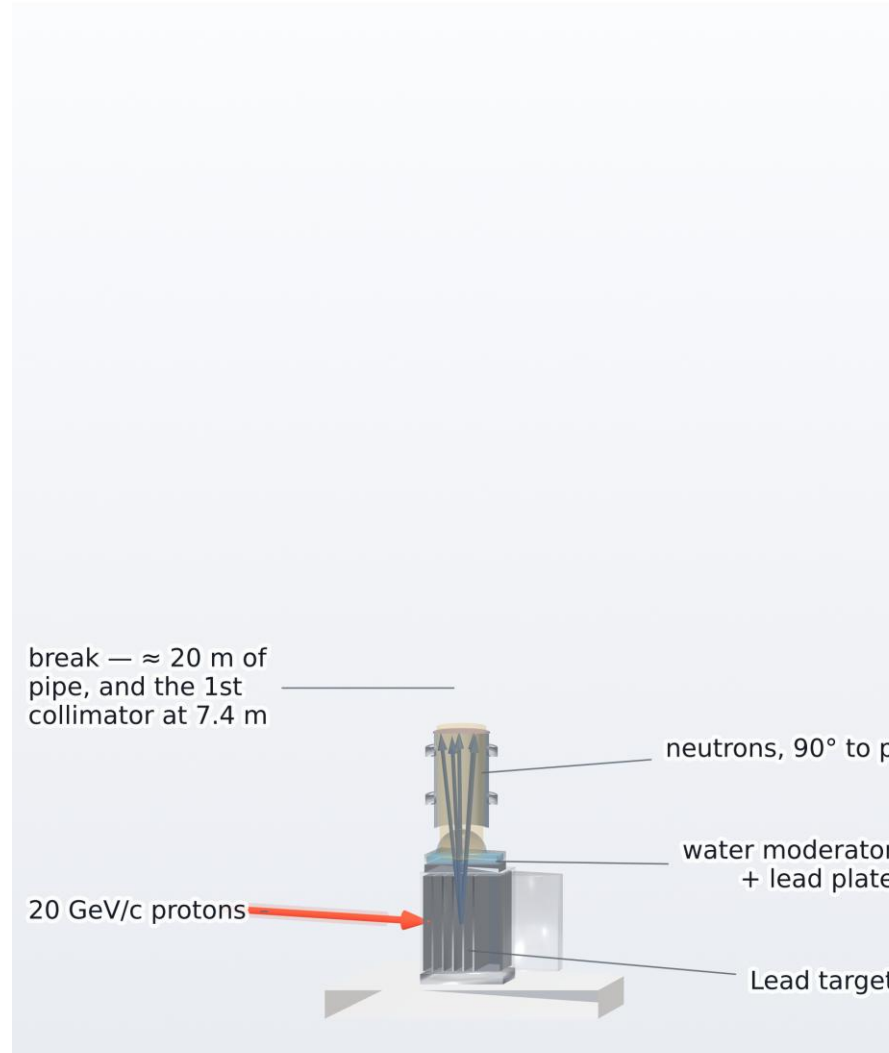
In-house render · Weiß (2015), Pavon-Rodriguez (2025)

- ▶ 20 GeV/c PS protons hit a lead spallation target

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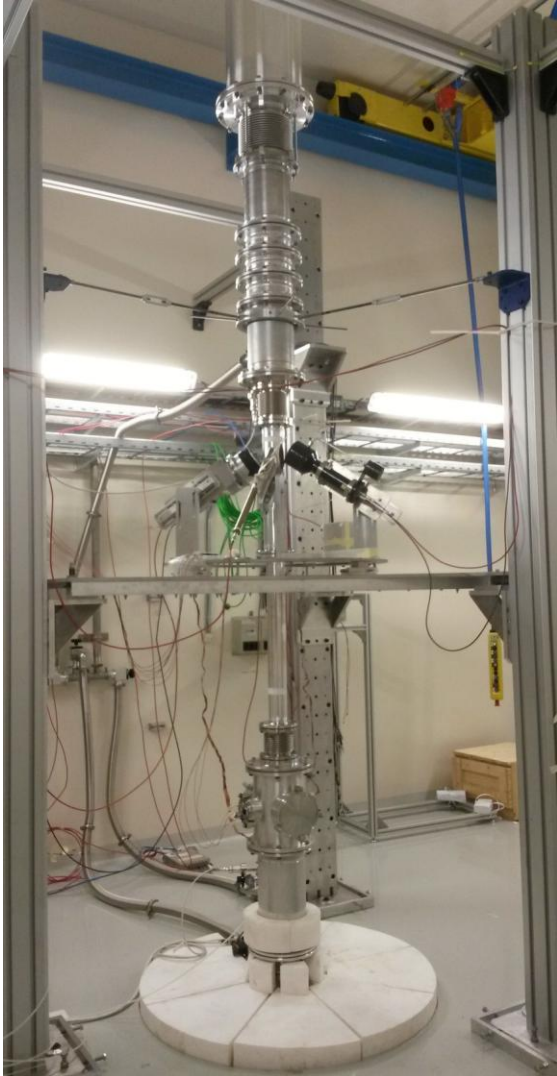
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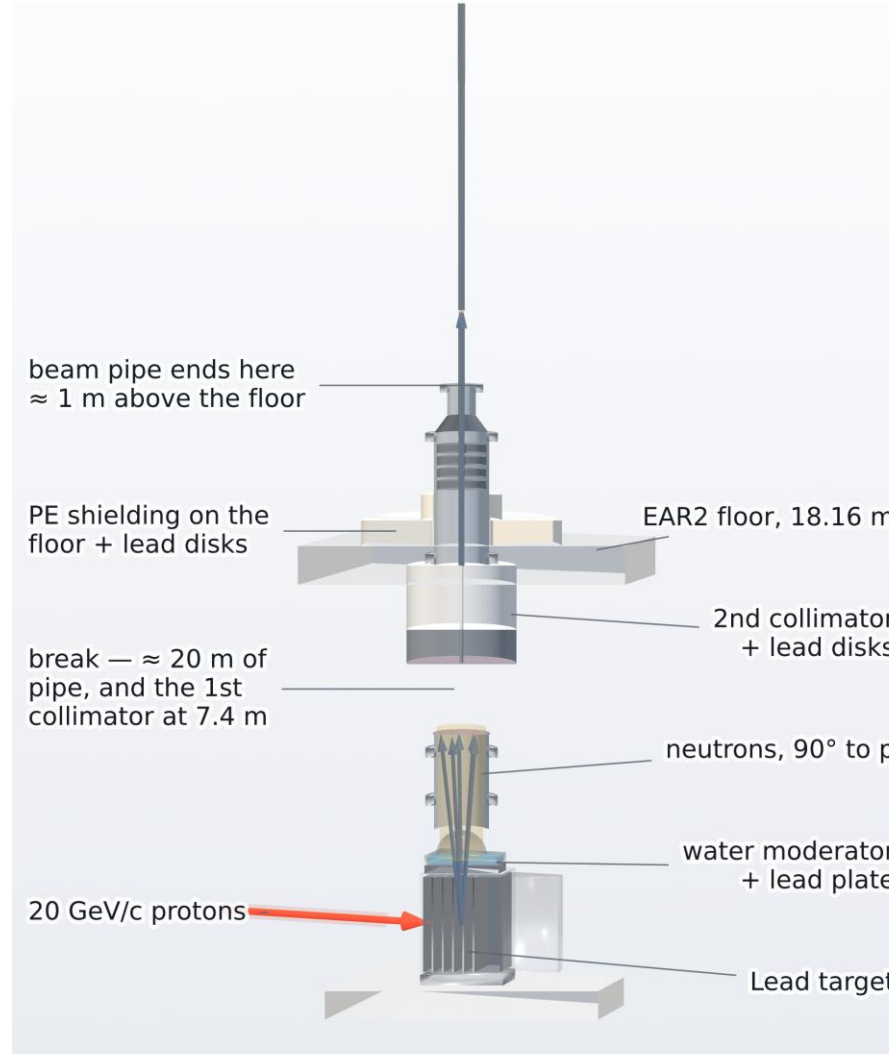
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- ▶ 20 GeV/c PS protons hit a lead spallation target
- ▶ Neutrons ejected at **90°** climb **~20 m** to EAR2

n_TOF EAR2: a vertical neutron beam line at CERN



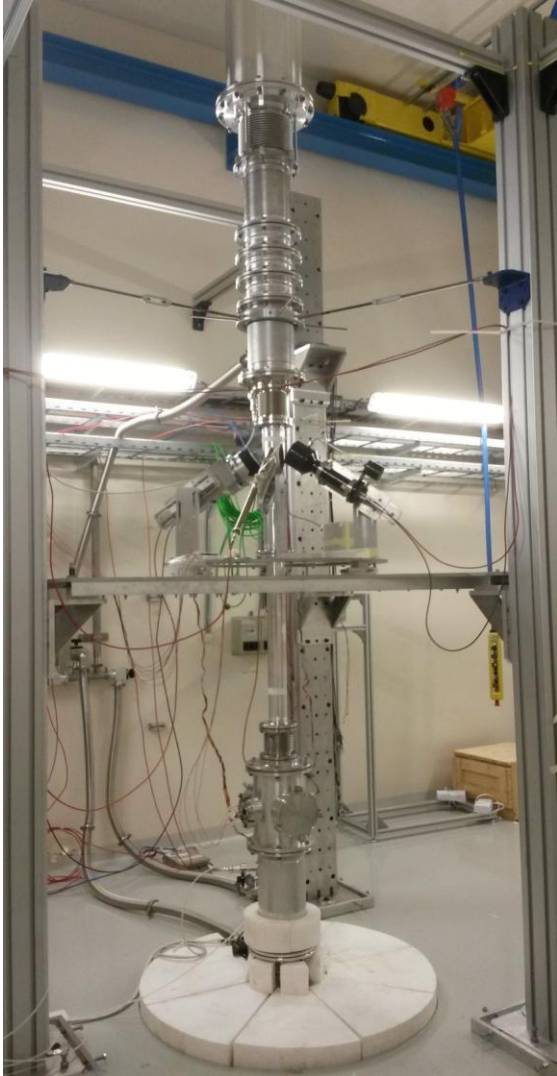
The EAR2 hall — © CERN



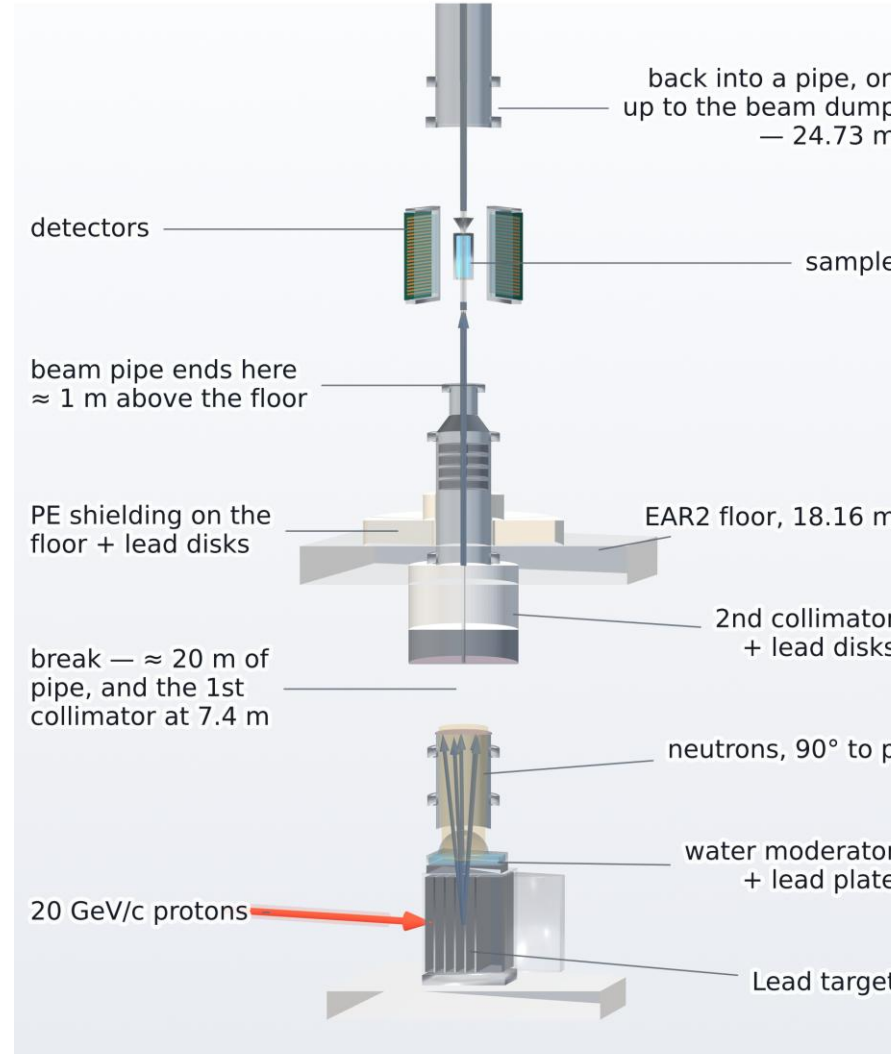
In-house render · Weiß (2015), Pavon-Rodriguez (2025)

- ▶ 20 GeV/c PS protons hit a lead spallation target
- ▶ Neutrons ejected at 90° climb ~20 m to EAR2
- ▶ Collimated to a ≈ 3 cm pencil
- ▶ The pipe ends: the beam crosses the hall in air

n_TOF EAR2: a vertical neutron beam line at CERN



The EAR2 hall — © CERN

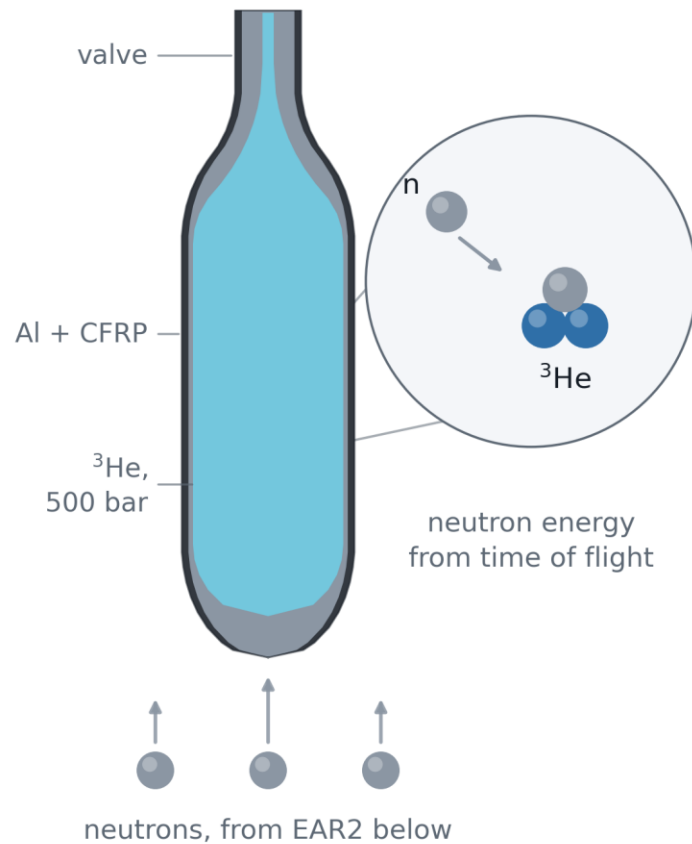


In-house render · Weiß (2015), Pavon-Rodriguez (2025)

- ▶ 20 GeV/c PS protons hit a lead spallation target
- ▶ Neutrons ejected at 90° climb ~20 m to EAR2
- ▶ Collimated to a ≈ 3 cm pencil
- ▶ The pipe ends: the beam crosses the hall in air
- ▶ Measurement station sits in that open beam
- ▶ The highest instantaneous flux at n_TOF

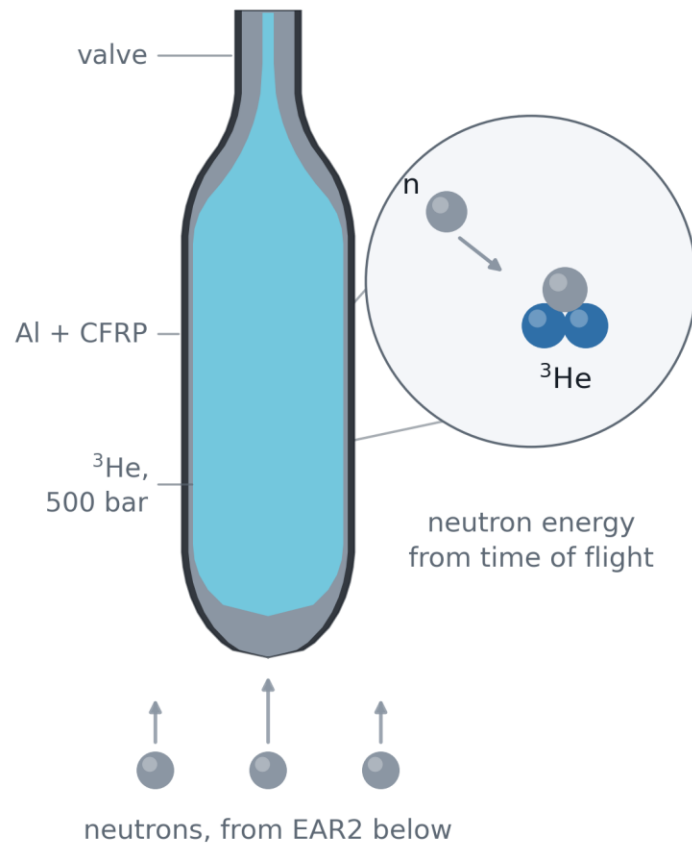
Capture on ^3He and how X17 could come out

1. A neutron beam on ^3He

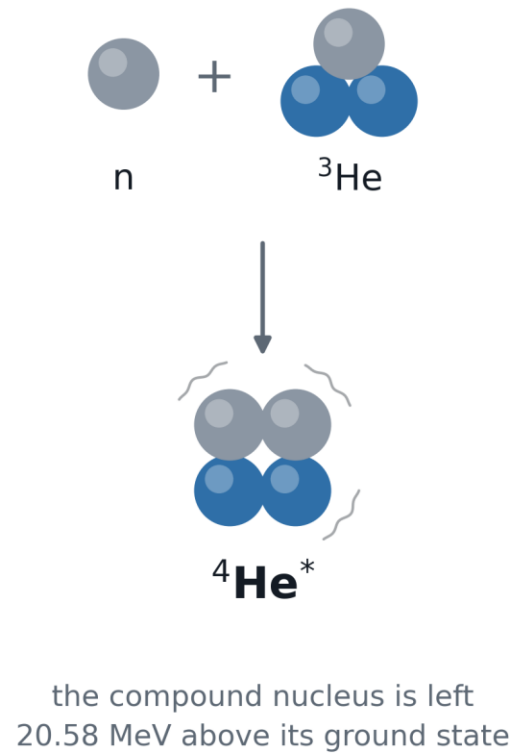


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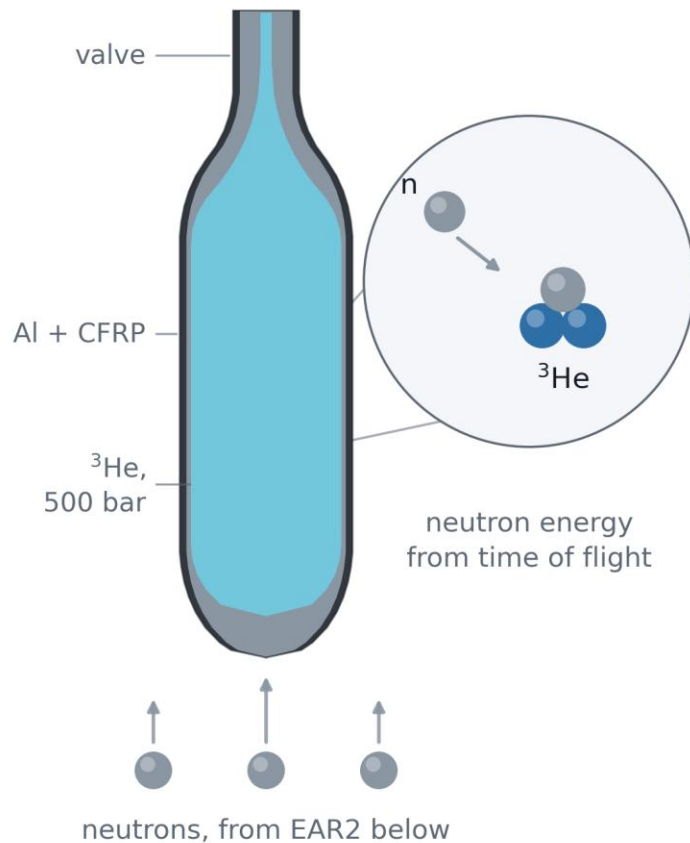


2. Capture makes $^4\text{He}^*$

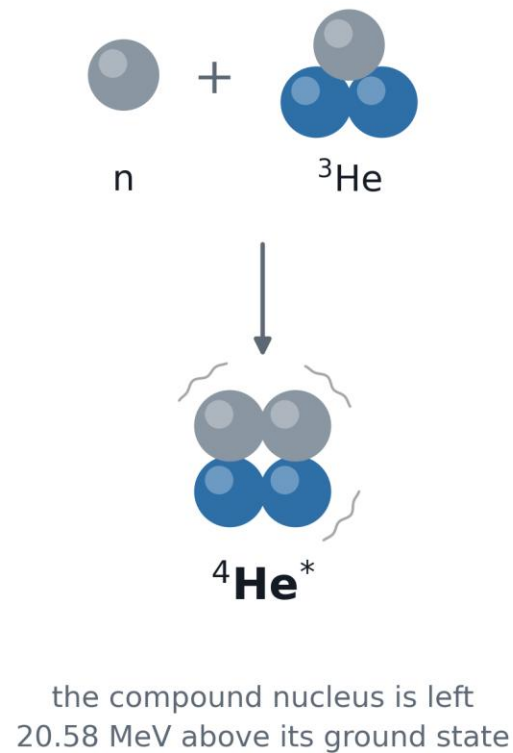


Capture on ^3He and how X17 could come out

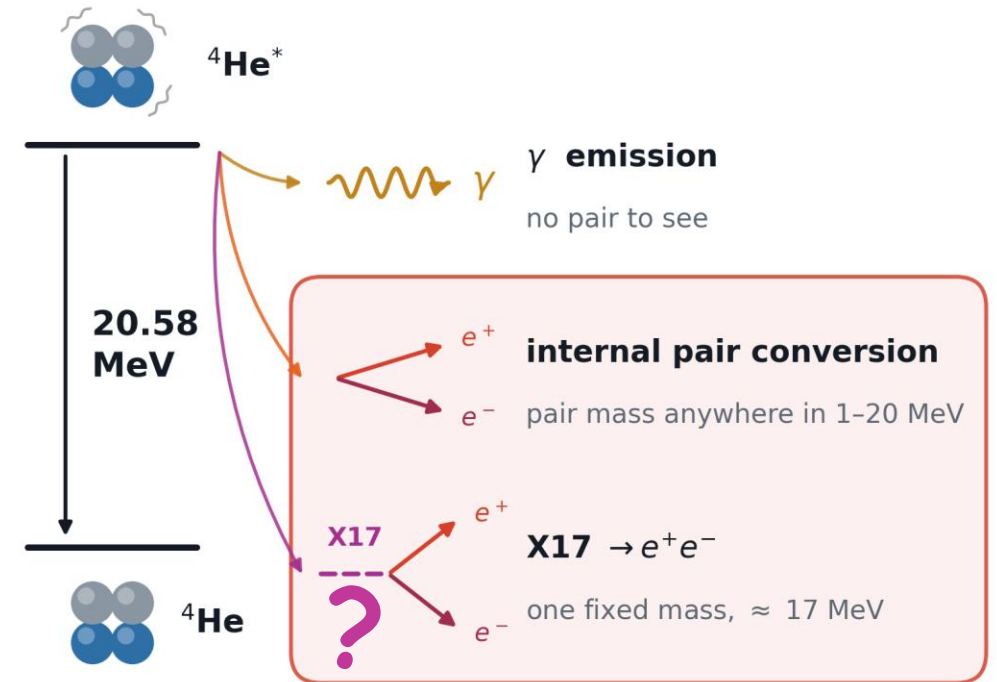
1. A neutron beam on ^3He



2. Capture makes $^4\text{He}^*$



3. Three ways to shed it

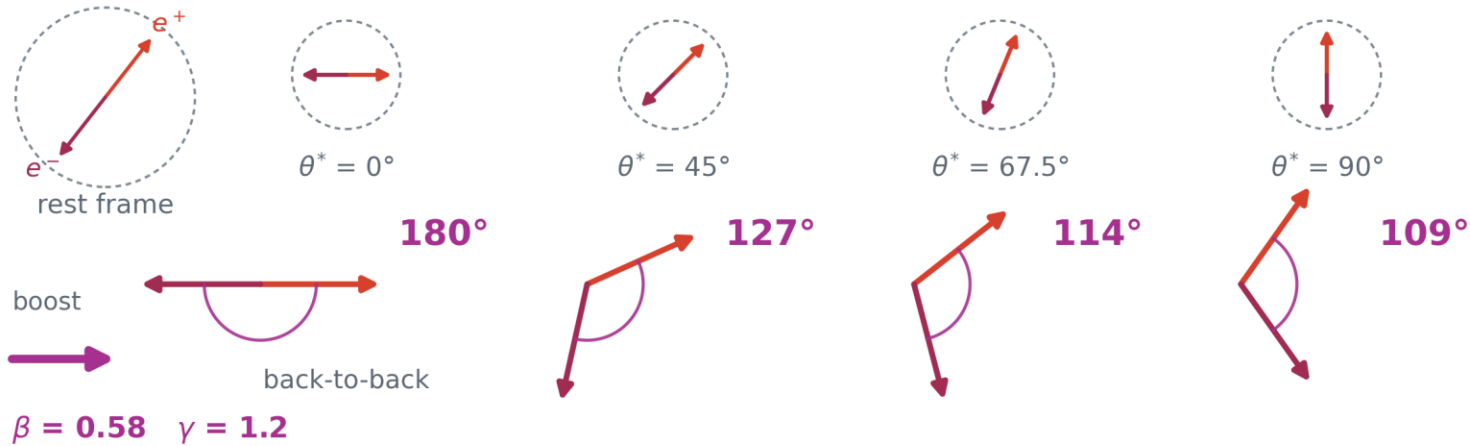


Detect the e^+e^- pair!

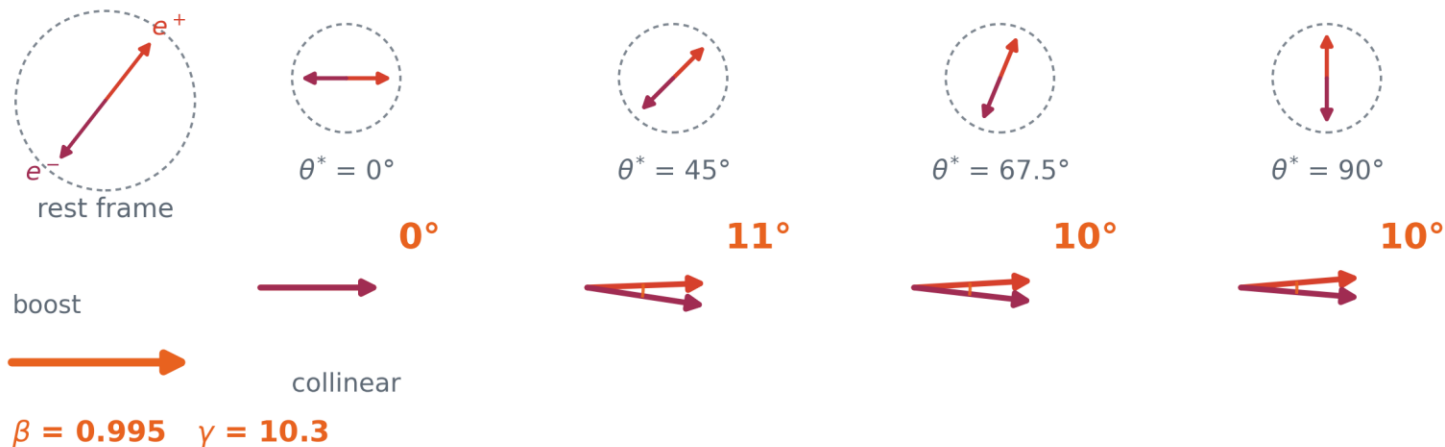
The opening angle observable -- X17 moves slow

4. The boost is what makes the difference

X17 — one mass, 16.8 MeV, heavy and slow



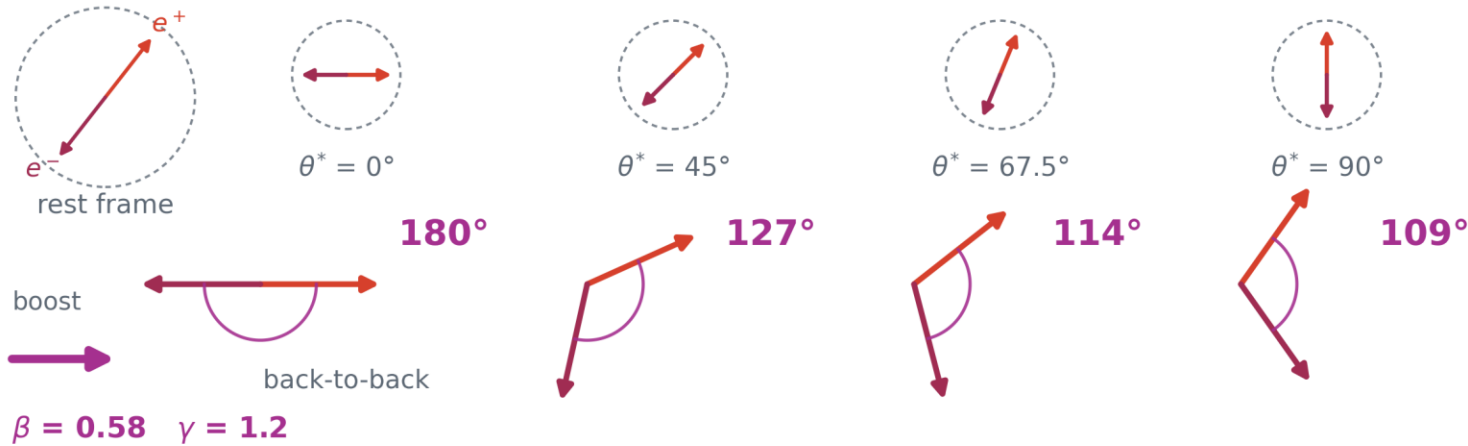
IPC — any mass, here 2 MeV, light and fast



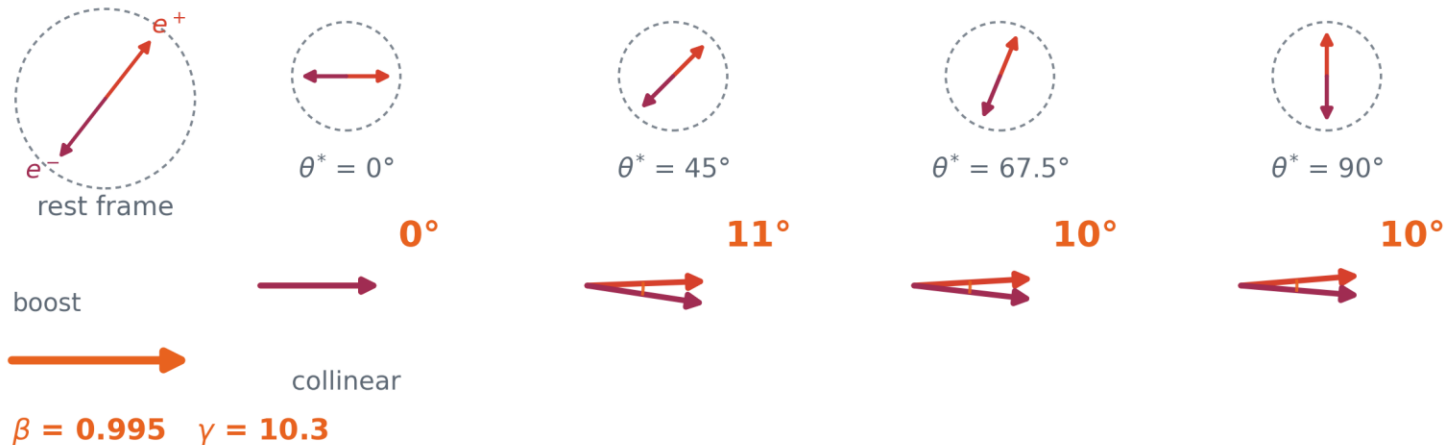
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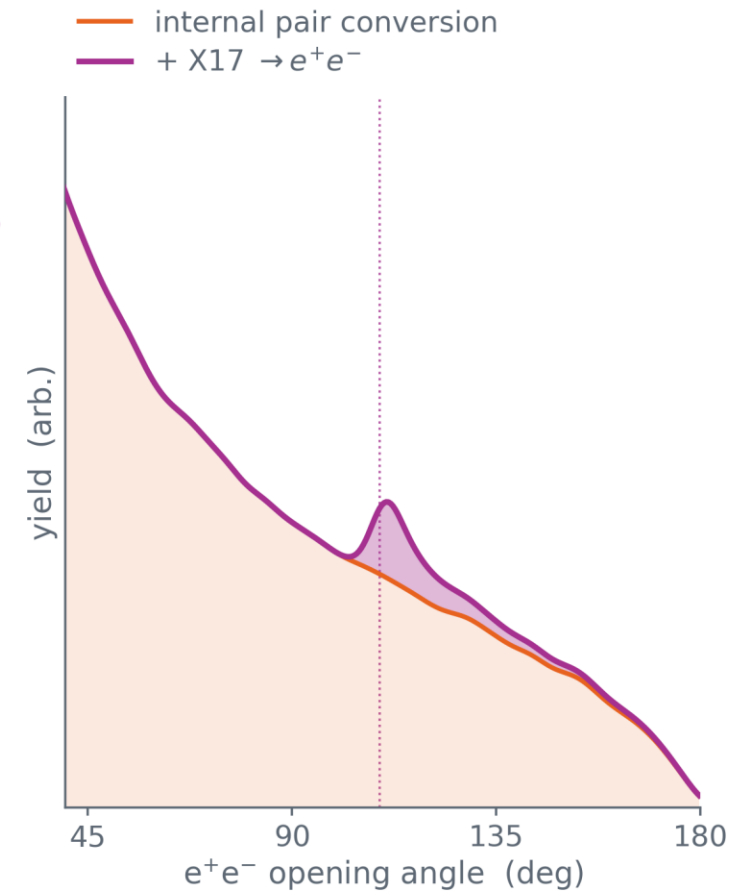
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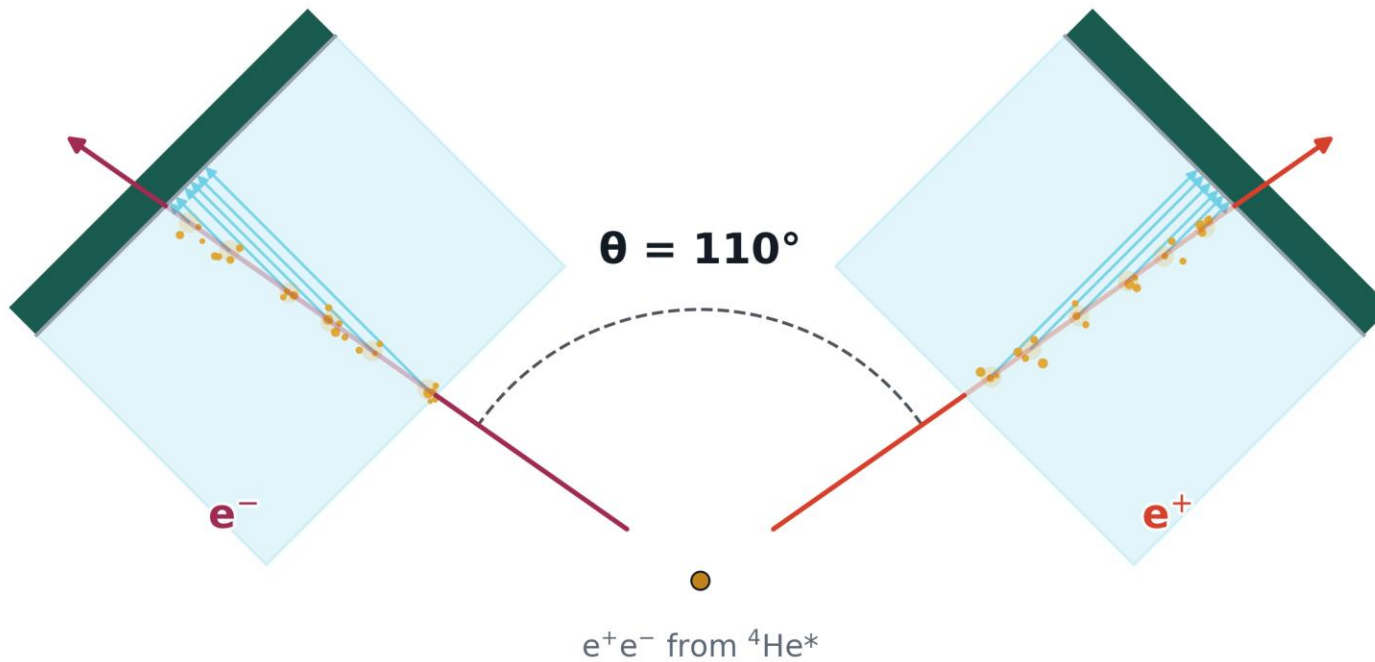
5. So this is what we look for



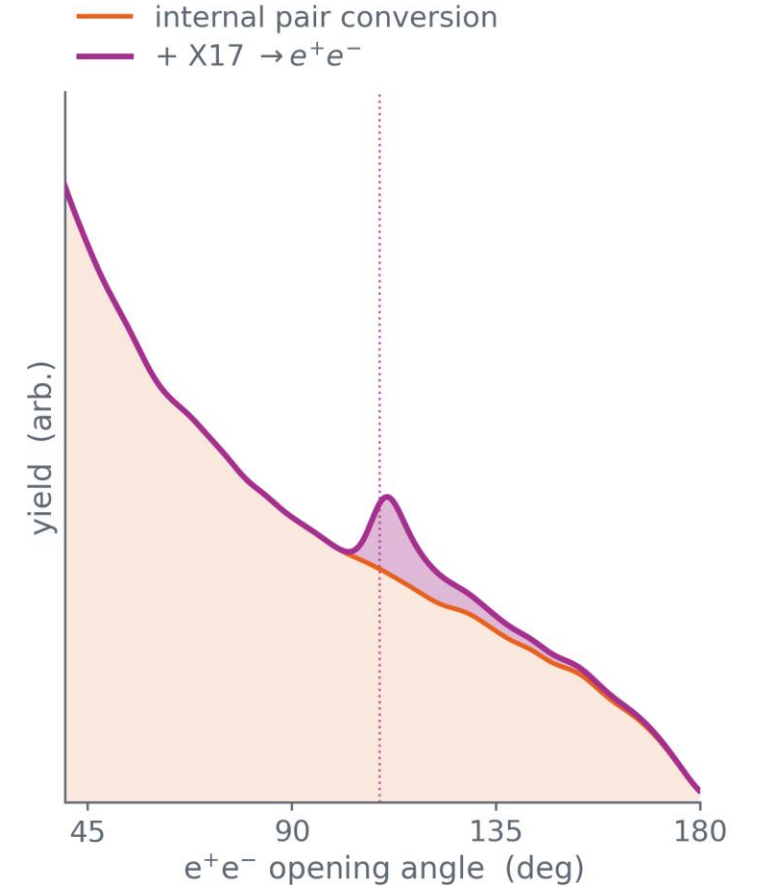
The opening angle observable -- X17 moves slow

6. Measure it with a μ TPC Micromegas

Micromegas μ TPC
30 mm drift gap



5. So this is what we look for

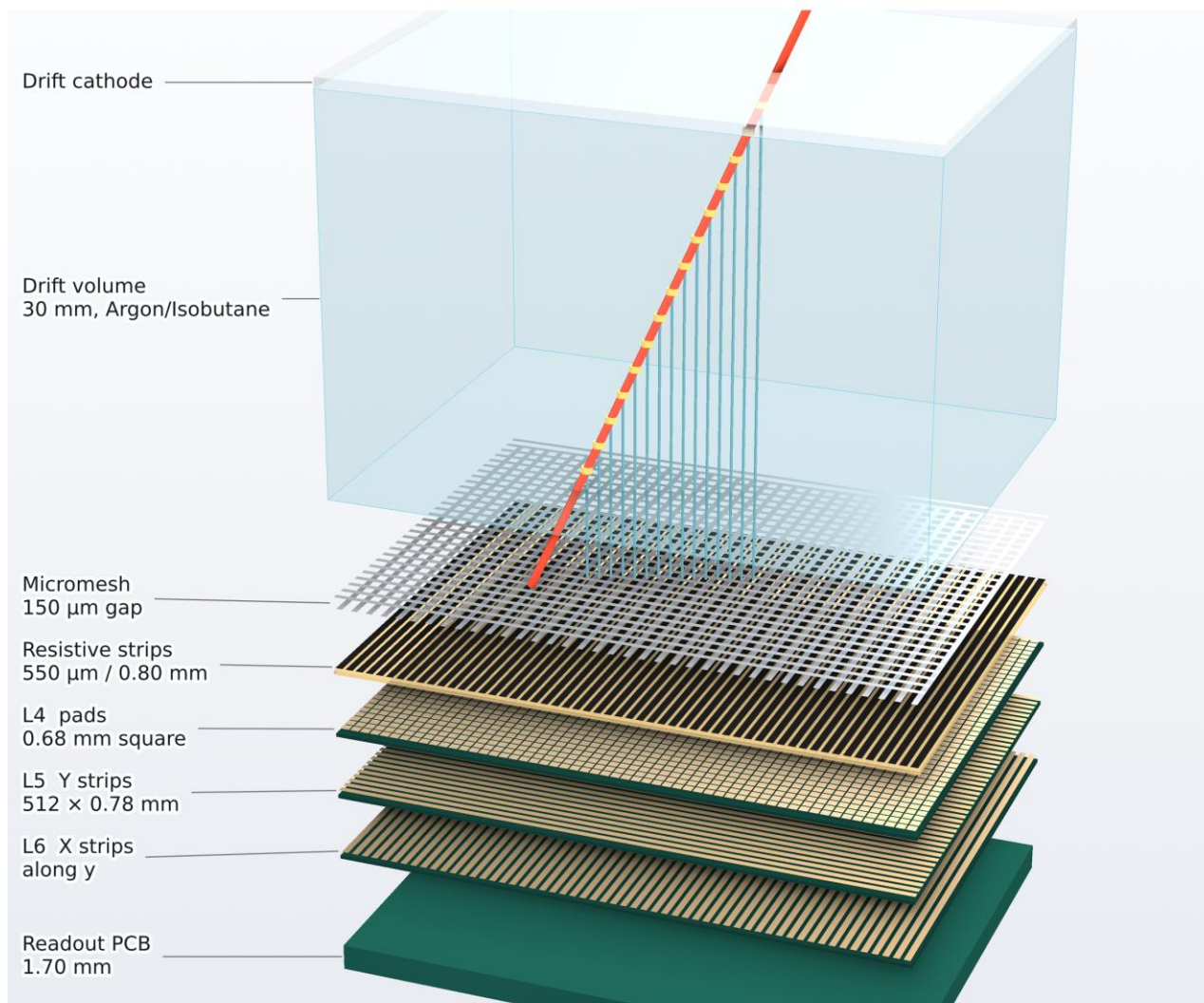


The Micromegas μ TPC

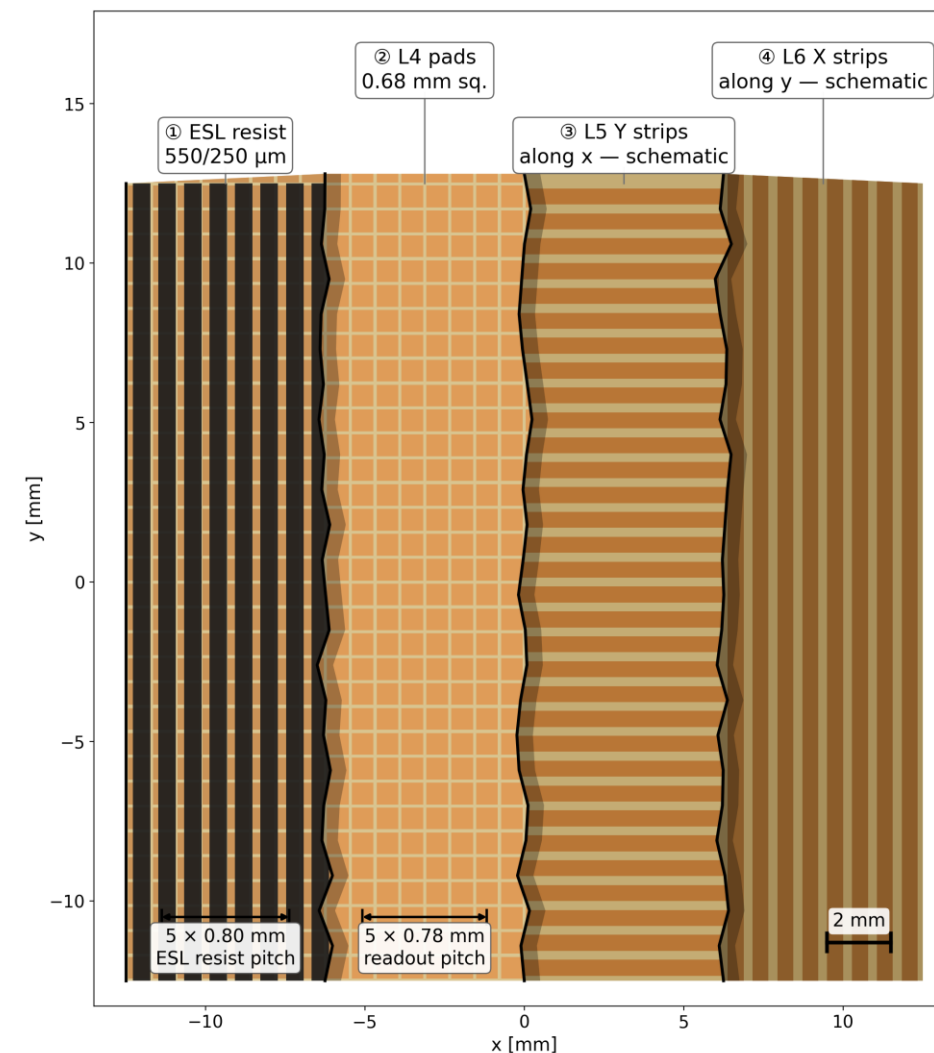
A large-area resistive-strip detector, built to measure the opening angle of the pair.

The Micromegas μ TPC design

One chamber, exploded along the drift axis
a 44×34 mm window at the real pitch



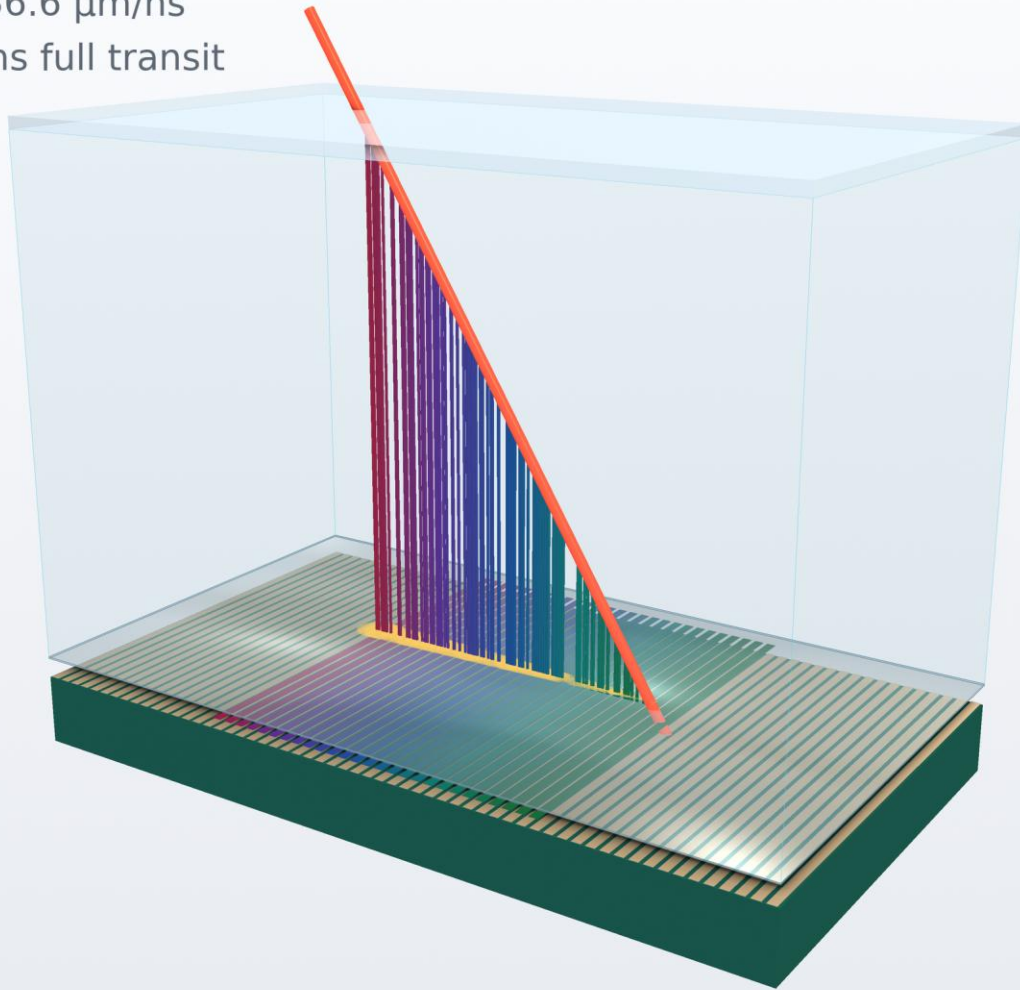
MX17 readout board, four layers peeled back
25 $\times$ 25 mm of the 470 mm board



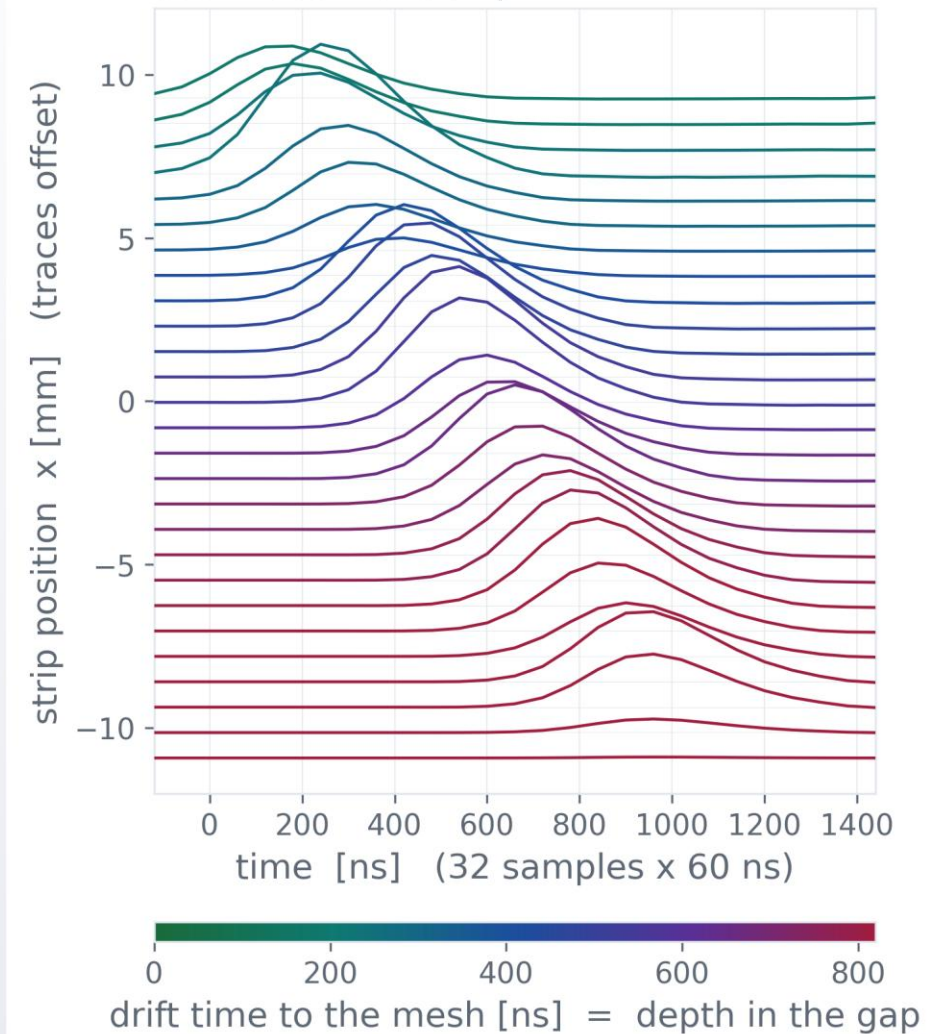
Micro-TPC operation

30 mm gap · 333 V/cm
 $v = 36.6 \mu\text{m/ns}$
 820 ns full transit

Simulated event

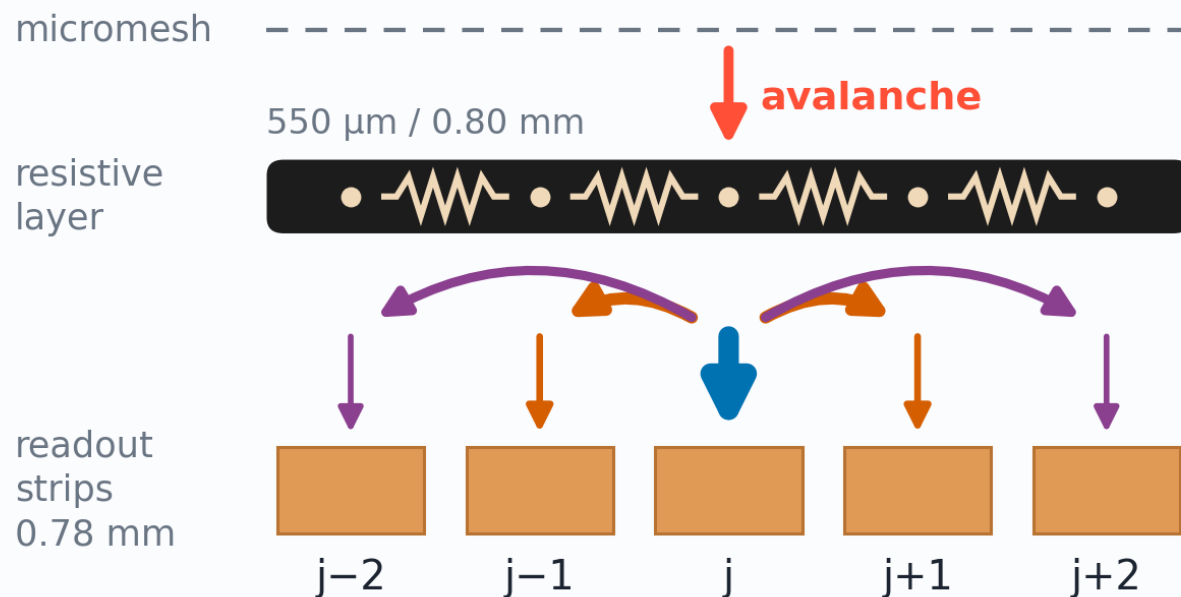


each trace = one strip



Resistive strips spread and delay charge

What happens on the resistive layer



own charge

± 1 neighbour — late by 166 ns

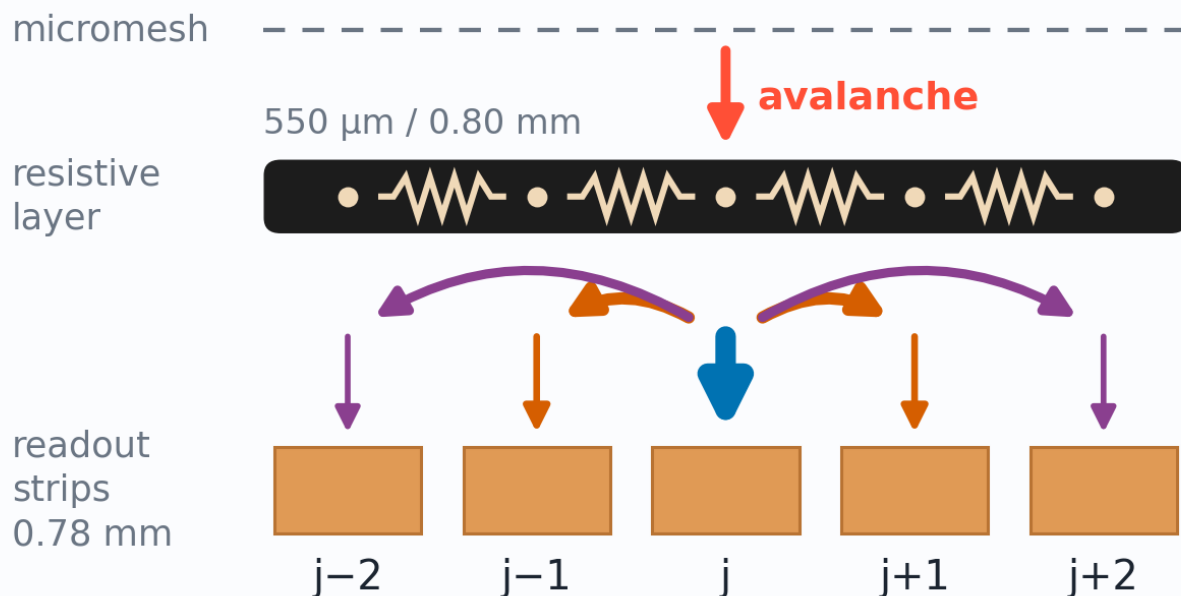
± 2 — 333 ns

Charge spreads through the resistive strips

A neighbor's copy arrives **late and dispersed**

Resistive strips spread and delay charge

What happens on the resistive layer



own charge

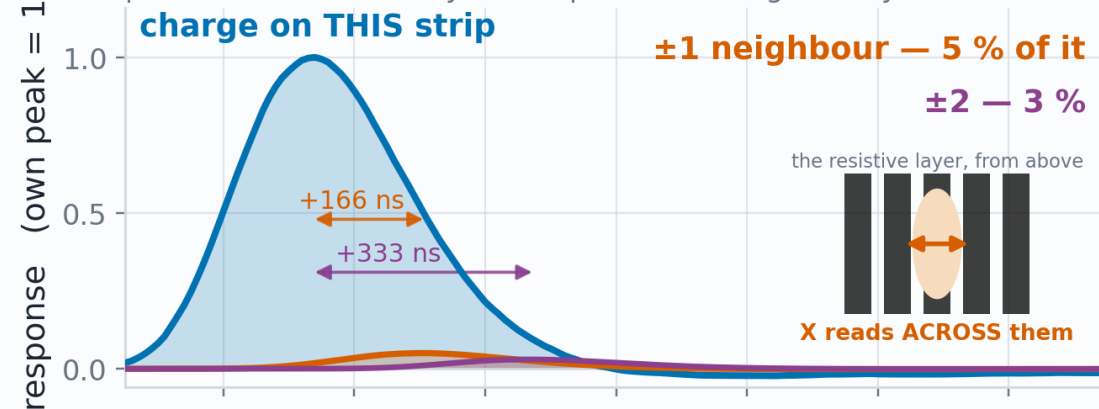
± 1 neighbour — late by 166 ns

± 2 — 333 ns

Charge spreads through the resistive strips
A neighbor's copy arrives **late** and **dispersed**

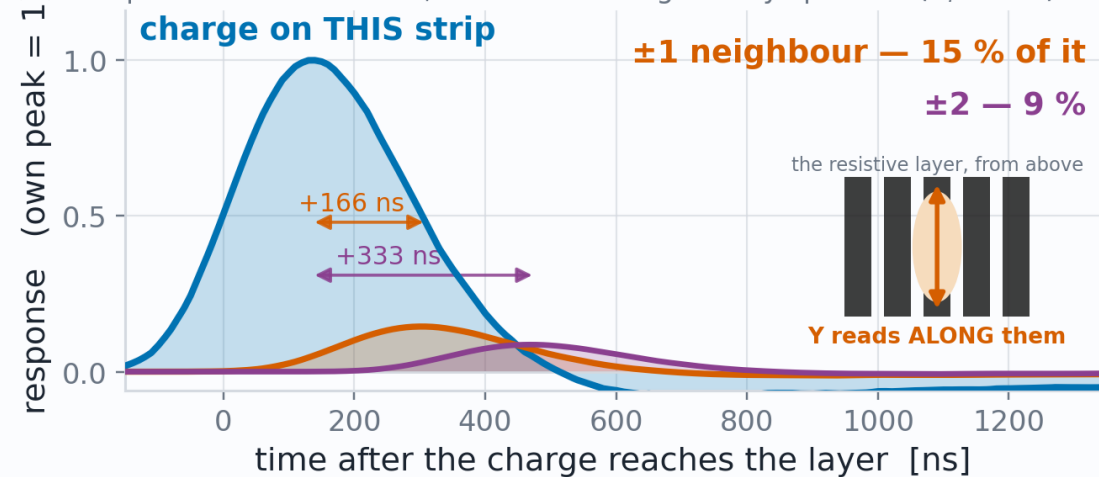
X plane

pitched **ACROSS** the layer's strips — the charge barely crosses them



Y plane

pitched **ALONG** them, where the charge really spreads ($k_Y = 2.9$)



The forward fit: assume a track and predict every waveform



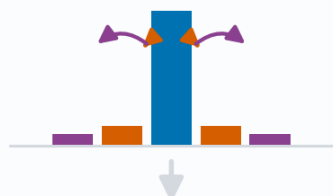
① the track, in 60 ns slices

$K = 18$ slices of the drift column. Each holds a charge $q_k \geq 0$ at transverse position $p_0 + w \cdot u_k$. The charges are solved, not searched.



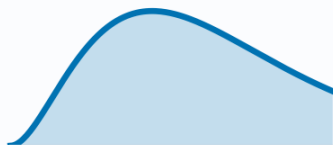
② geometric spread

What one slice puts on the layer, integrated over the 0.78 mm pitch. Its width is transverse diffusion plus the amplification width.



③ charge spread

Every strip's charge appears again on ± 1 and ± 2 — and LATE, by 166 and 333 ns. Per plane: Y 15 % / 9 %, X 5 % / 3 %.



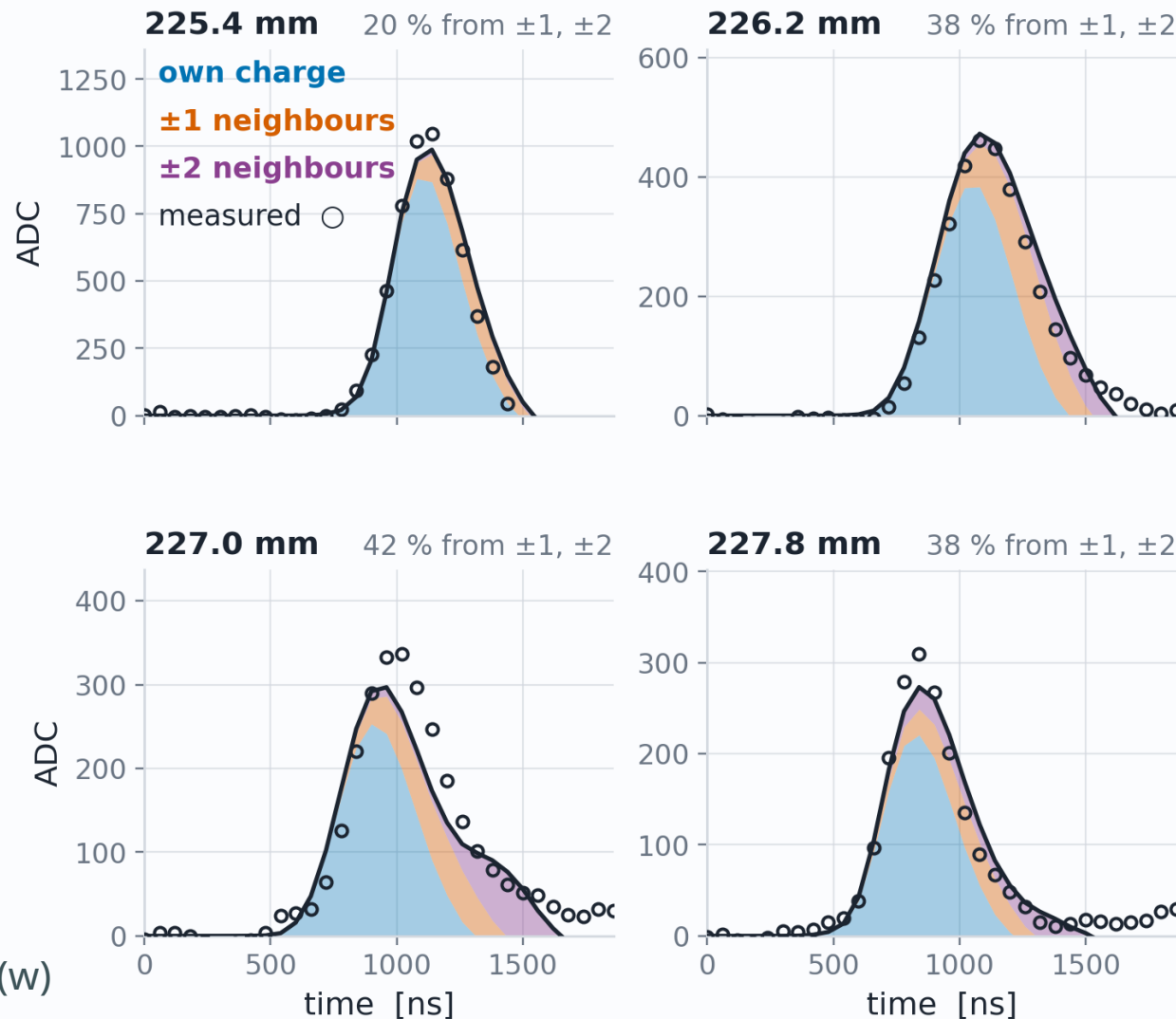
④ fold with response

The measured single-primary response of this plane, sampled at 32×60 ns — one complete predicted (strip $\times$ sample) window.

Three numbers are searched:

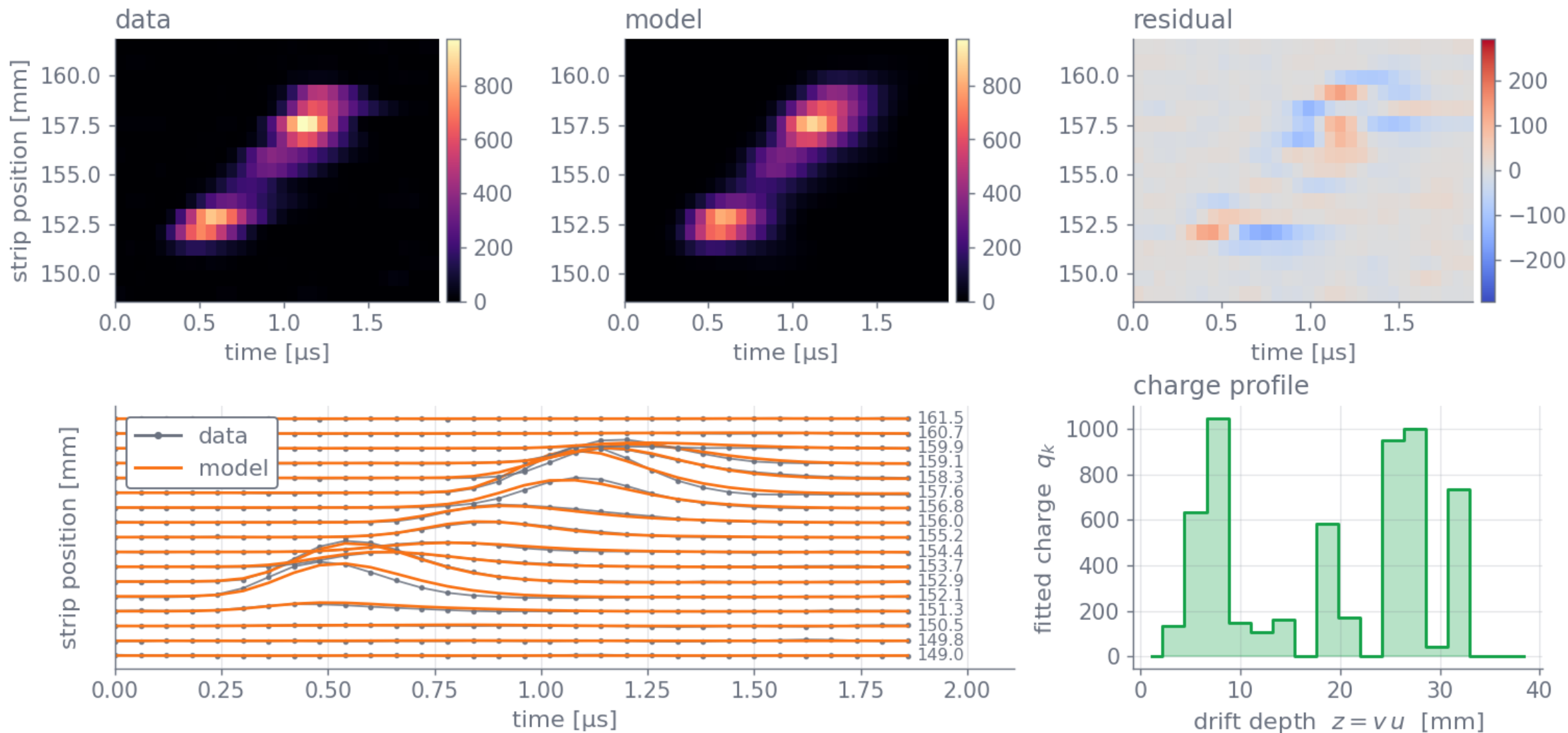
Mesh arrival position and time (p_0 , t_0) and transverse speed (w)

v_{drift} enters once at the end: $\tan \theta = w / v_{\text{drift}}$



Four consecutive strips of one real muon event

One muon through the forward fit



Characterization

How well the chamber actually performs,
measured with cosmic muons.

Characterization on the Saclay cosmic bench

A cosmic muon fires the scintillators at the top and bottom of the stack.

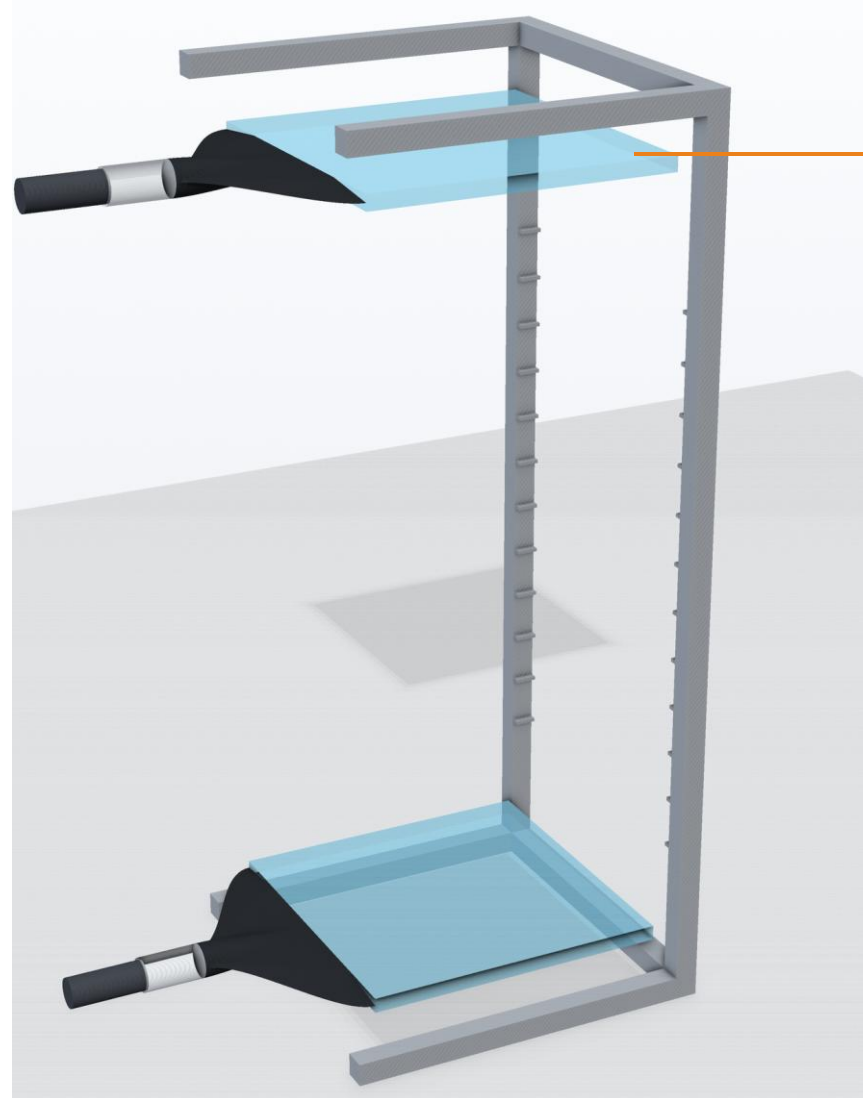
Four reference Micromegas reconstruct its track; the two MX17 detectors under test sit in between.



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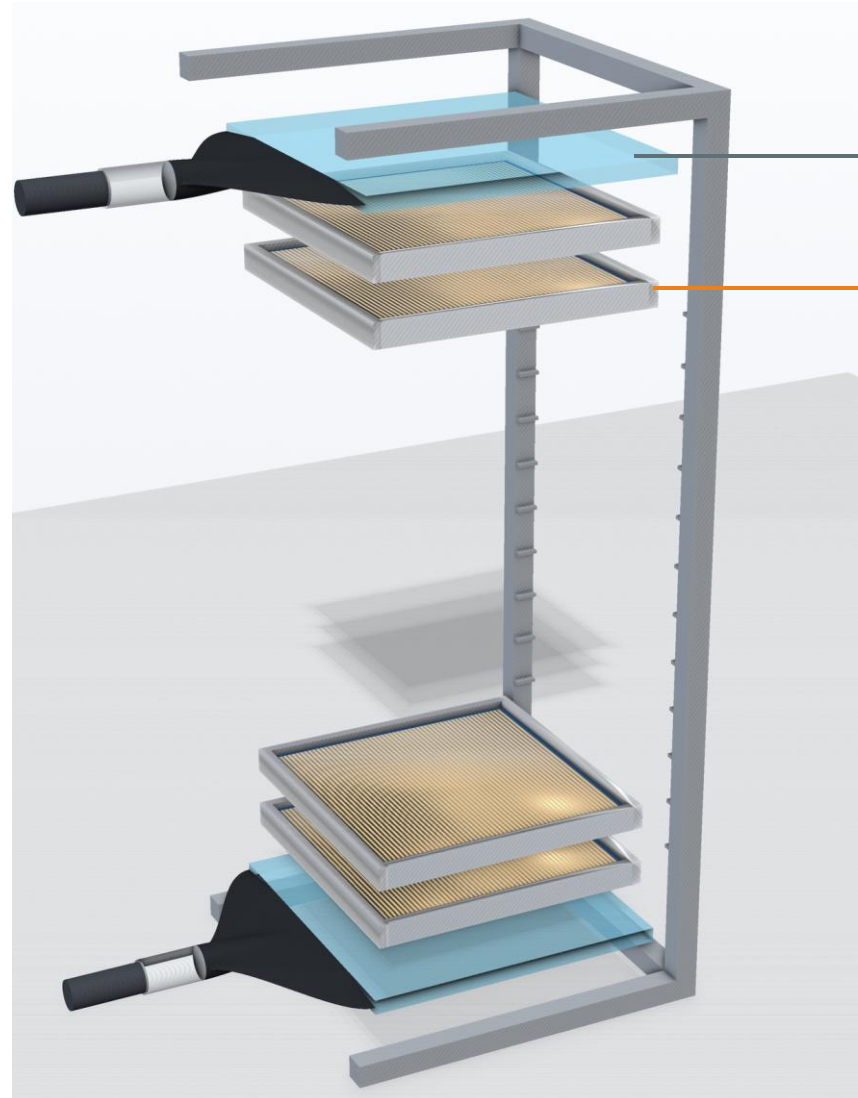


60 × 60 cm trigger scintillators, top and bottom

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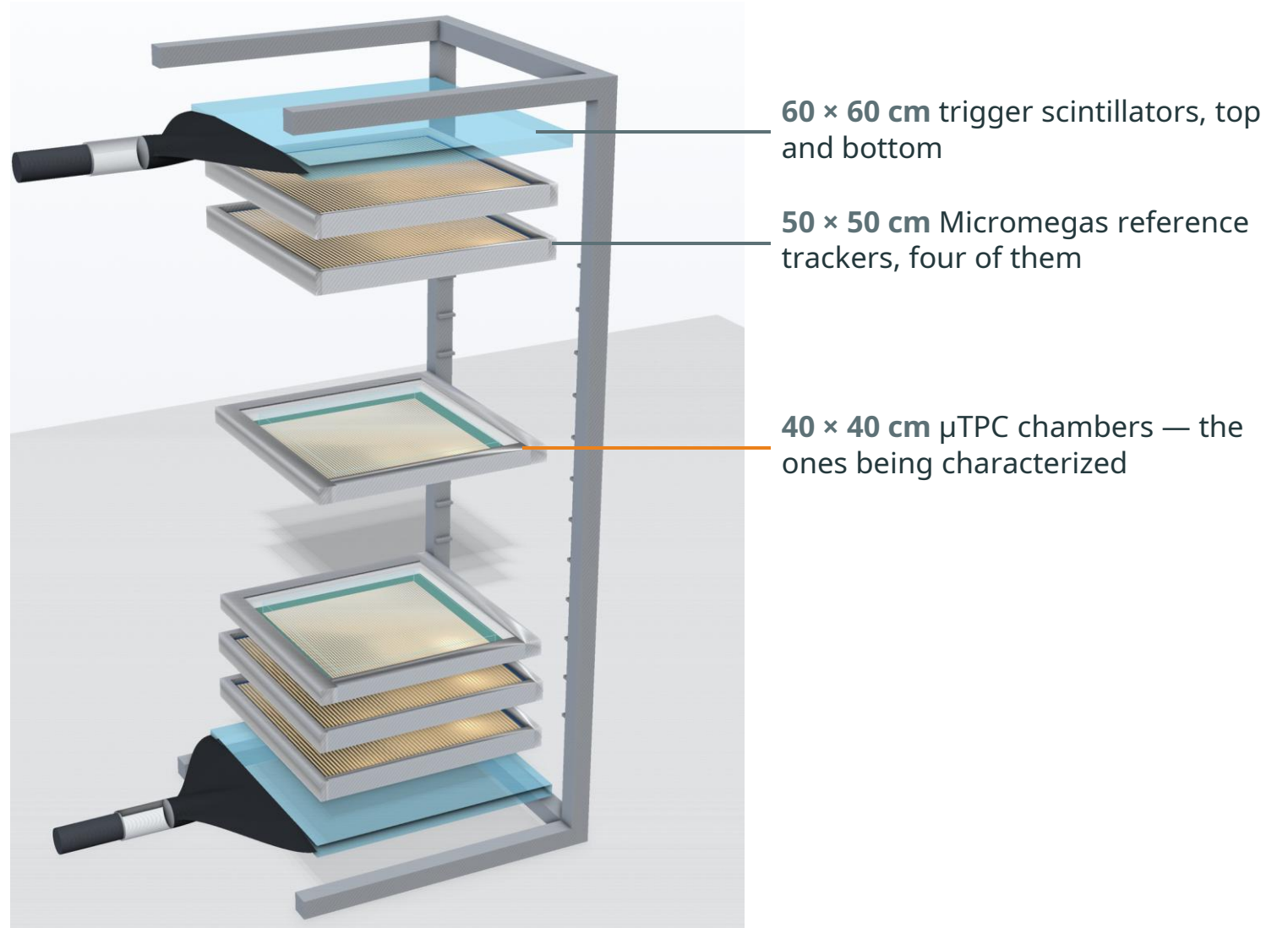
60 × 60 cm trigger scintillators, top and bottom

50 × 50 cm Micromegas reference trackers, four of them

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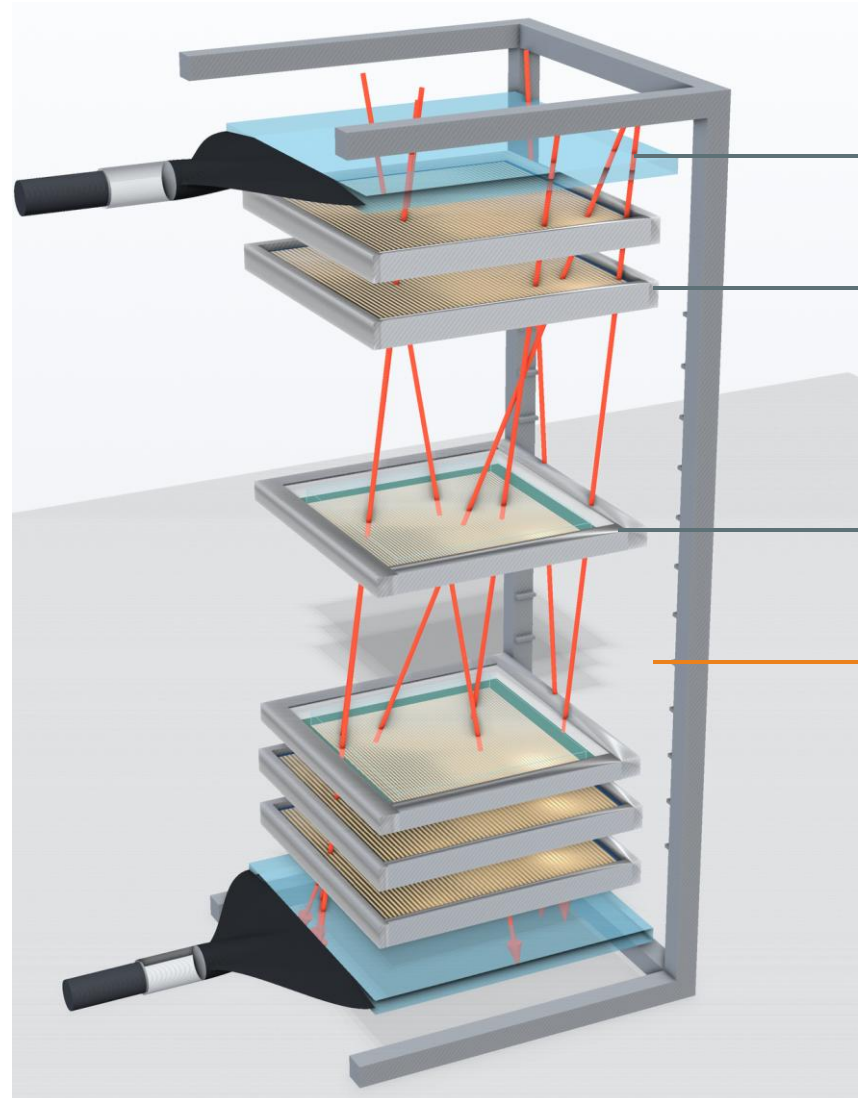
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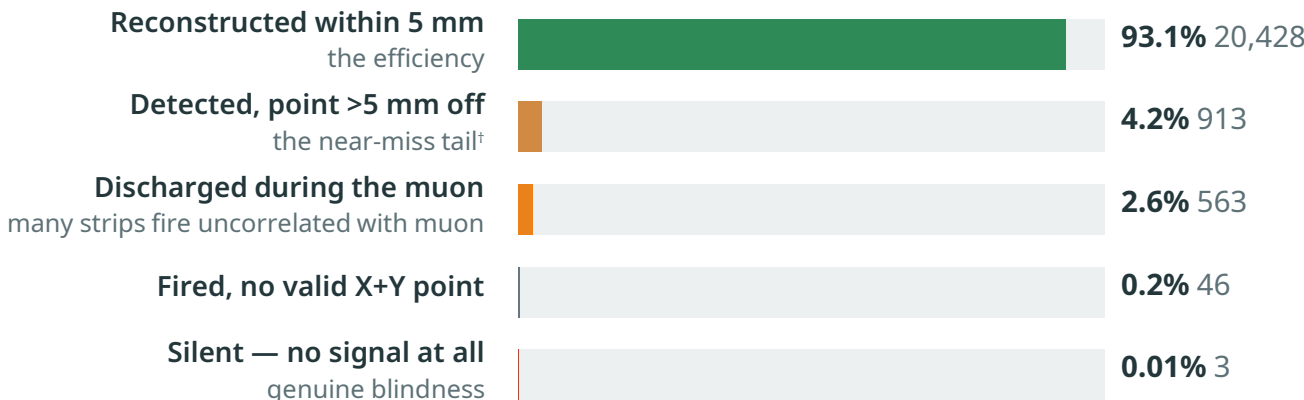
50 × 50 cm Micromegas reference trackers, four of them

40 × 40 cm μ TPC chambers — the ones being characterized

Cosmic muons — two slots measured at once, against the same reference

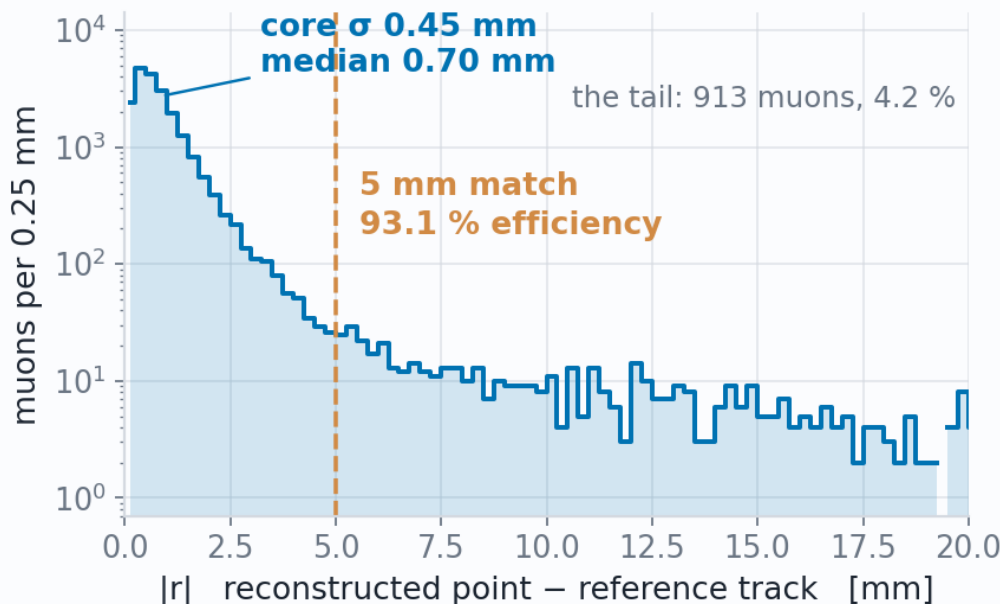
Efficiency: 93% on the best chamber

Where every crossing muon goes

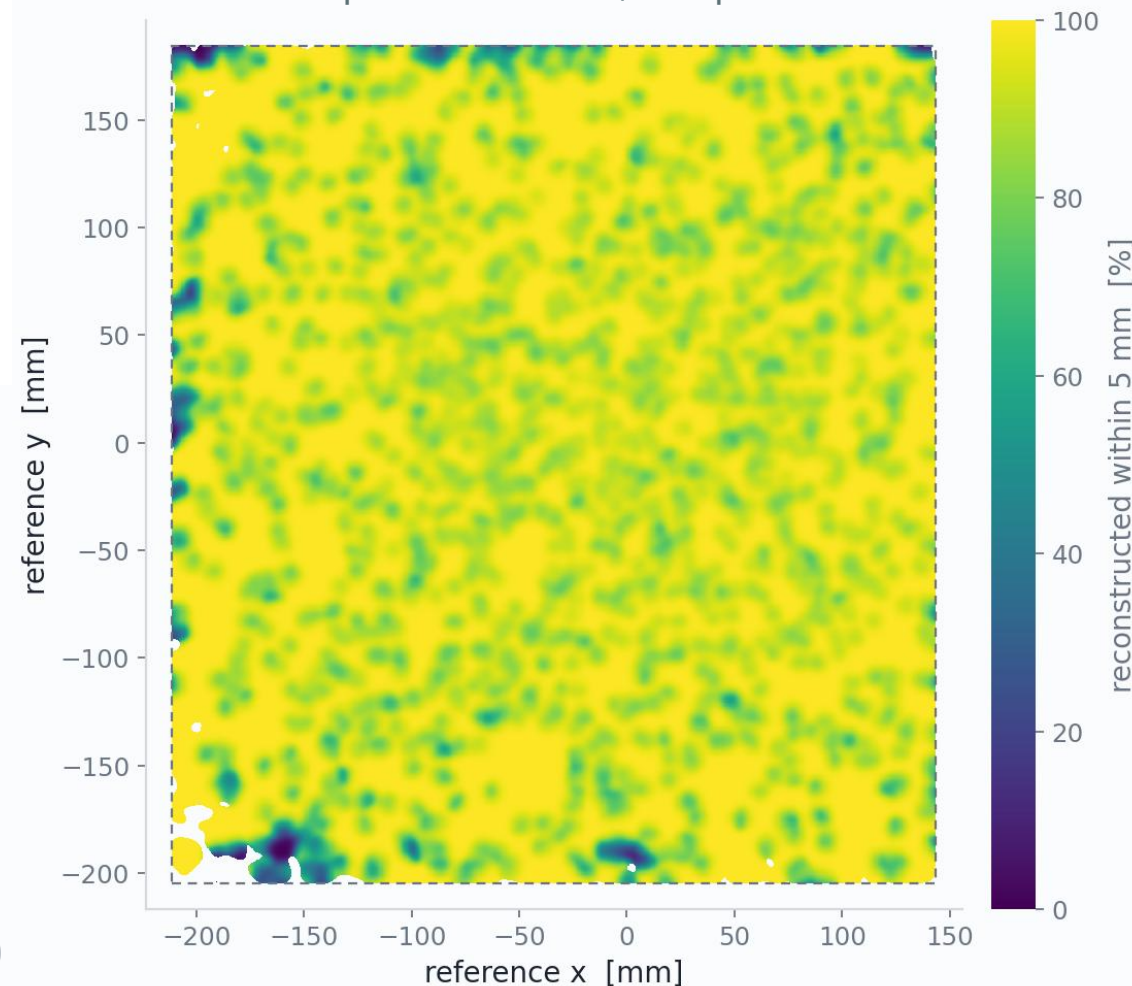


WHY 0.45 mm

Reference trackers
resolve to ~500 μm

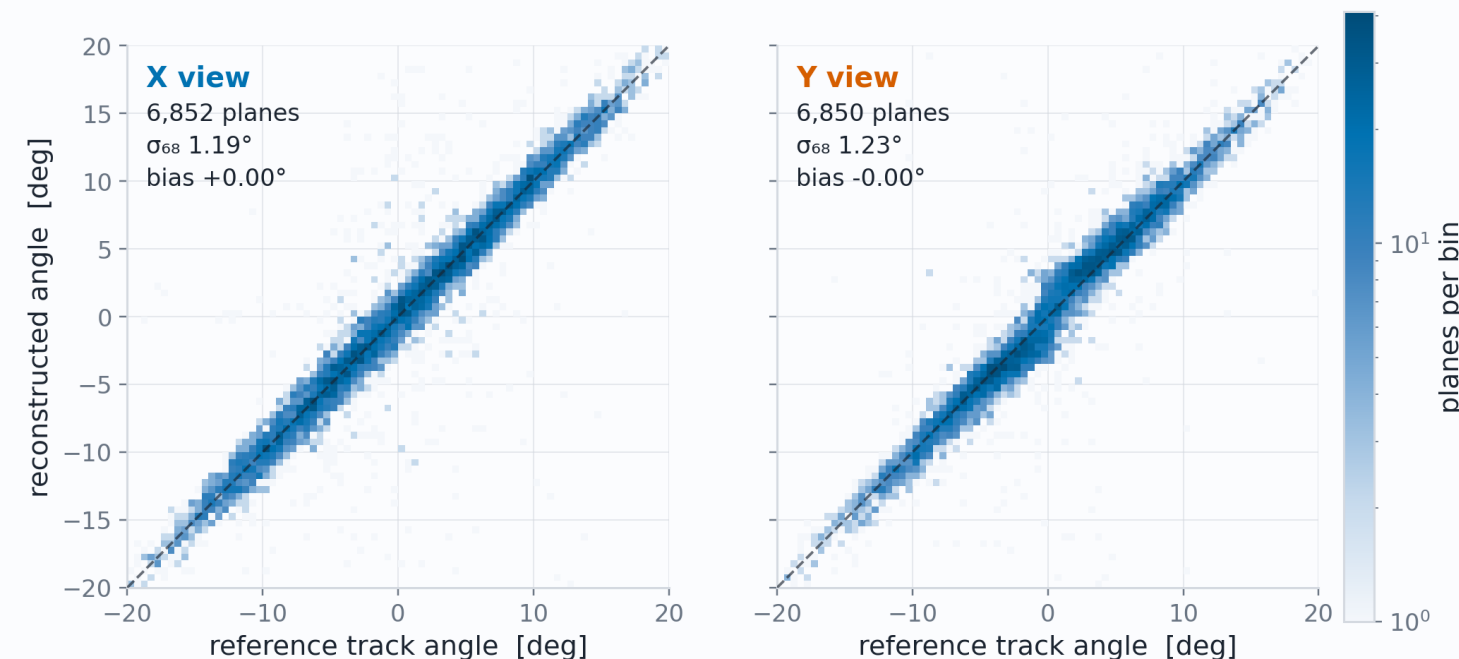


Flat across the chamber face
a circle swept over the face, 500 μm at a time

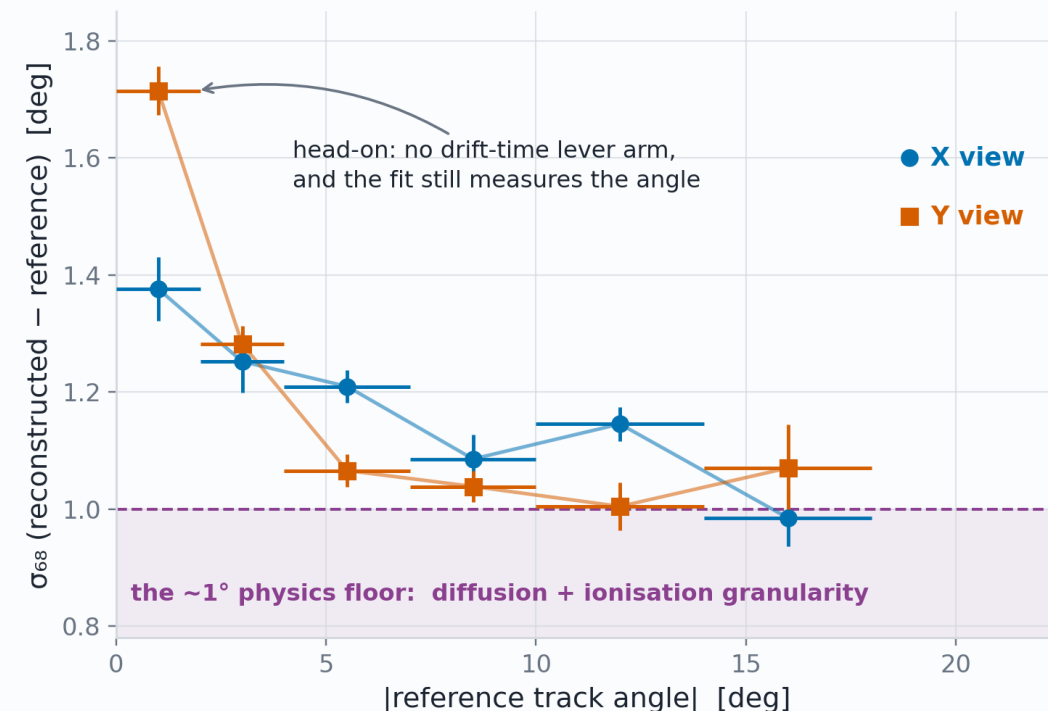


Adequate angular, spatial and temporal resolution

Reconstructed against reference, per plane



Head on tracks reasonable and angled tracks approach physics floor



1.19 / 1.23°

angle per plane, X / Y (σ_{68}), bias < 0.1°

180 μm

position at normal incidence, ± 0.035 mm conservatively

33 ns

plane-to-plane timing, ≈ 1.1 mm of drift

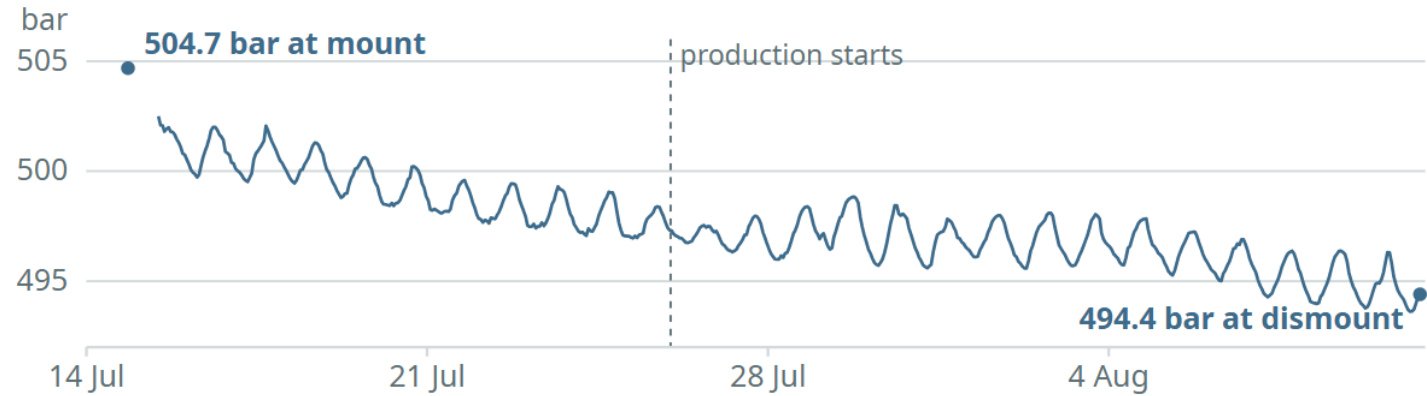
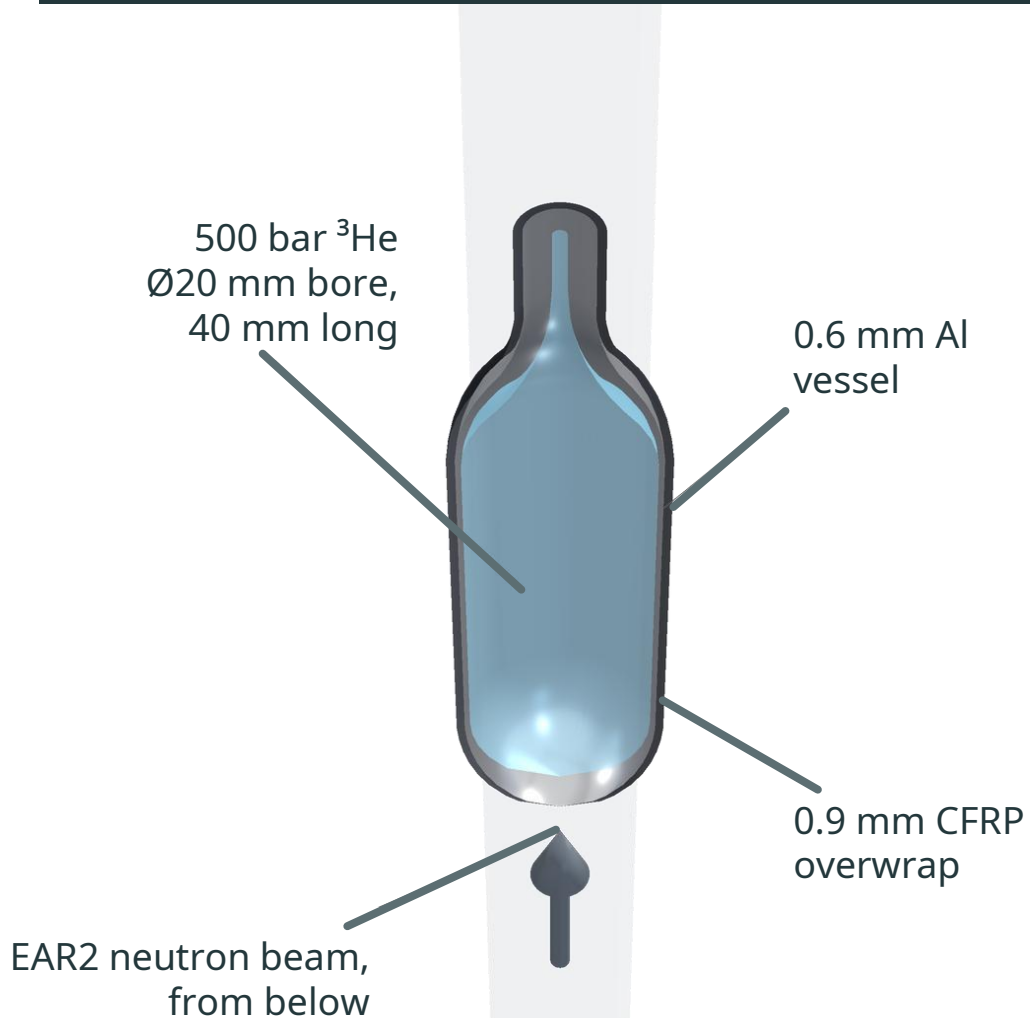


The n_TOF Search



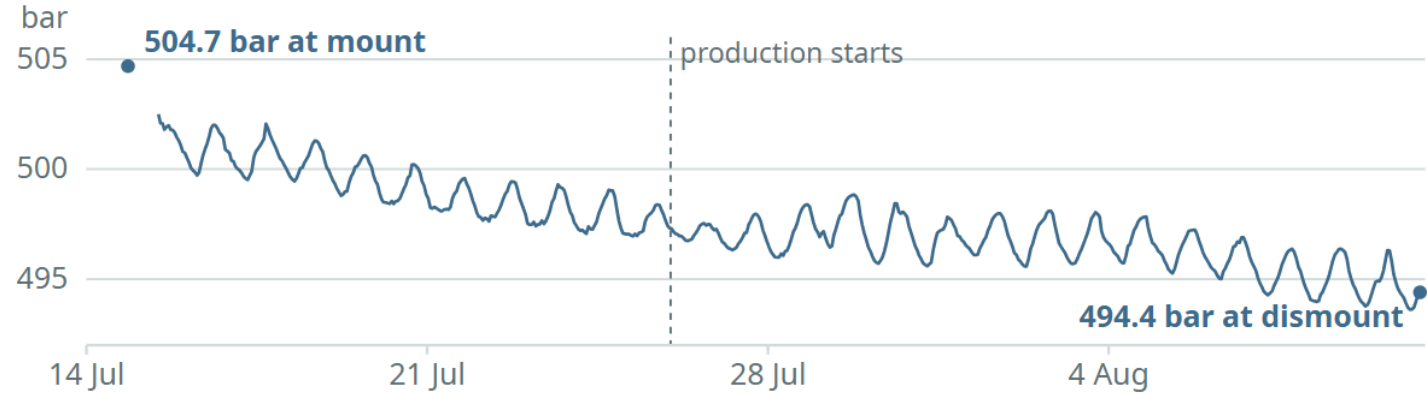
Four of these chambers, on a high intensity neutron beam at
EAR2.

The vessel, as built: 500 bar of ^3He in the beam



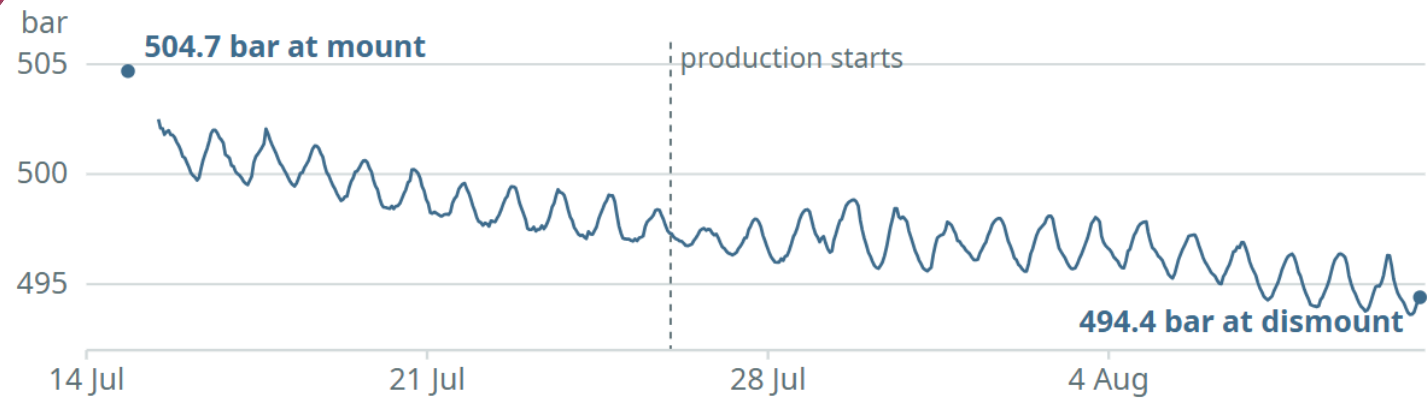
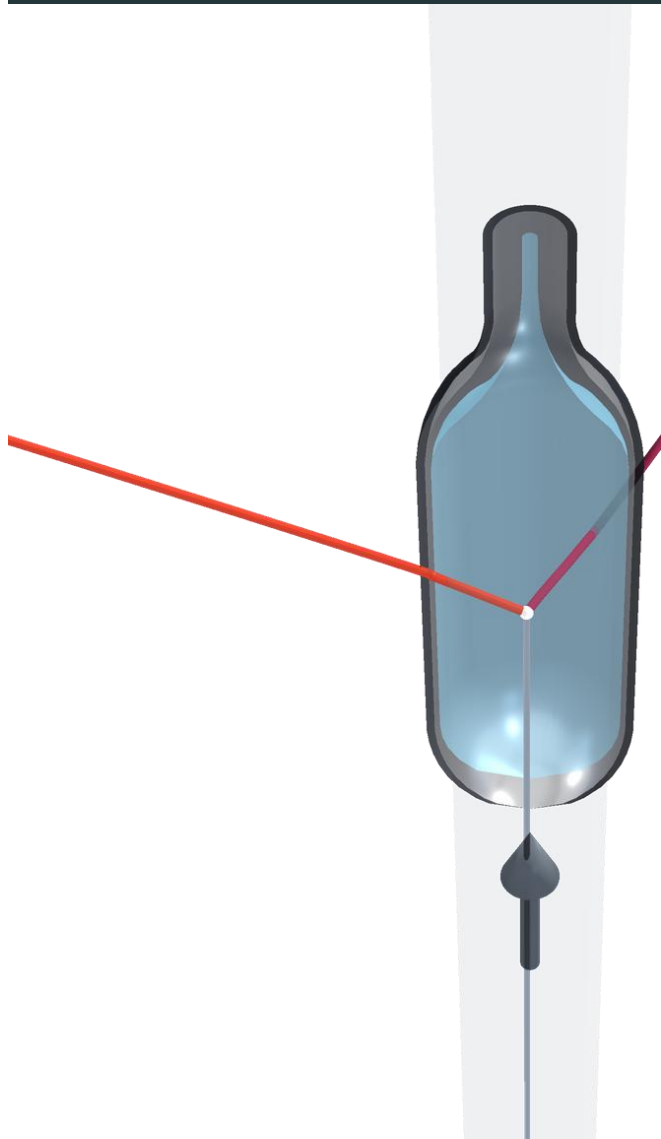
- ▶ **500 bar of ^3He** in as thin a capsule as we can build — least scattering on the way out, most interaction on the way in.
- ▶ Ø20 mm bore, 40 mm of gas, domed both ends — **walls of 0.6 mm Al + 0.9 mm CFRP.**

A neutron arrives



- ▶ 500 bar of ^3He in as thin a capsule as we can build — least scattering on the way out, most interaction on the way in.
- ▶ The **neutron beam arrives from below** and interacts in the ^3He gas volume.

...and a pair leaves

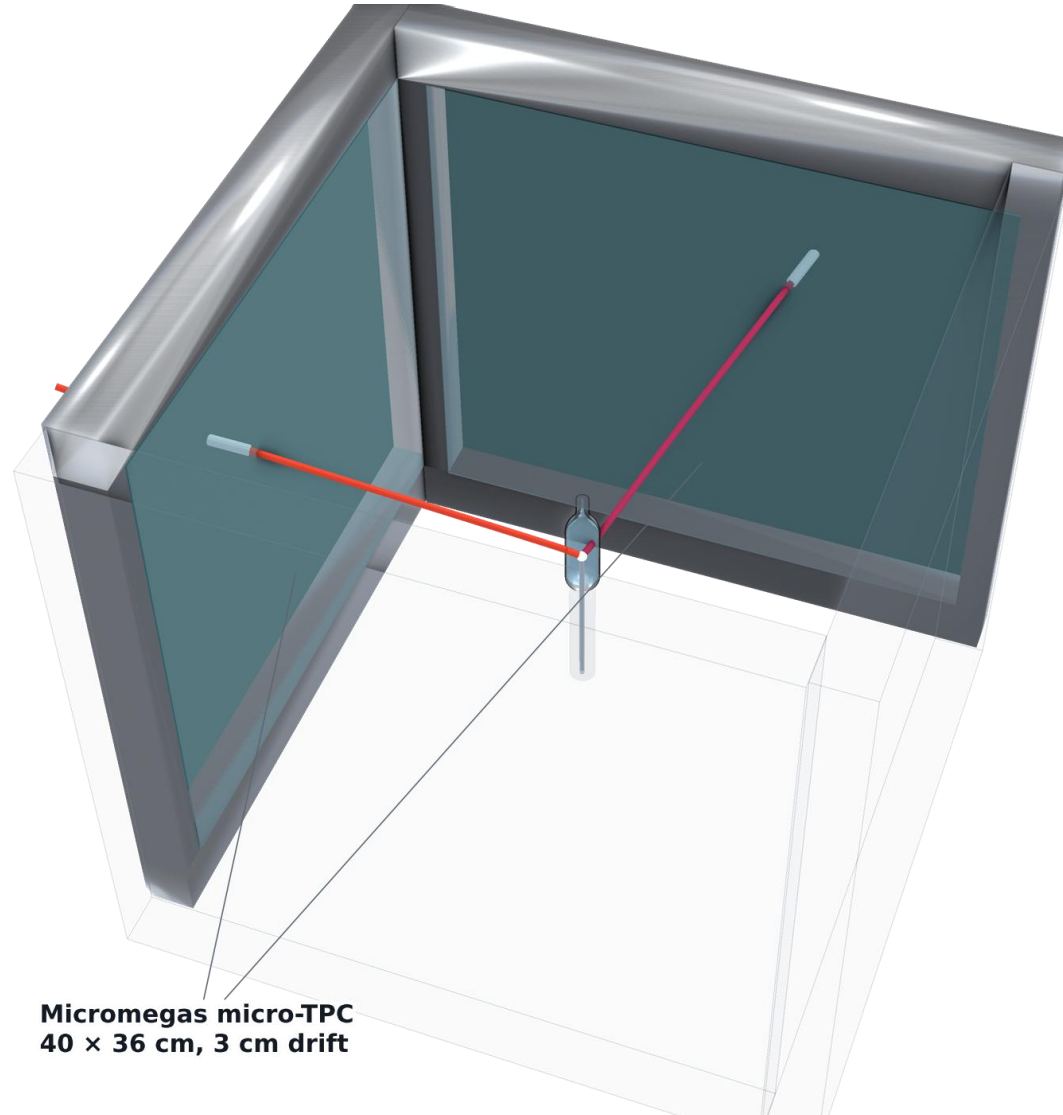


- ▶ **500 bar of ^3He** in as thin a capsule as we can build — least scattering on the way out, most interaction on the way in.
- ▶ The **neutron beam** arrives from below and interacts in the ^3He gas volume.
- ▶ Some probability of **internal pair conversion** — or of **X17** → **e^+e^-** — leaving the capsule, with some scattering in its wall.

— e^- — e^+ — neutron — beam

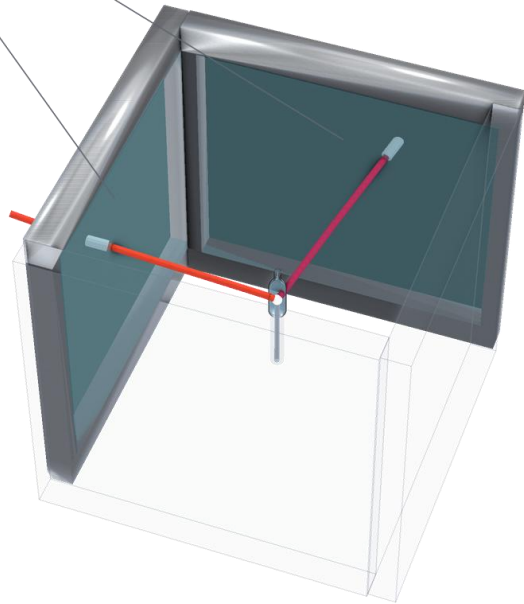
Now build something that can catch them

- ▶ Reduce **material budget** until the track angles have been recorded with four Micromegas μ TPCs



+ four Micromegas micro-TPCs

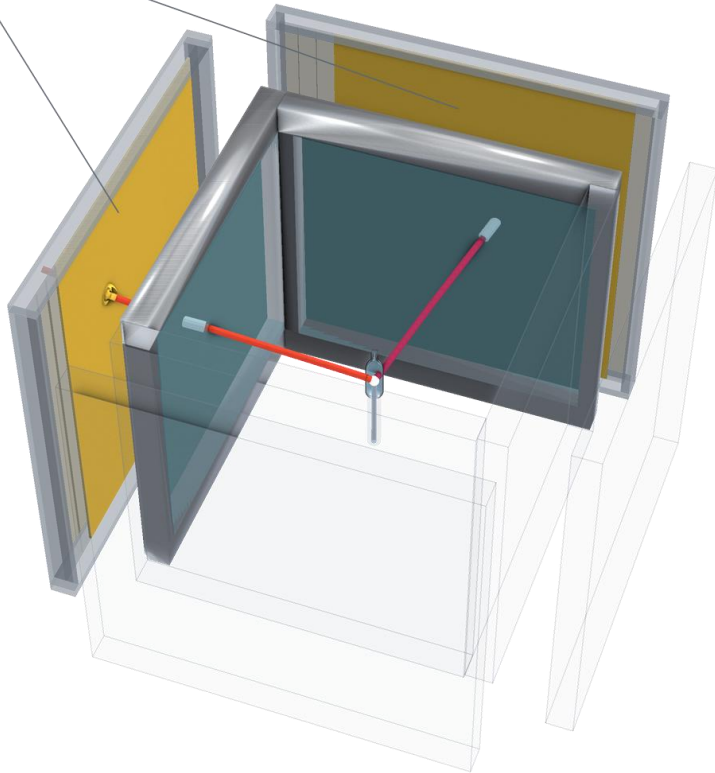
Micromegas micro-TPC
40 × 36 cm, 3 cm drift



- ▶ Reduce **material budget** until the track angles have been recorded with four Micromegas μ TPCs
- ▶ Need an **external trigger** for these trackers

+ the SiPM trigger wall

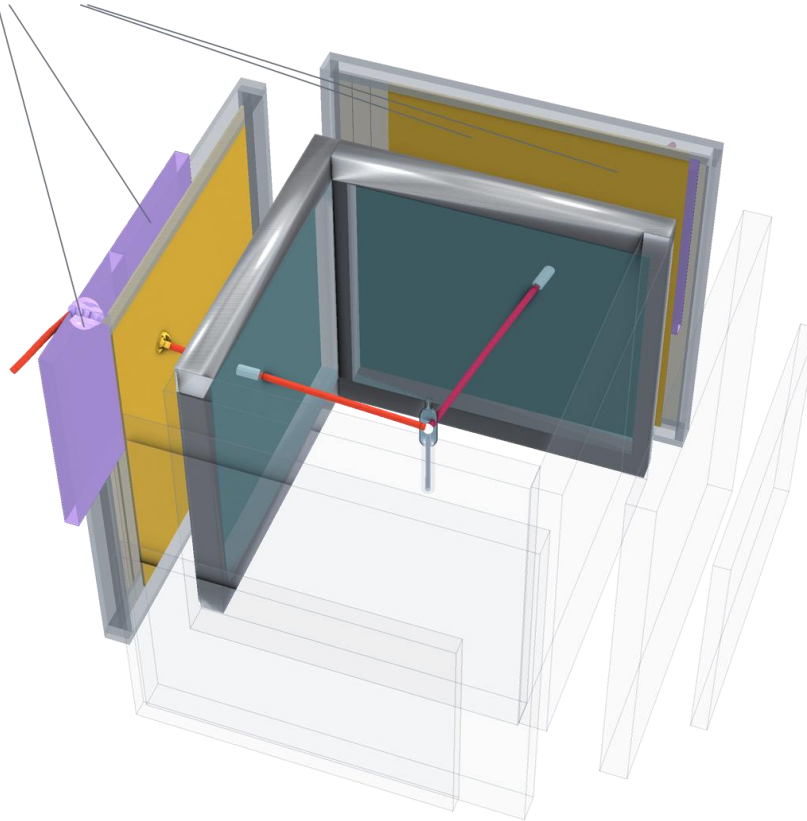
SiPM trigger wall
50 × 50 cm, 20 bars



- ▶ Reduce **material budget** until the track angles have been recorded with four Micromegas μ TPCs
- ▶ Need an **external trigger** for these trackers
- ▶ Thin vertical scintillator bars readout by SiPMs: **register the hit** without stopping the electron or the positron. Precise timing.

+ the plastic scintillators

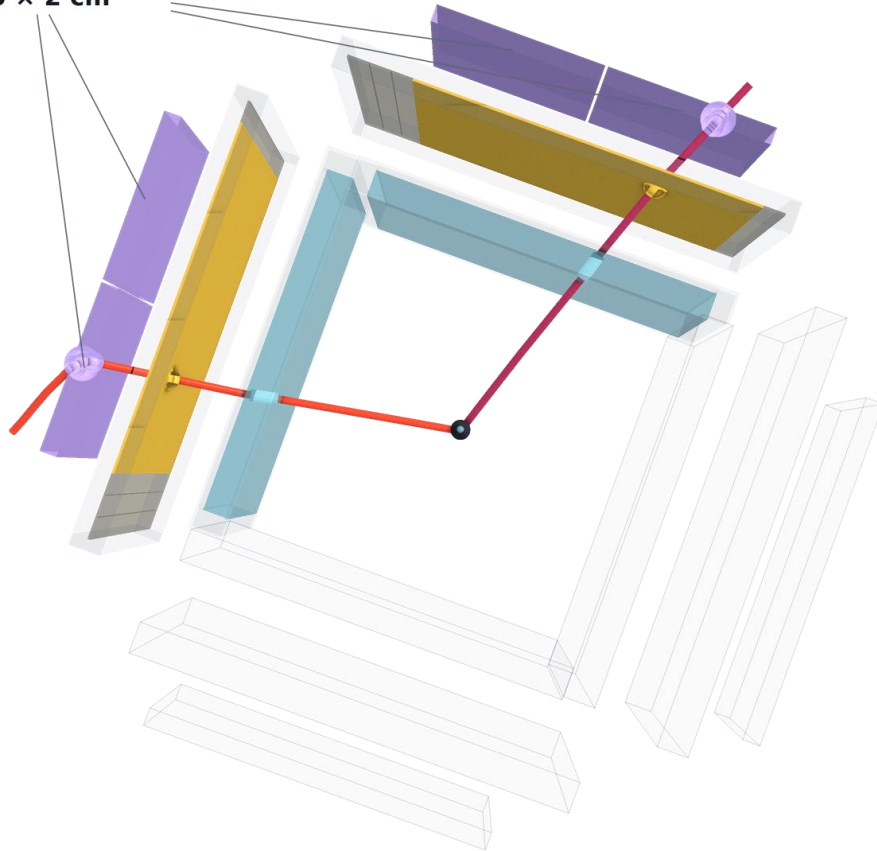
Plastic scintillators
20 × 30 × 2 cm



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- ▶ 2cm thick plastic scintillators **complete the coincidence** with the trigger wall.

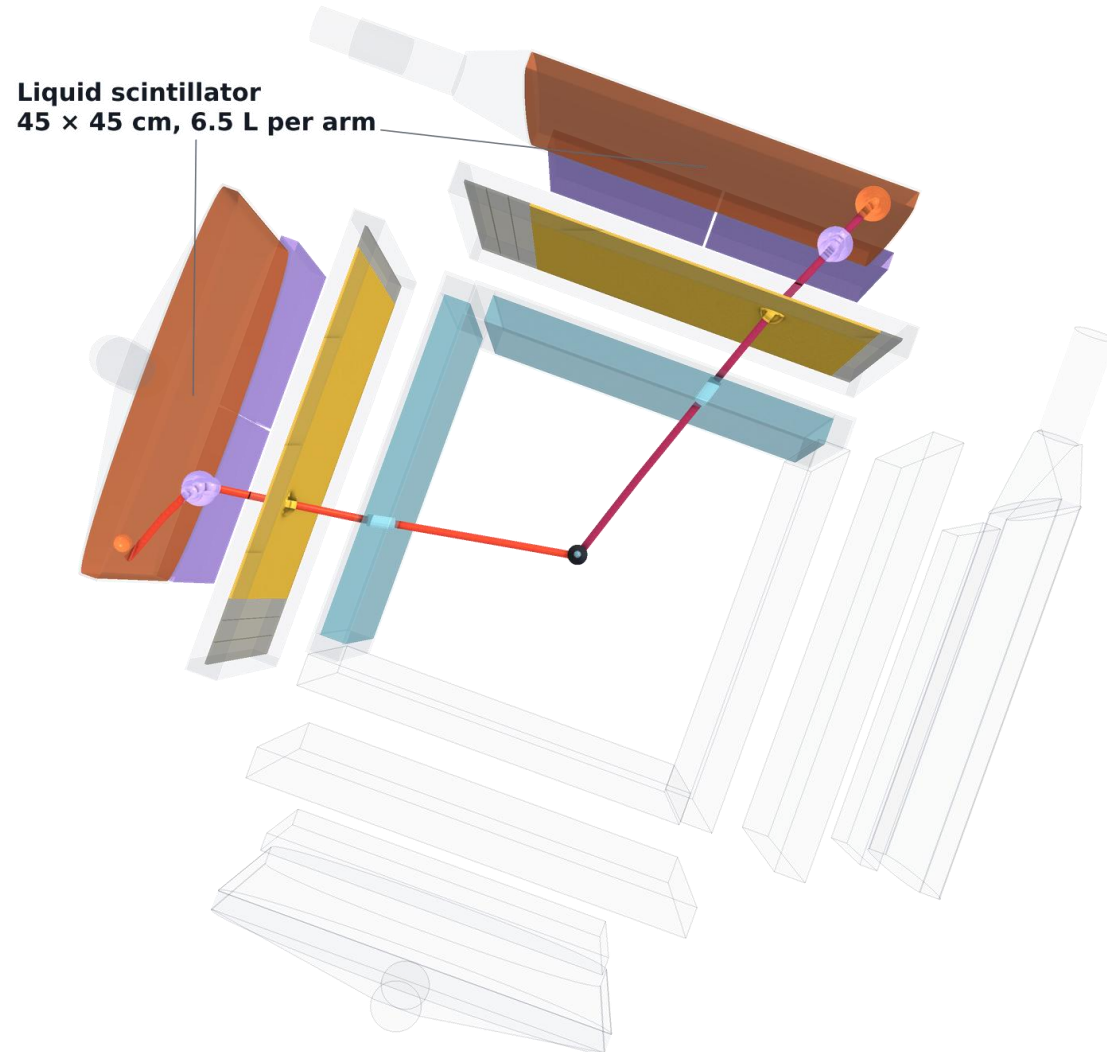
The same event from above — where the pair goes

Plastic scintillators
20 × 30 × 2 cm



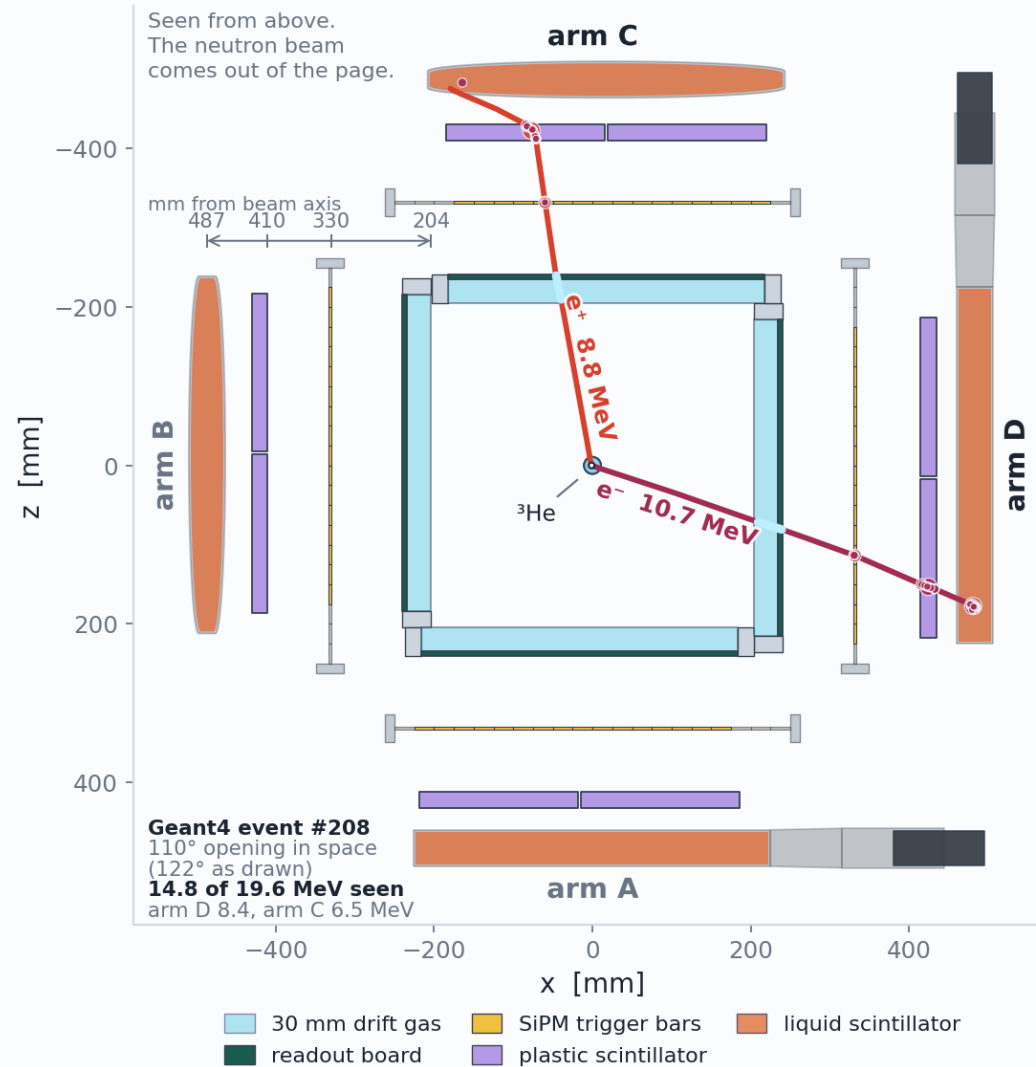
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- ▶ 2cm thick plastic scintillators **complete the coincidence** with the trigger wall.
- ▶ Also provides stopping for **calorimetry**

+ liquid-scintillator calorimetry — the full setup



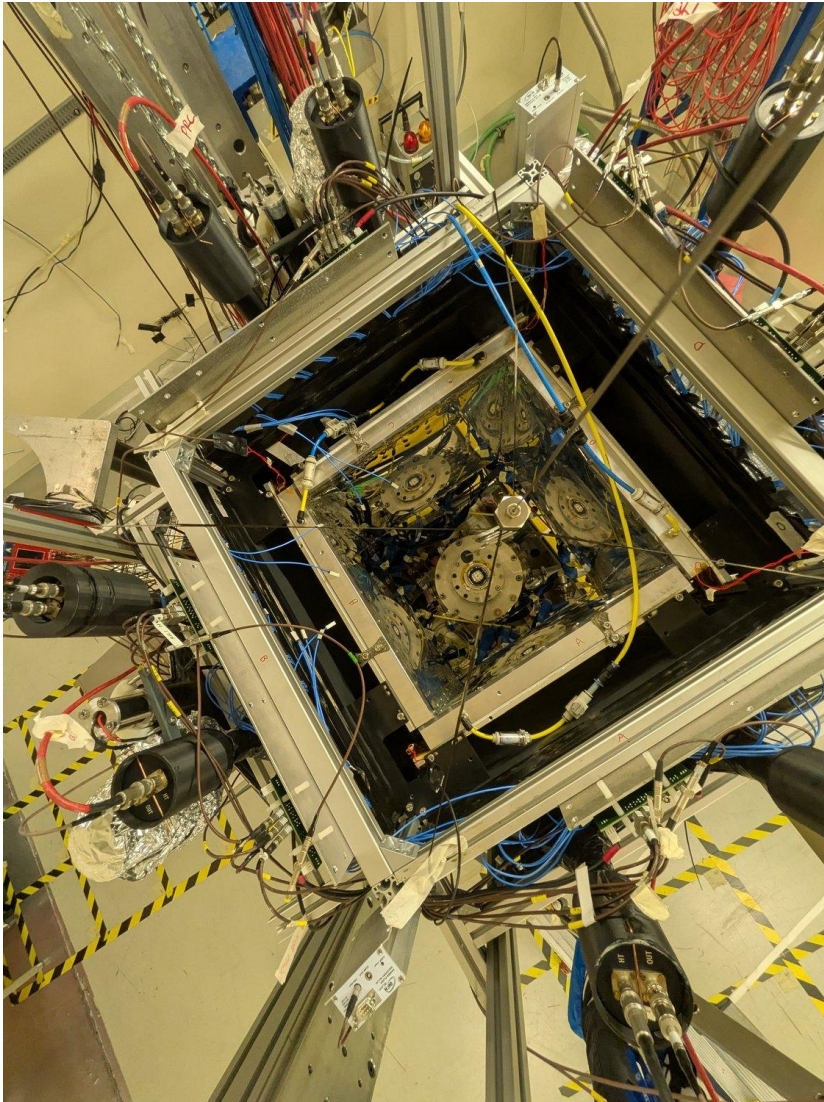
- ▶ Reduce **material budget** until the track angles have been recorded with four Micromegas μ TPCs
- ▶ Need an **external trigger** for these trackers
- ▶ Thin vertical scintillator bars readout by SiPMs: **register the hit** without stopping the electron or the positron. Precise timing.
- ▶ 2cm thick plastic scintillators **complete the coincidence** with the trigger wall.
- ▶ Also provides stopping for **calorimetry**
- ▶ Add 2cm of **liquid scintillator** to stop the pair completely

The same setup from above



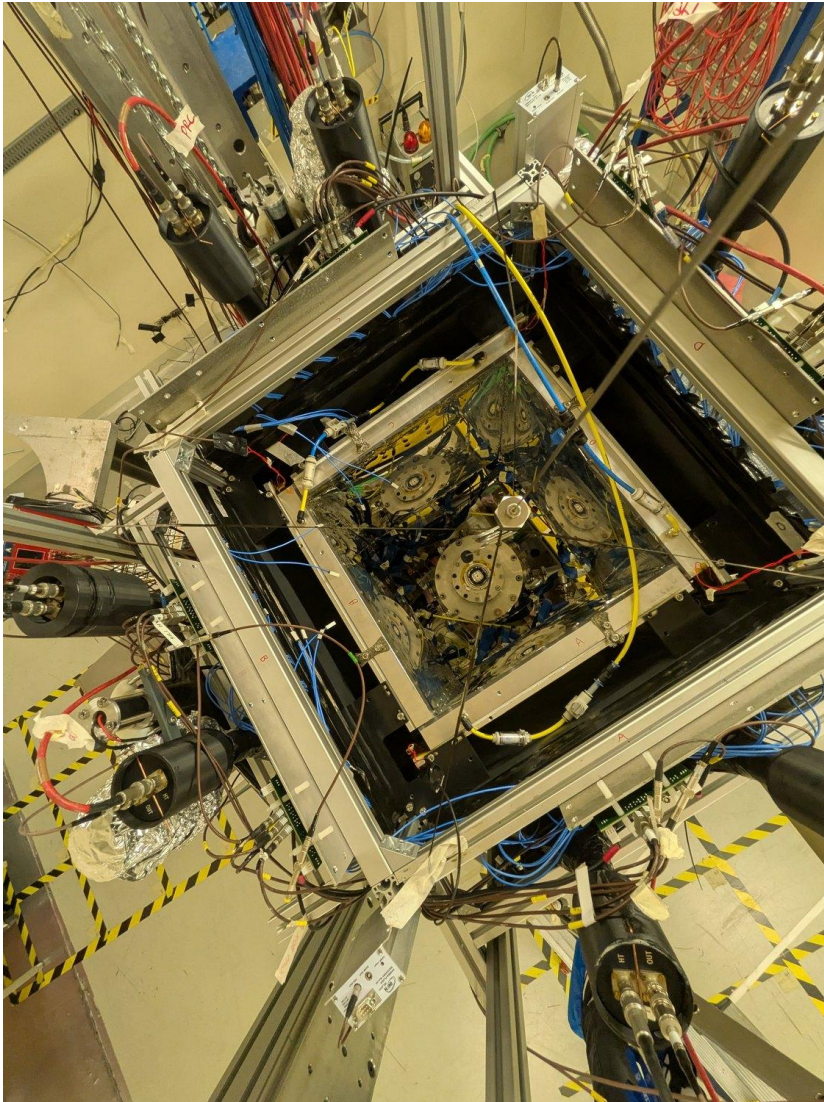
- ▶ **Opening angle** from the Micromegas, before anything else touches the pair.
- ▶ **Trigger and timing** from the wall and the plastics.
- ▶ **Calorimetry** from the full scintillator stack.

A look at the actual experiment – ran this July+August

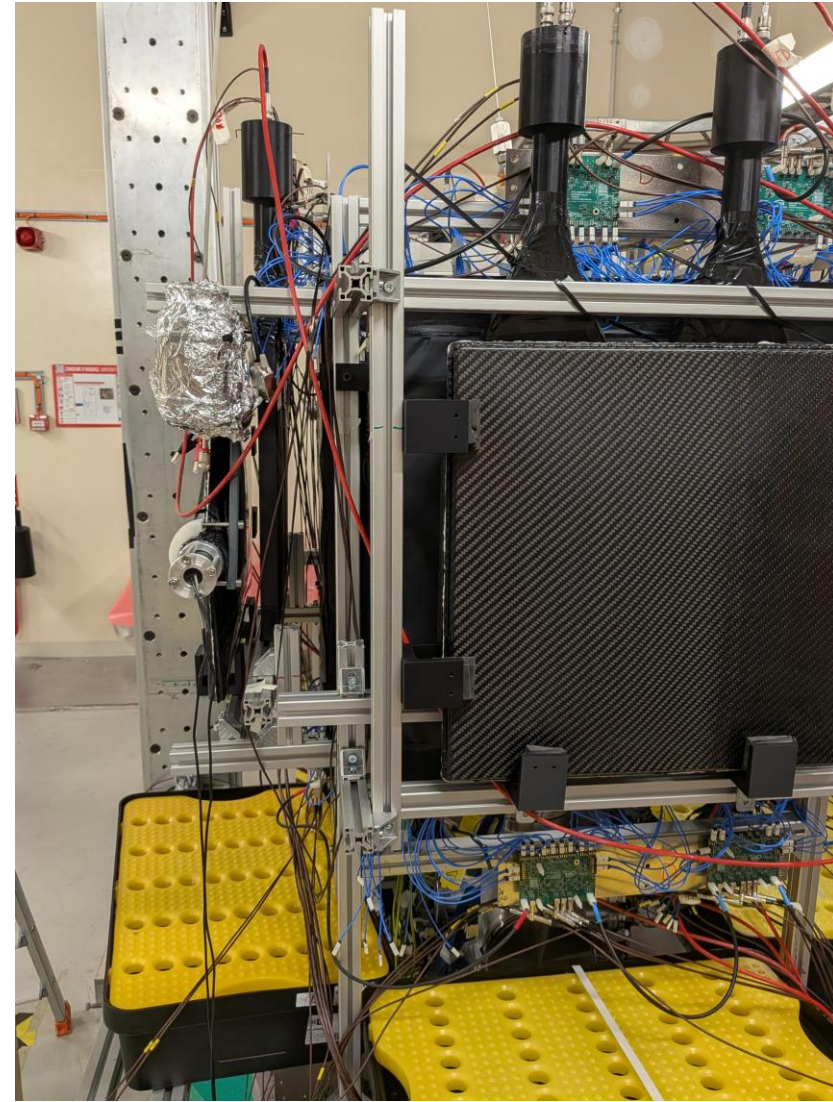


Looking down into the station · n_TOF EAR2, 10 August 2026

A look at the actual experiment – ran this July+August



Looking down into the station · n_TOF EAR2, 10 August 2026

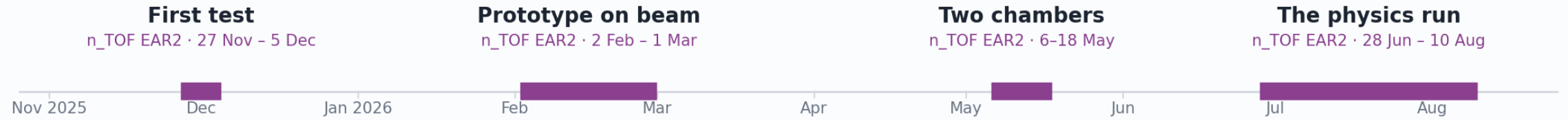


One arm, from outside · n_TOF EAR2, 10 August 2026

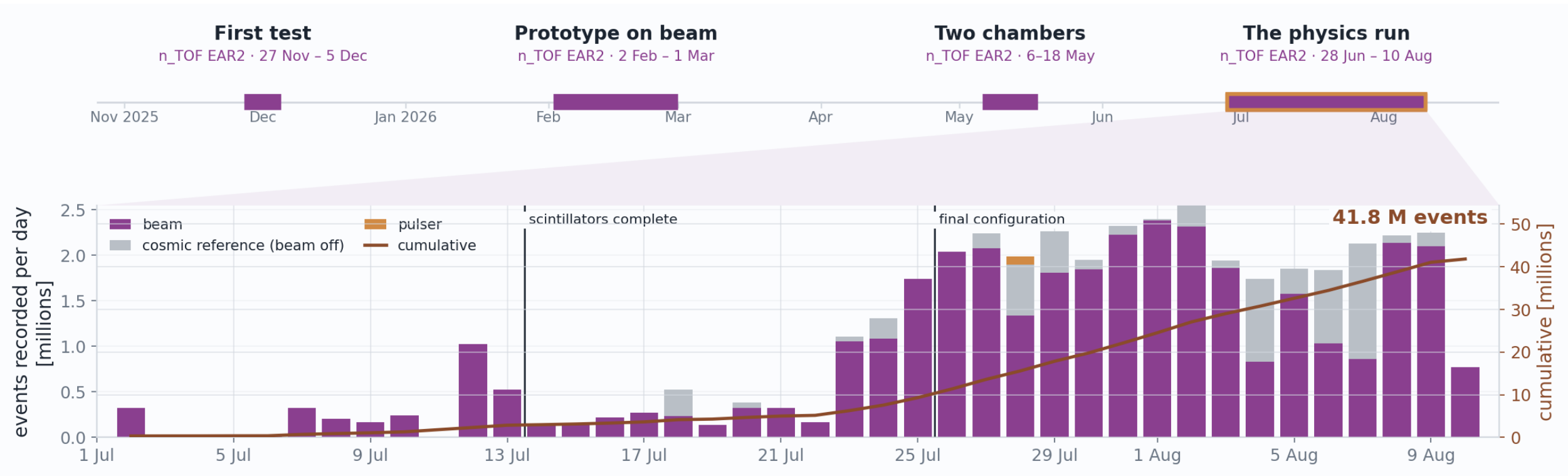
Measurement Status

What the campaign actually recorded, and the problems we faced.

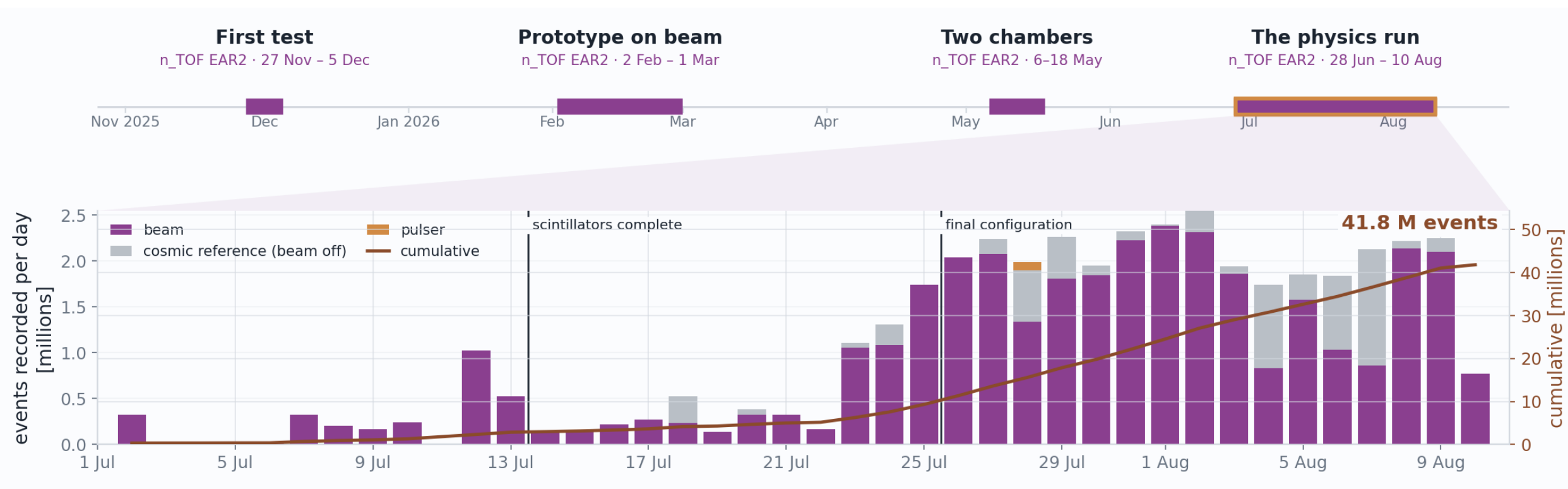
Experiment timeline and statistics



Experiment timeline and statistics



Experiment timeline and statistics



41.8 M

events recorded — 35.9 M on beam, 5.8 M cosmic reference

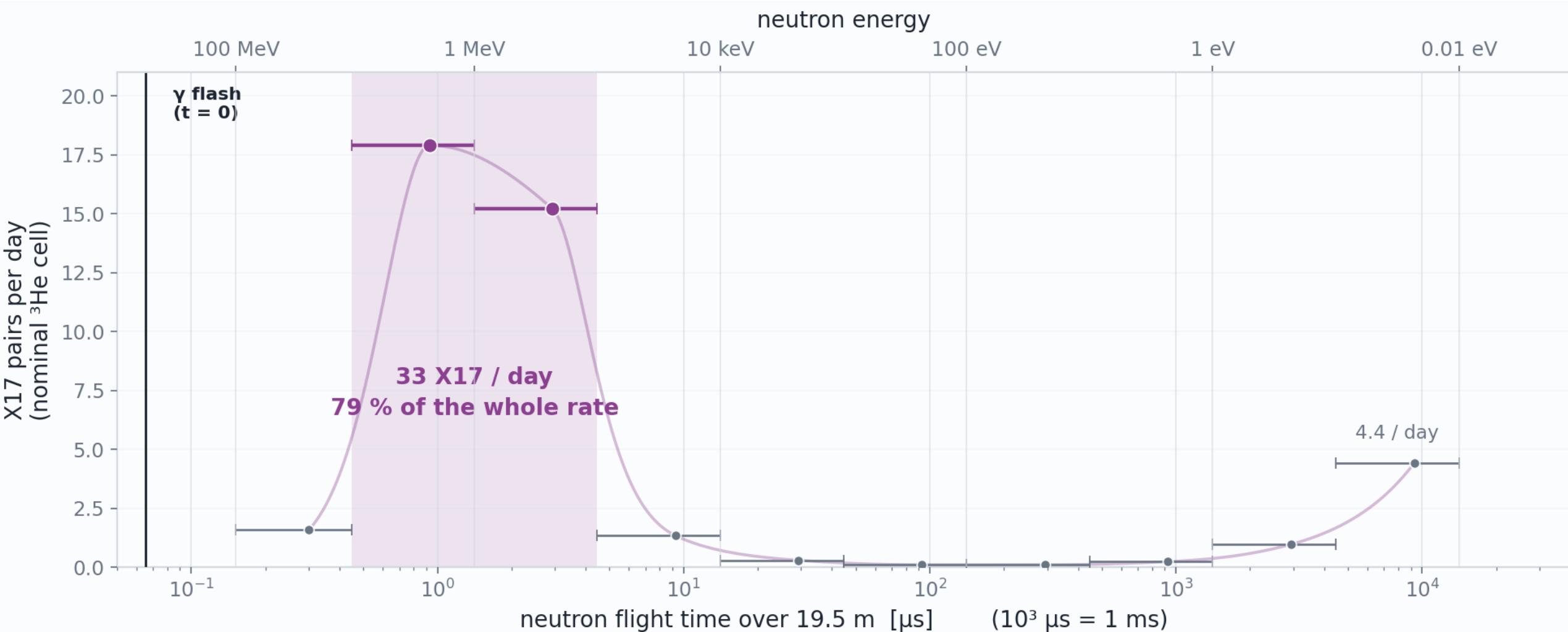
17.9 TB

on EOS, across 162 runs / 2,705 sub-runs

436,373

beam pulses on target over 44 days

Almost all of the X17 rate is in the MeV



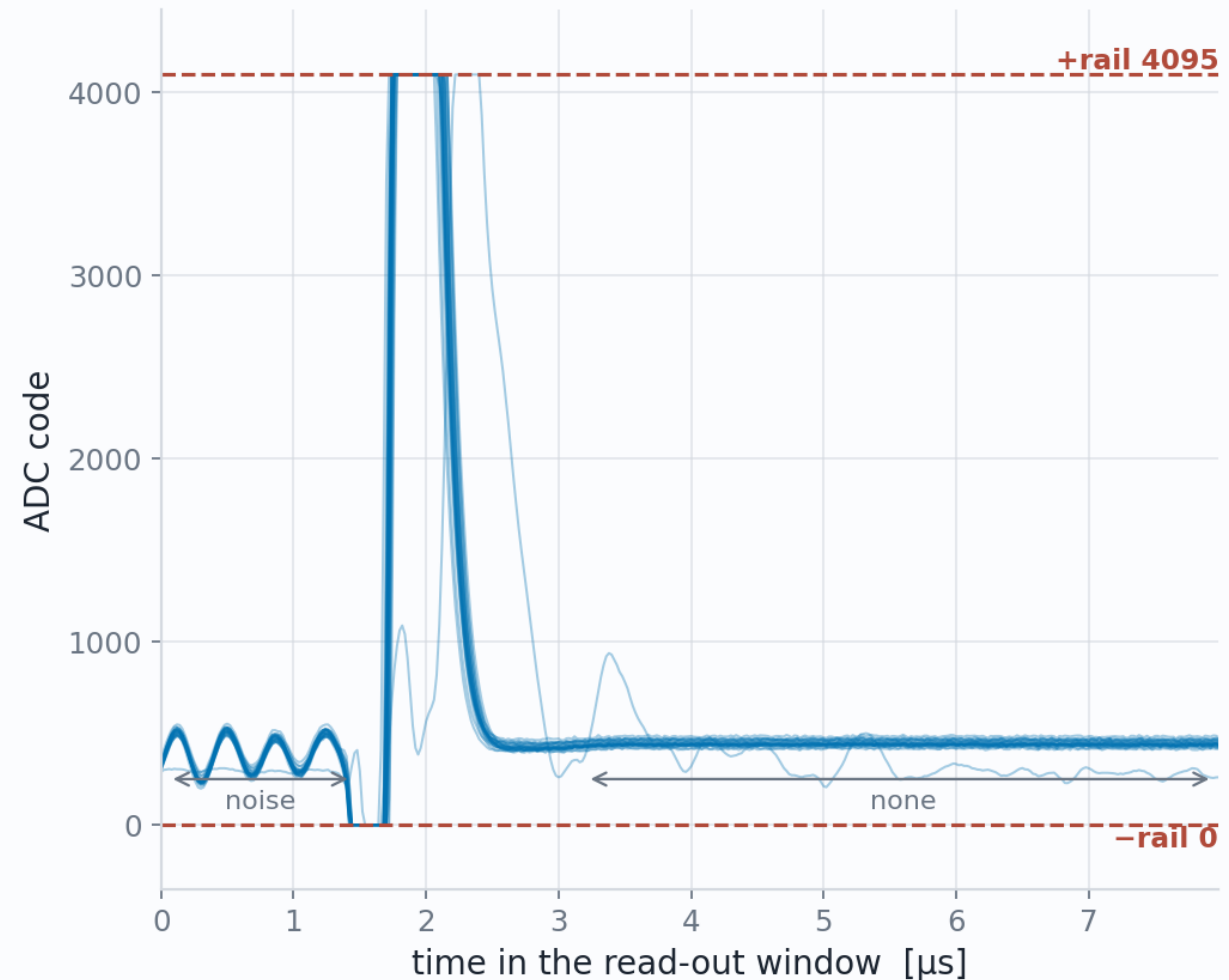
Our read-out is DREAM — and the γ flash rails every channel of it

- ▶ **DREAM** — *Dead-timeless Read-out Electronics ASIC for Micromegas*, developed at **CEA Saclay** and built for the **CLAS12** Micromegas vertex tracker.
- ▶ **512 channels per front-end unit** (8 ASICs $\times$ 64); **two units per chamber**, 1,024 channels for each MX17.
- ▶ Every channel: a **charge-sensitive preamplifier and shaper**, full scale switchable **50–600 fC**, into a switched-capacitor analogue memory.
- ▶ Sampled here at **20 ns**, digitized on an external trigger.

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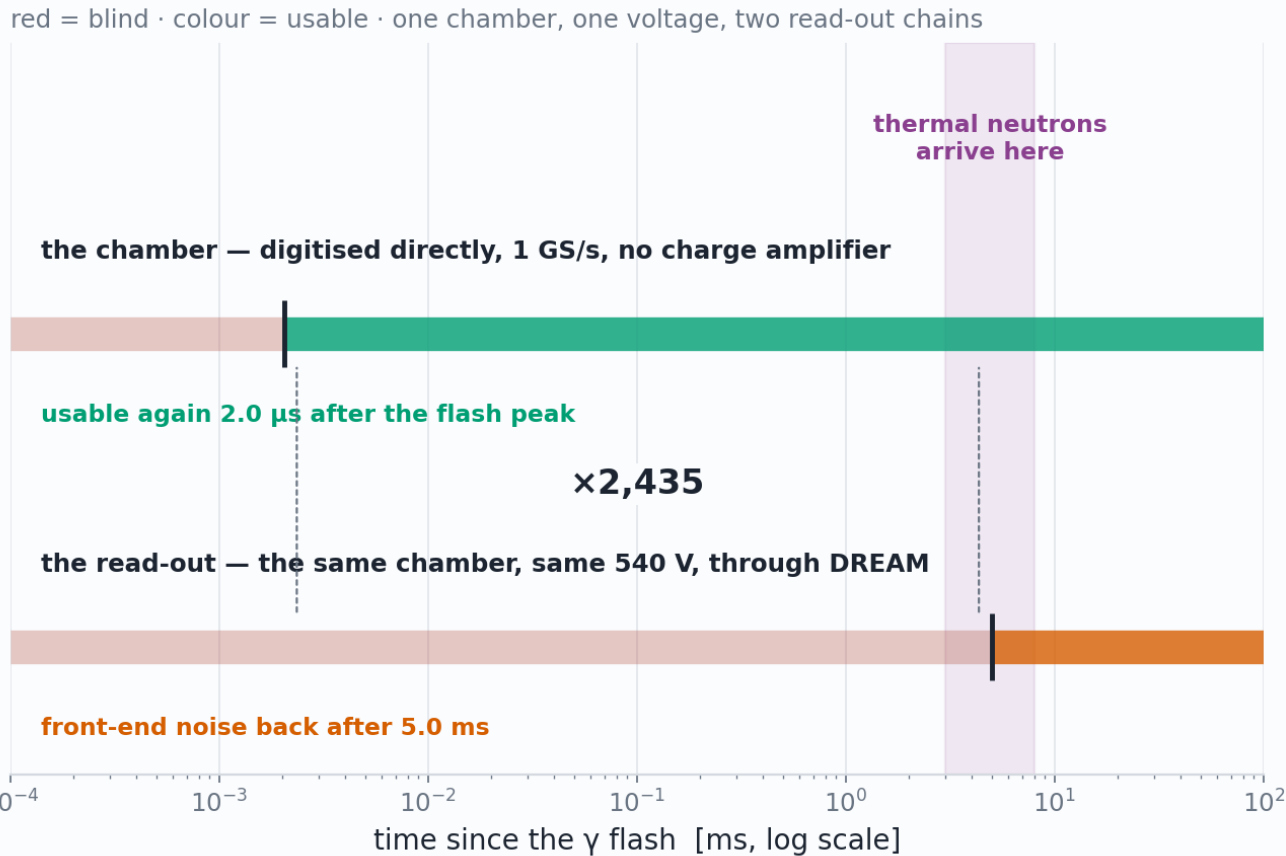
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- ▶ Sampled here at **20 ns**, digitized on an external trigger.
- ▶ **February 2026, the first prototype on beam:** saturated — and saturated in a way that hides itself. The ADC rails to **+4095**, crosses to **0**, and settles onto a flat baseline within **$\sim 3 \mu\text{s}$** .

One flash, the 24 hottest strips

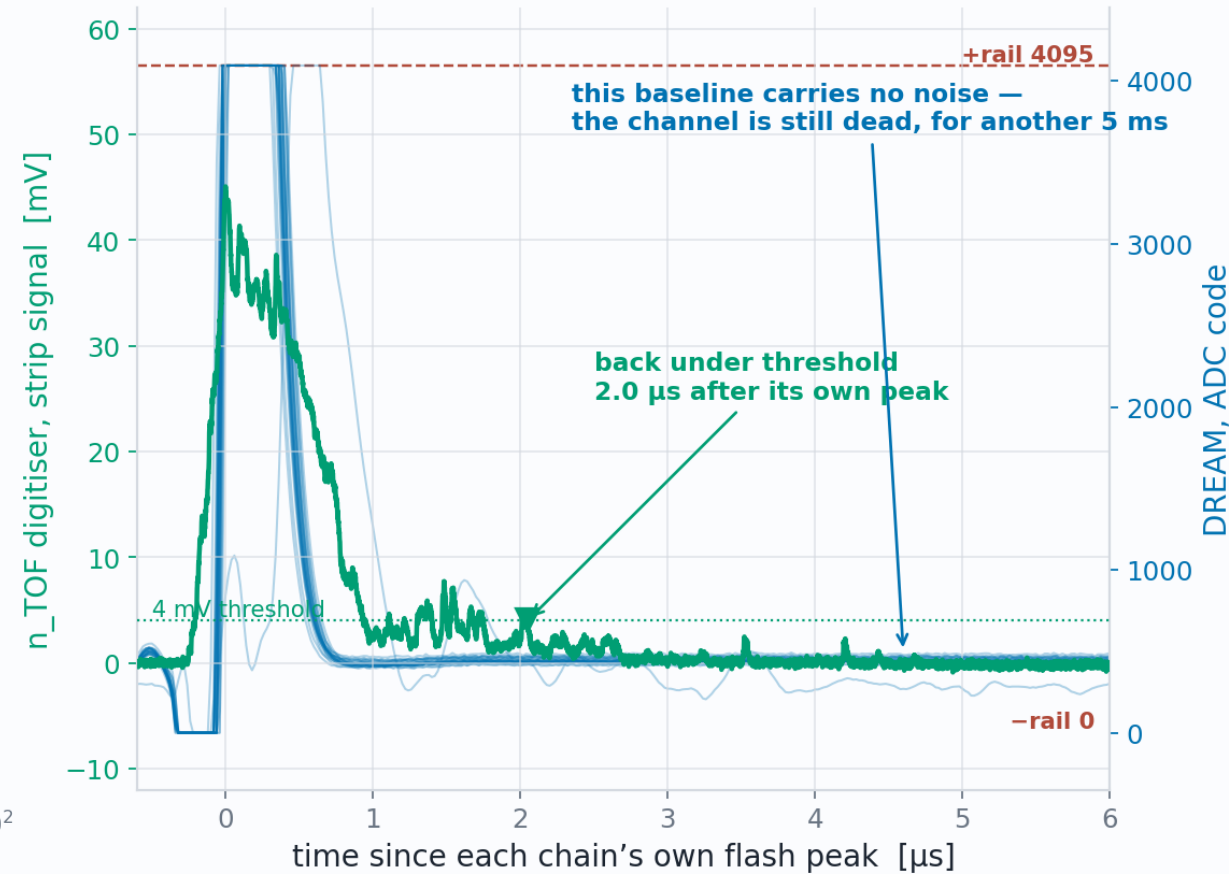


The detector is fine. The DREAM DAQ is not.

When each DAQ is usable again

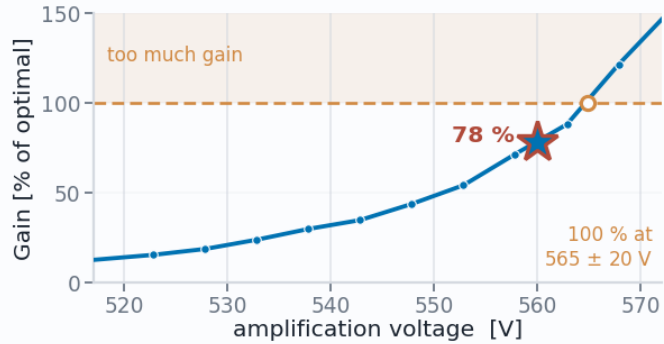


The same flash, seen by both DAQ aligned on their own peaks

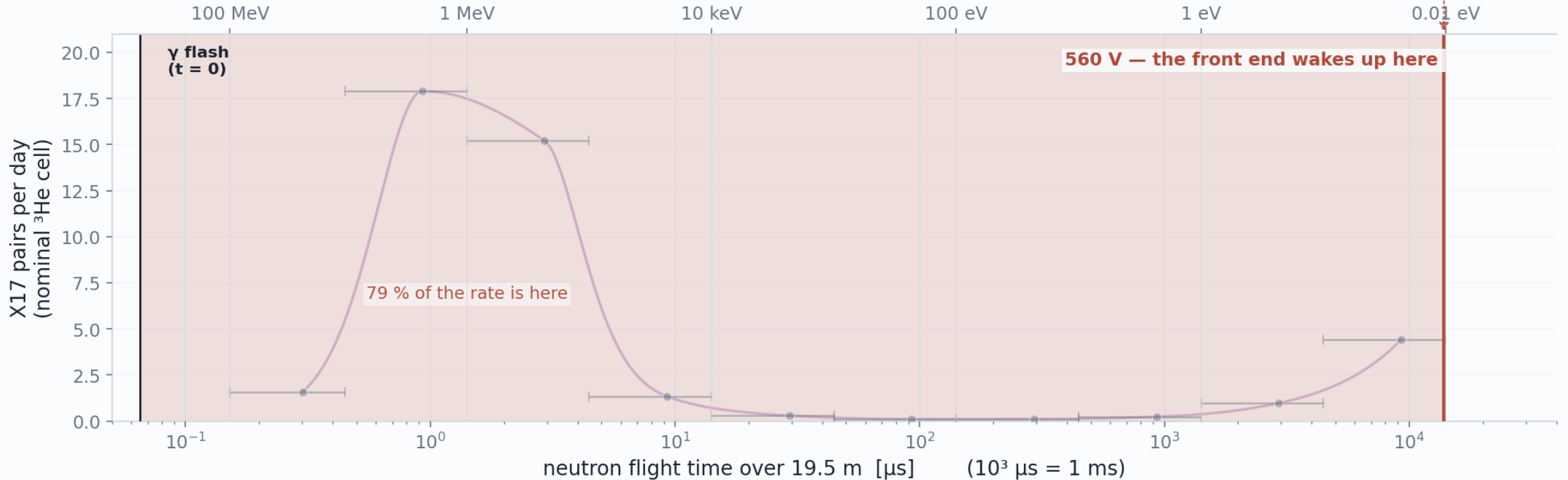
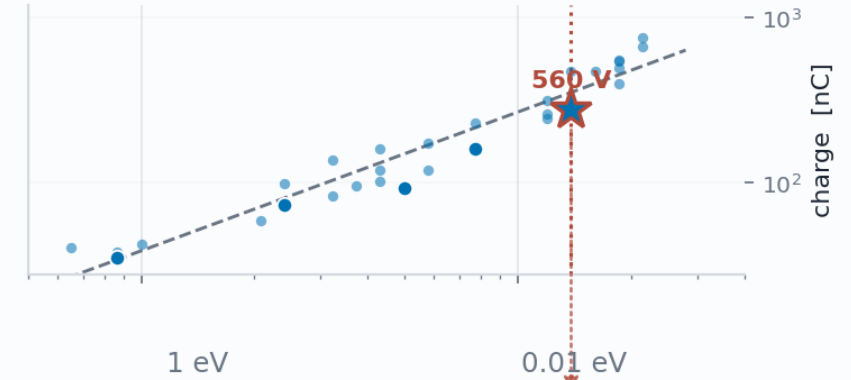


Detector A at 540V

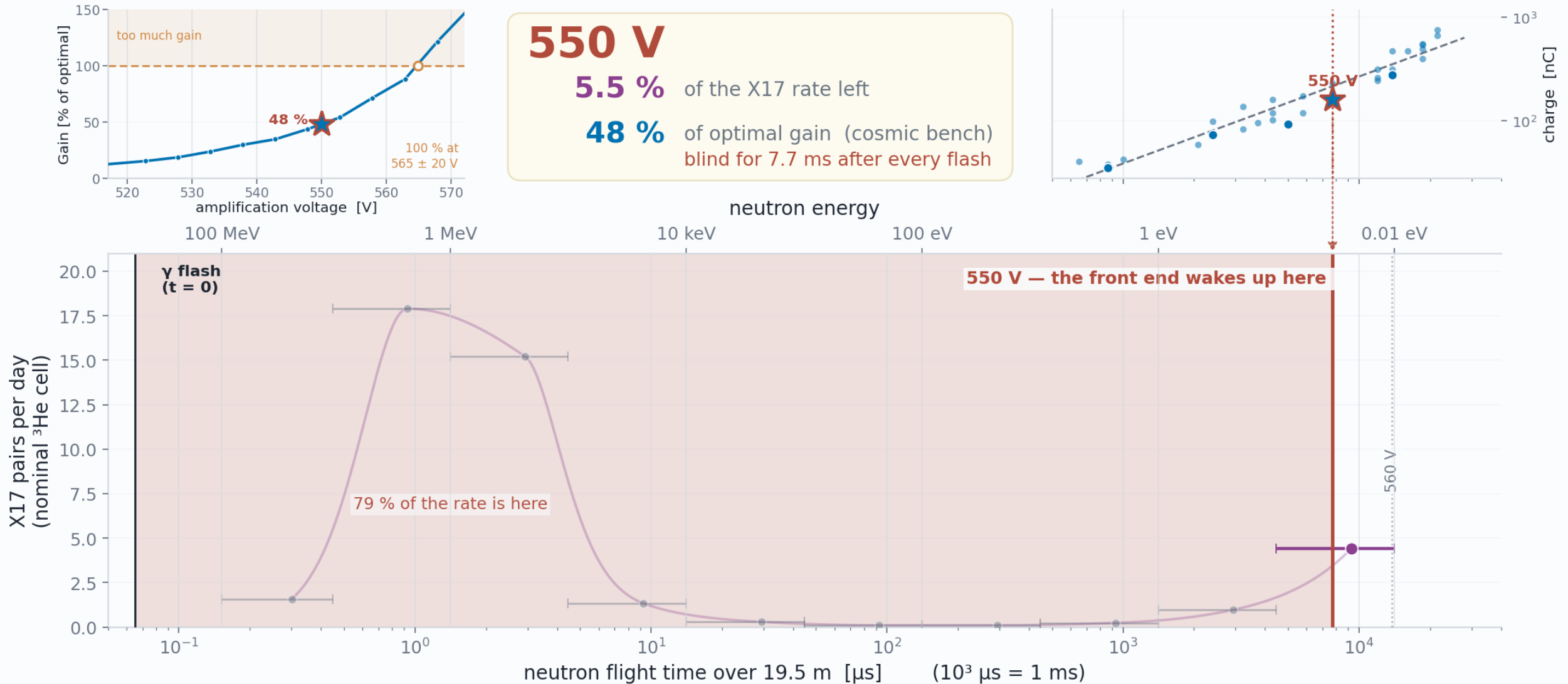
Every volt of gain costs milliseconds of beam



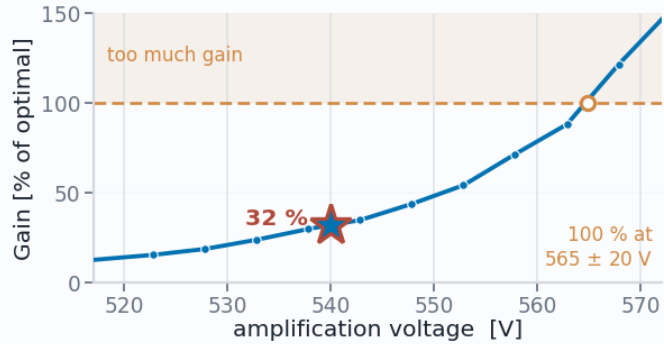
560 V where the gain wants to be
0.1 % of the X17 rate left
78 % of optimal gain (cosmic bench)
 blind for 13.9 ms after every flash



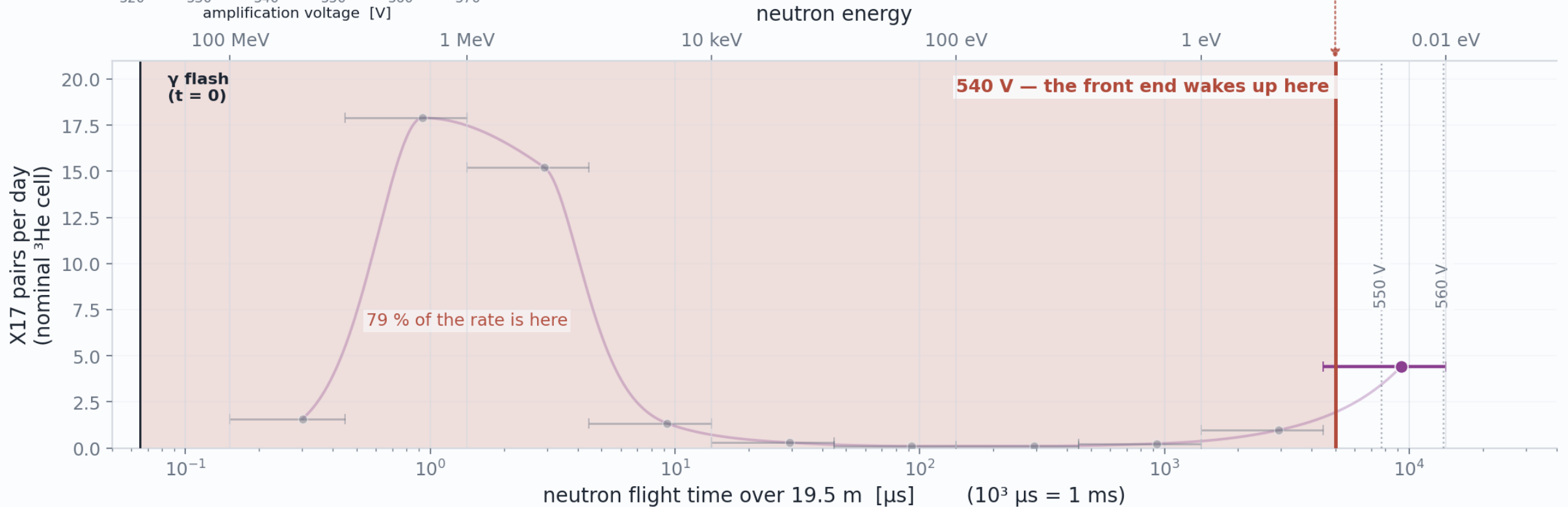
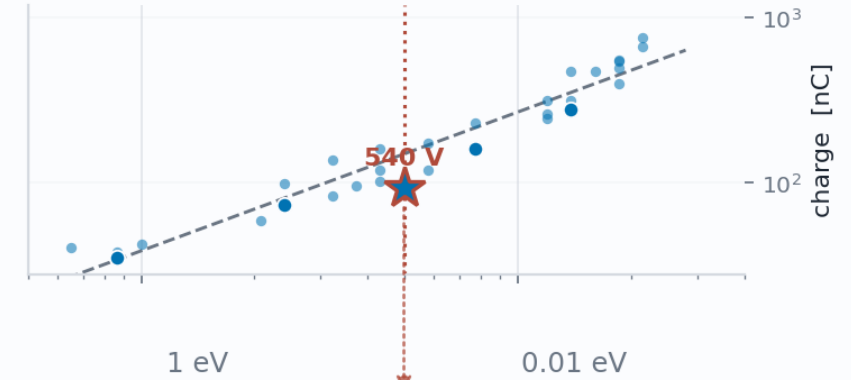
Every volt of gain costs milliseconds of beam



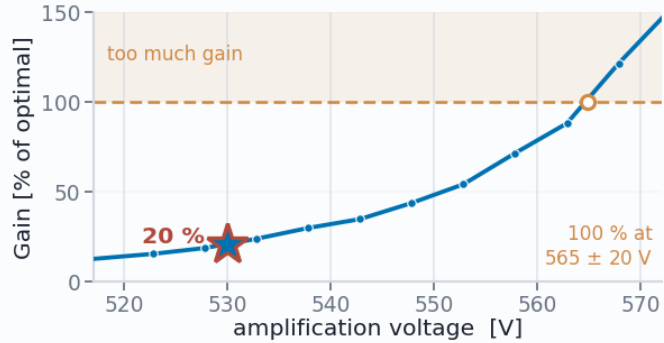
Every volt of gain costs milliseconds of beam



540 V where we ran
9.5 % of the X17 rate left
32 % of optimal gain (cosmic bench)
 blind for 5.0 ms after every flash



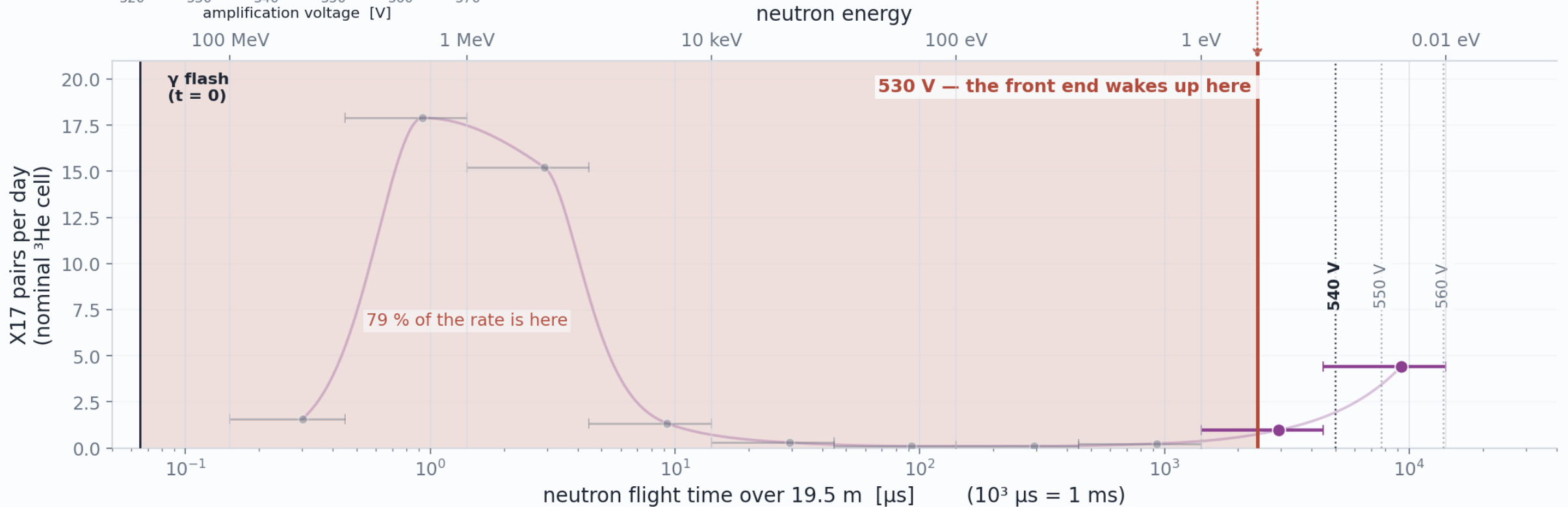
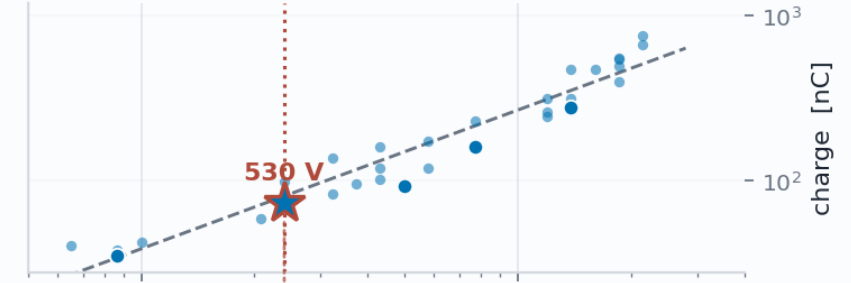
Every volt of gain costs milliseconds of beam



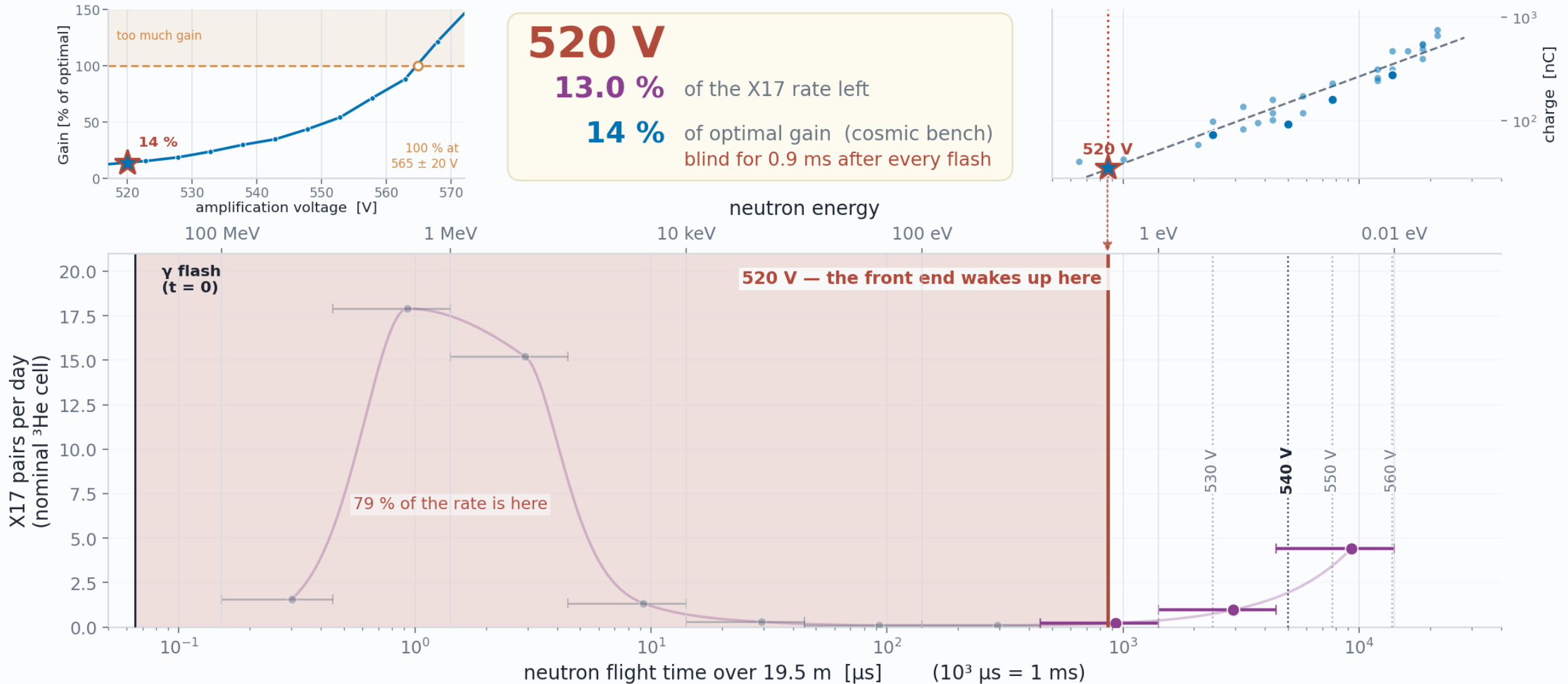
530 V

11.7 % of the X17 rate left

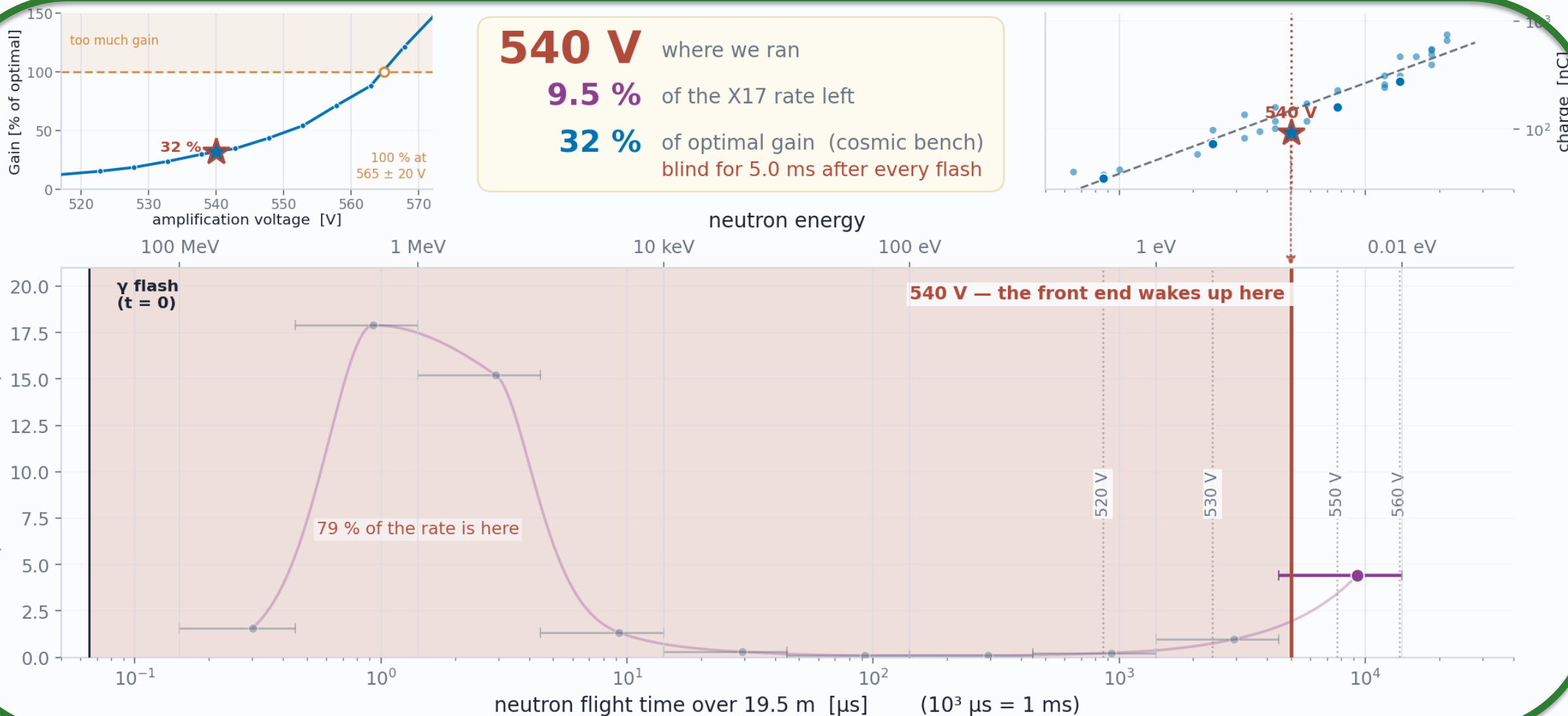
20 % of optimal gain (cosmic bench)
blind for 2.4 ms after every flash



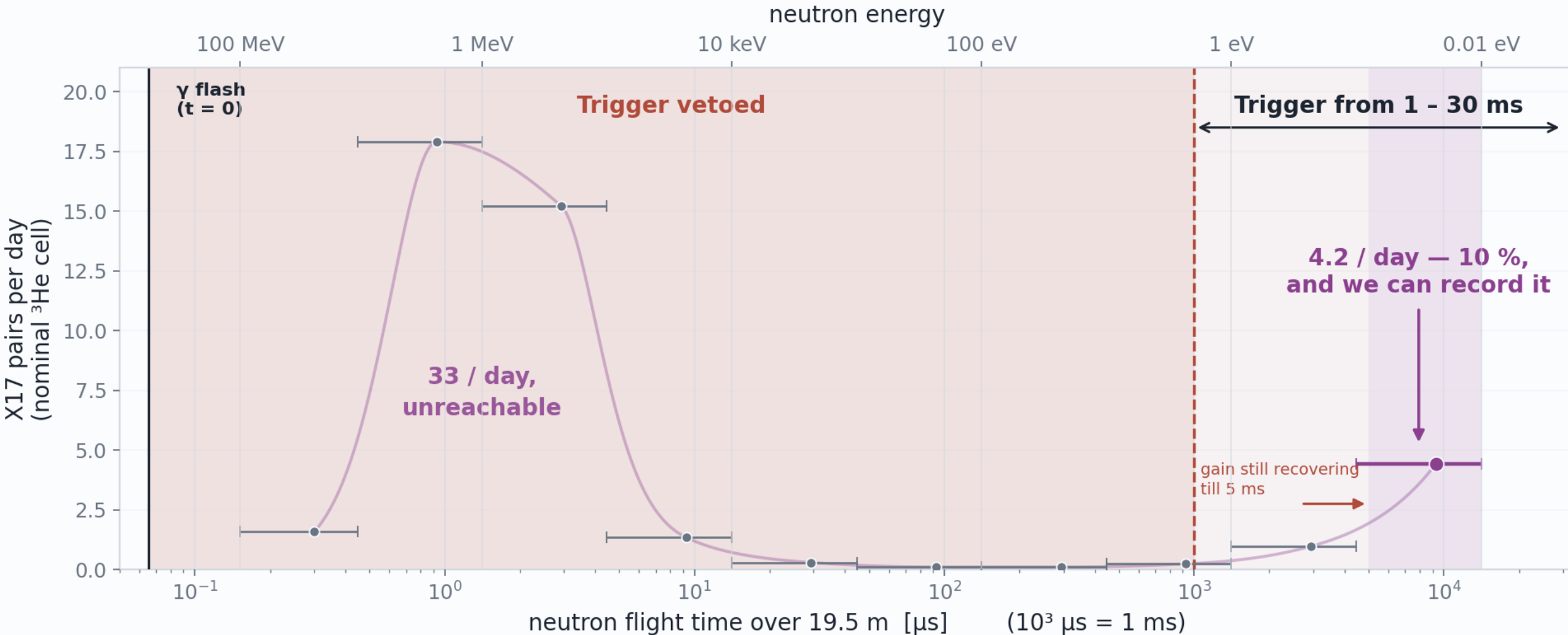
Every volt of gain costs milliseconds of beam



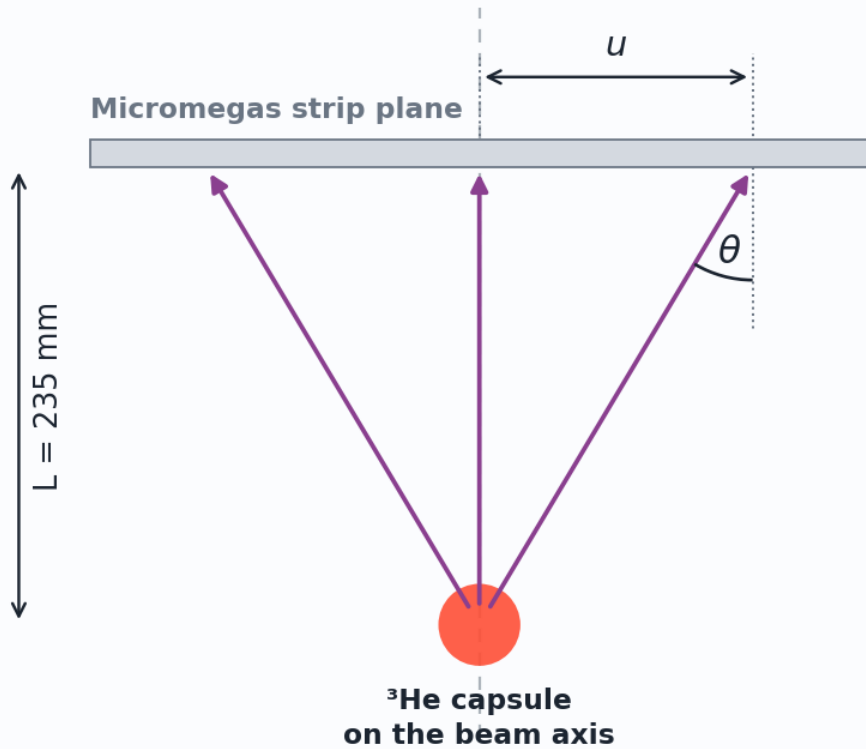
Every volt of gain costs milliseconds of beam



So we made a measurement at thermal energies

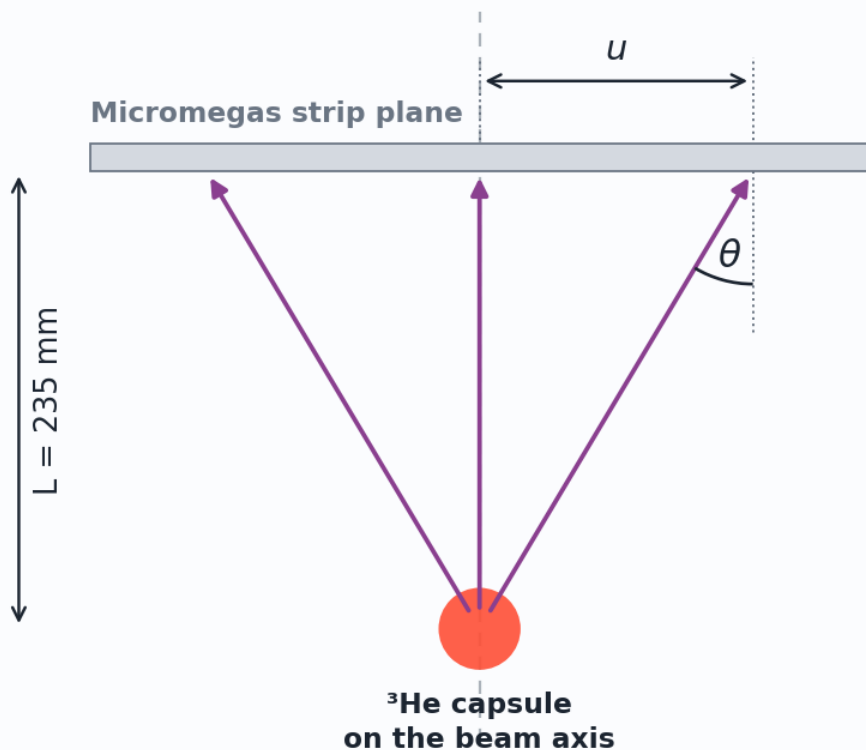


The angle a track arrives with says where it came from



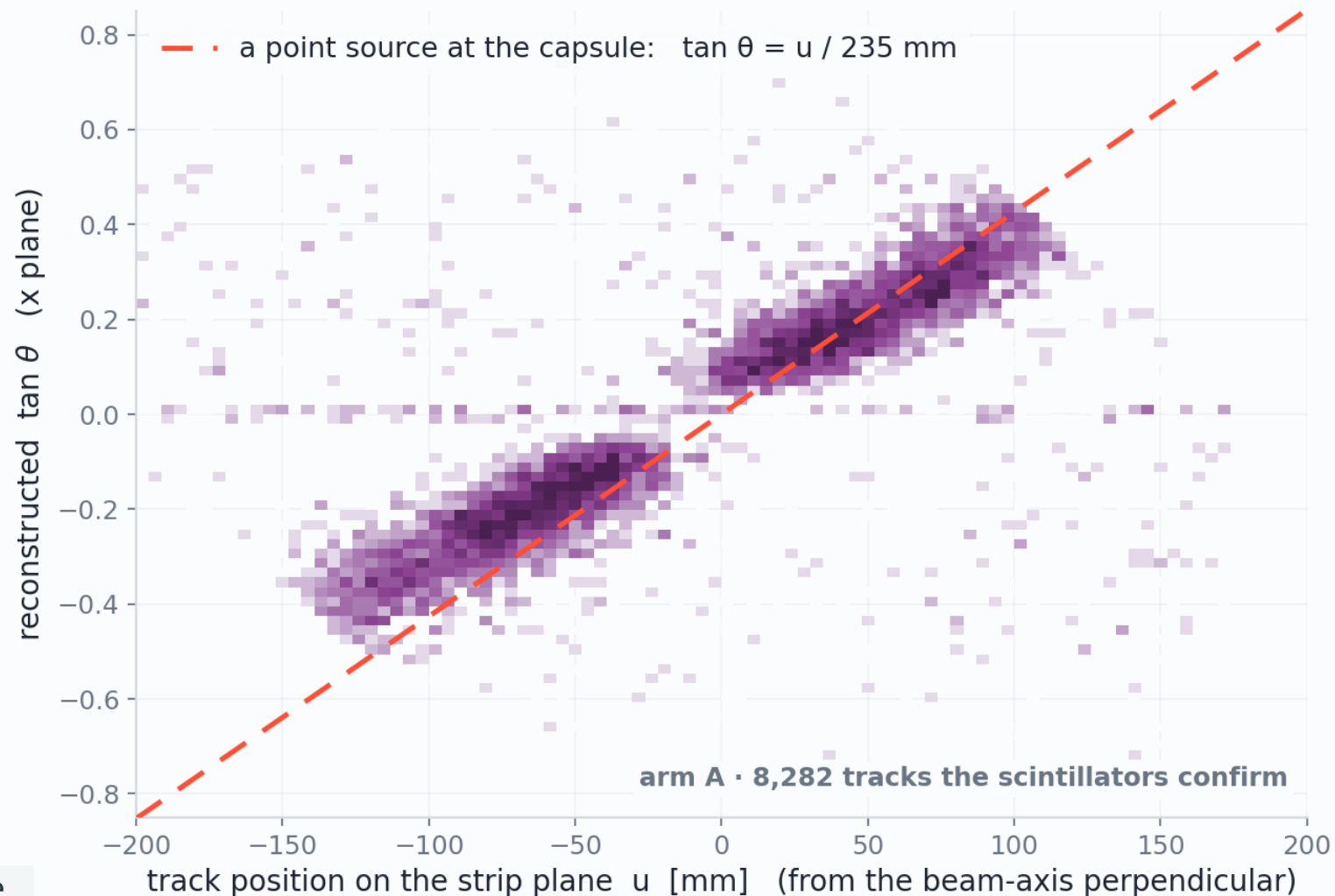
a point source can only arrive as $\tan \theta = u/L$

The angle a track arrives with says where it came from



a point source can only arrive as $\tan \theta = u / L$

- Correlation implies that tracks point back to capsule



Two opposing arms, seen from above – image of capsule

the ^3He bore, $r = 10$ mm

arm C

arm A

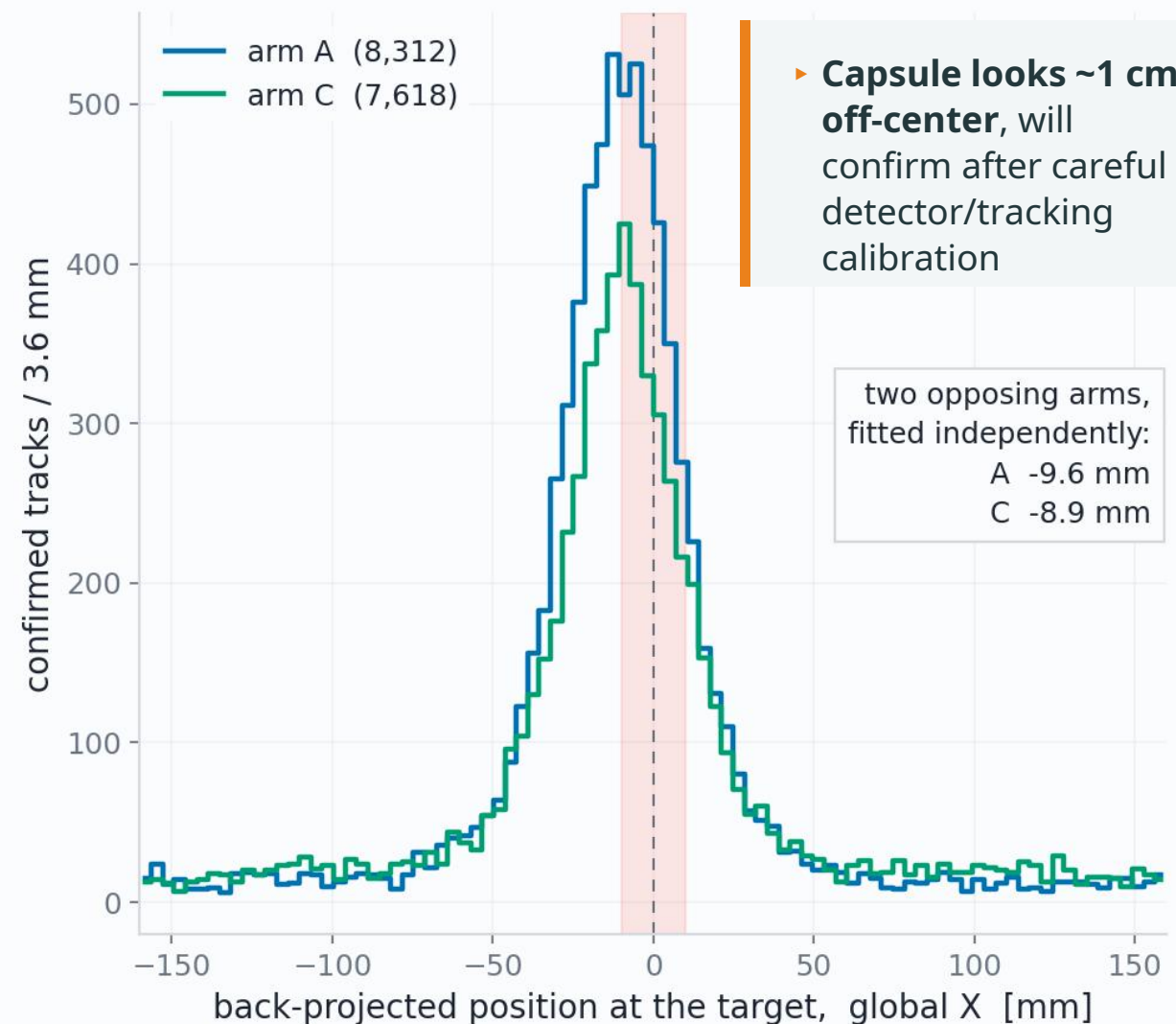
the ^3He capsule, seen end-on



100 mm

15,930 confirmed tracks · strip planes end to end

Other two detector more problematic – working to recover

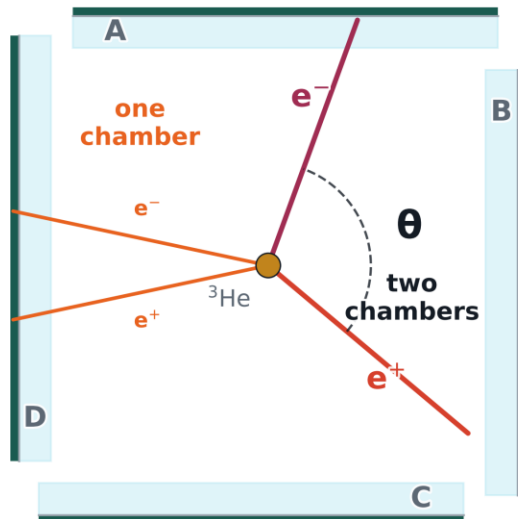


Summary and where we go from here

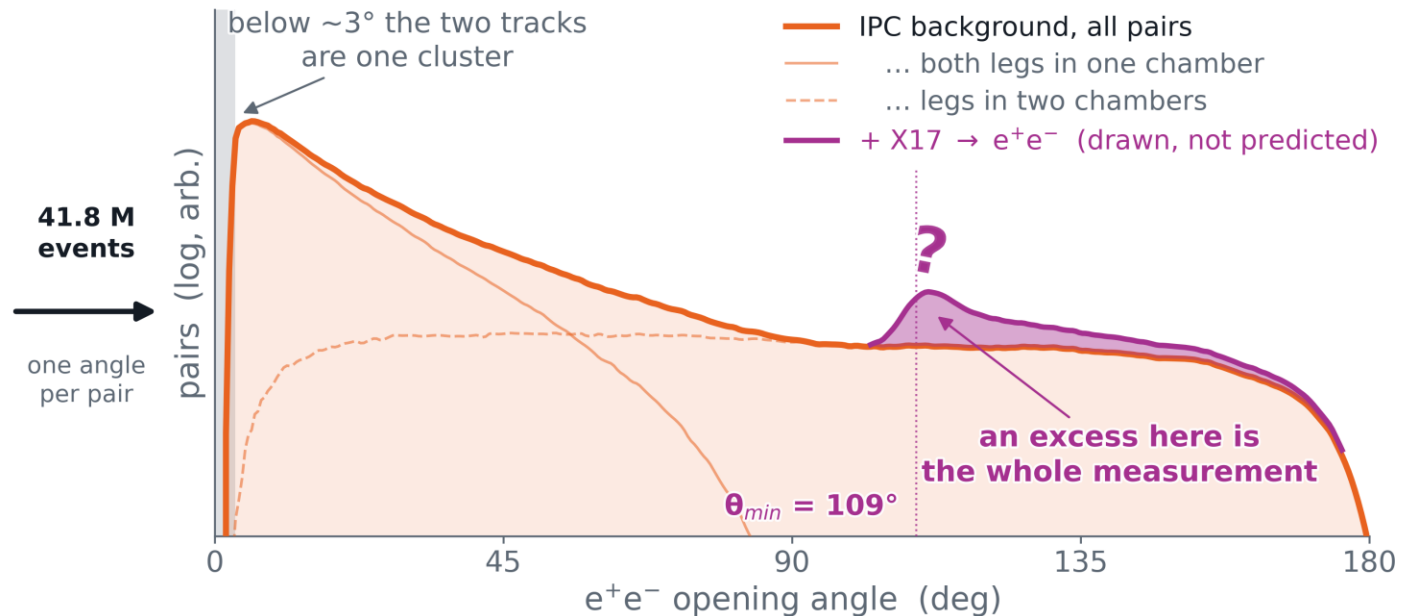
- ▶ Constructed Micromegas μ TPCs which measure charged track angle to $\sim 1^\circ$
- ▶ 2026 X17 campaign at n_TOF EAR2 complete: **41.8 M events, 17.9 TB, 44 days**
- ▶ **Thermal** window recorded

Next steps for the analysis

1. Find the two-track events



2. Histogram the opening angle



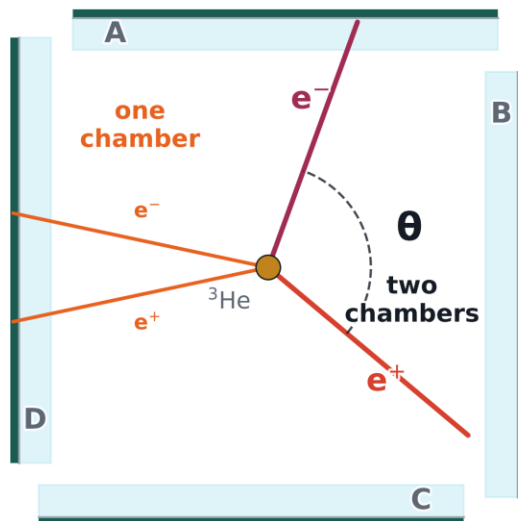
Summary and where we go from here

THANKS TO:

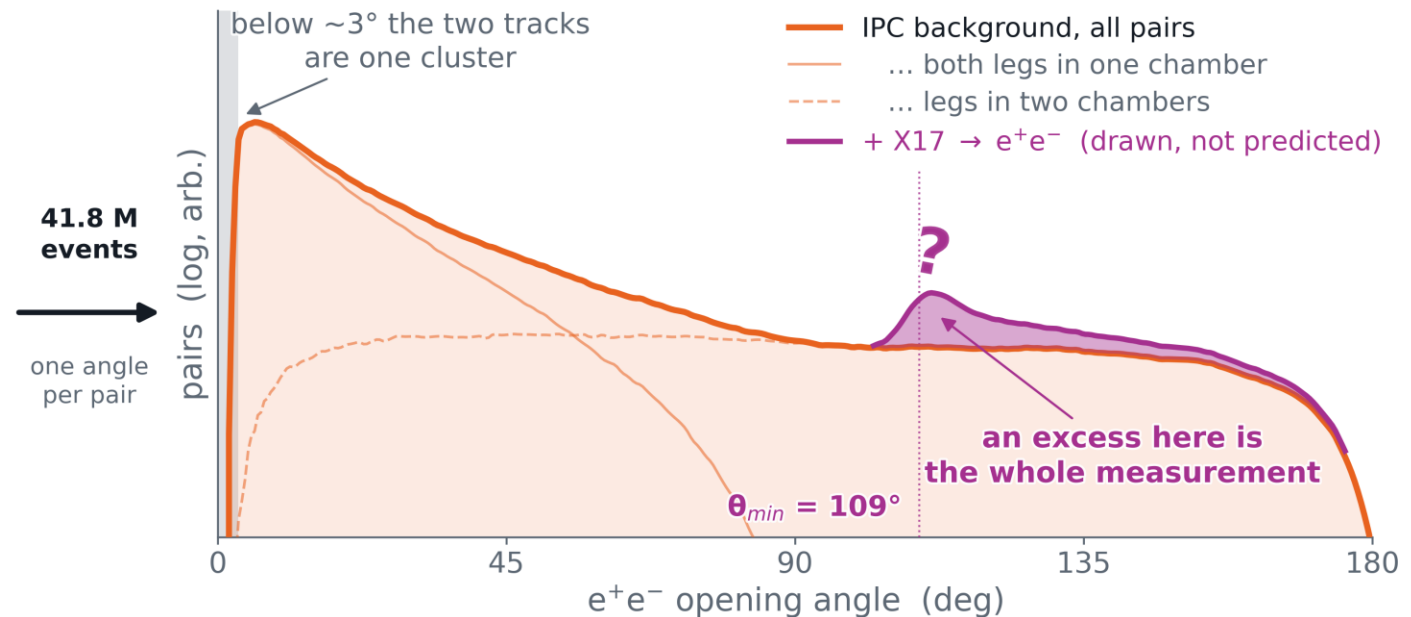
- **n_TOF** — the whole collaboration, and especially the local team, for making the measurement happen
- **EUROLABS** — for the funding that made this experiment possible
- **CERN GDD & MPGD** — for their indispensable support that enabled the measurement
- **CEA Saclay & Bulk Lab** — for help building the detectors and supplying the DAQ
- **MPGD 2026** — for giving me the opportunity to present these results

Next steps for the analysis

1. Find the two-track events



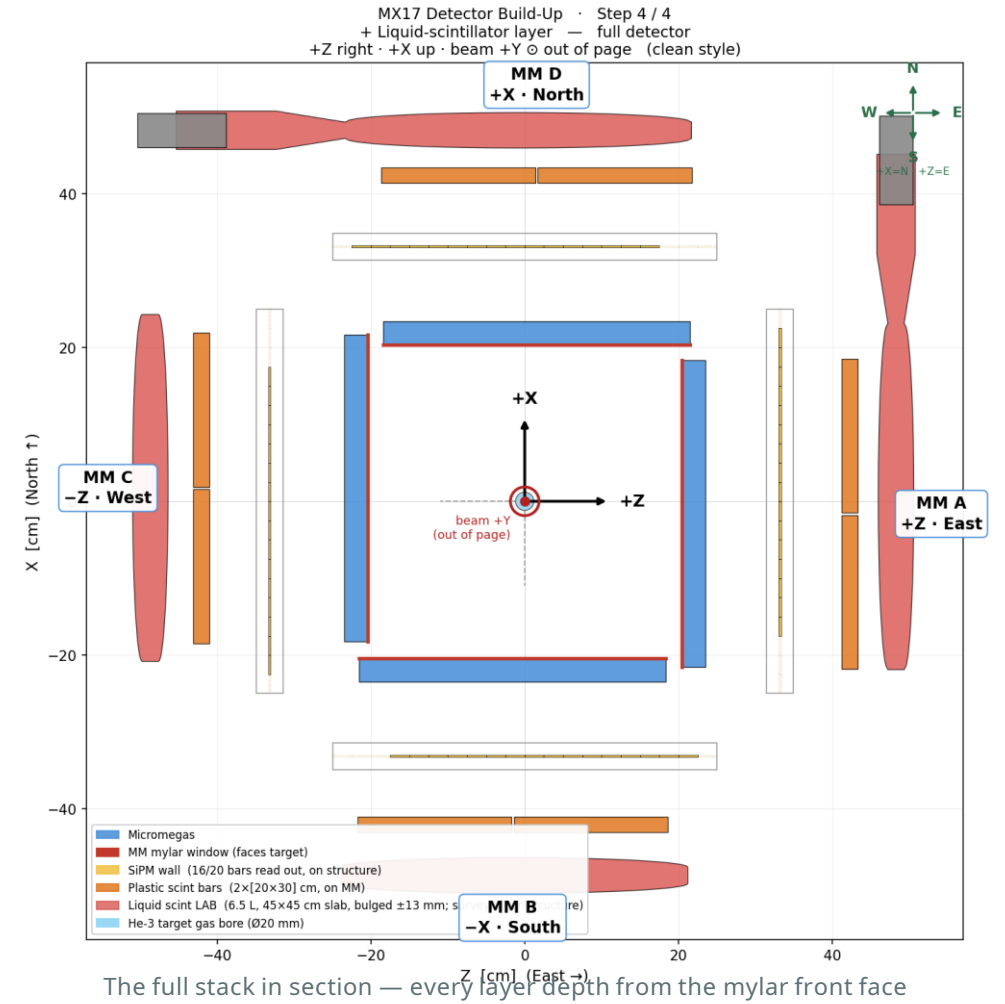
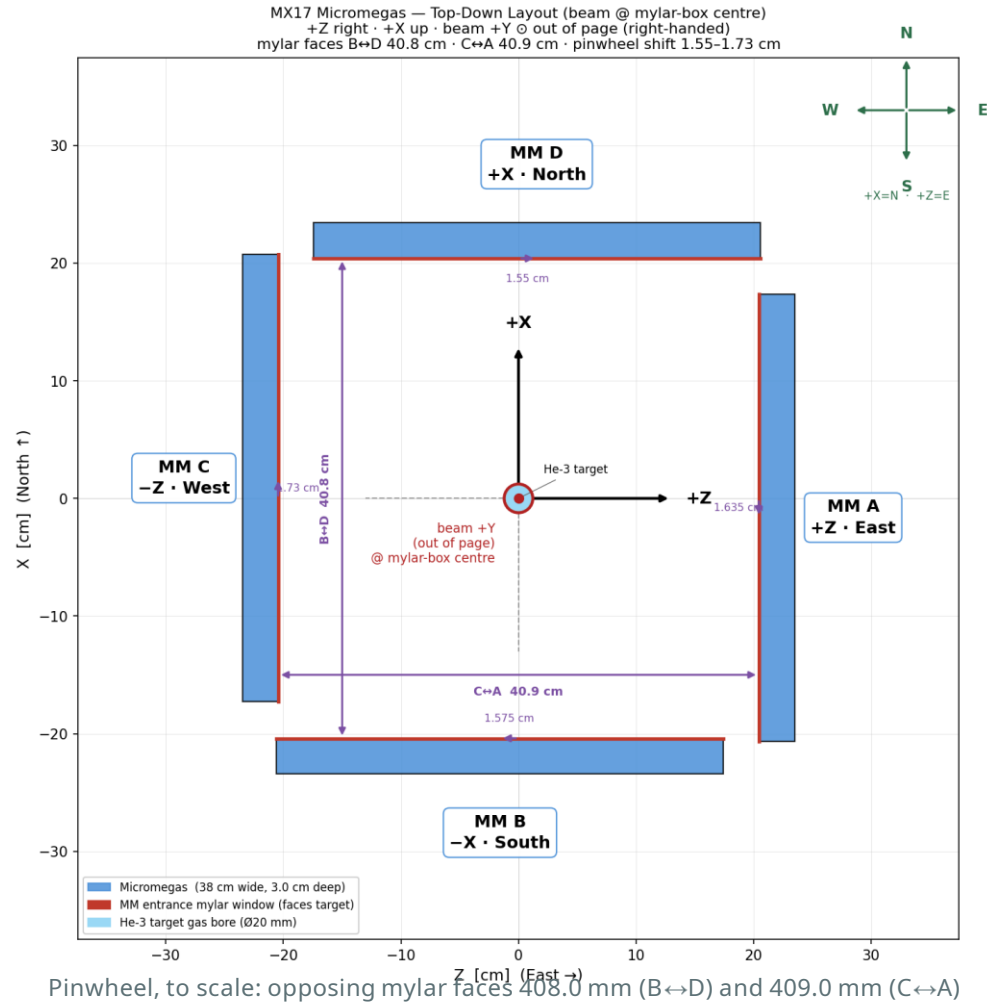
2. Histogram the opening angle



Thank you! Questions?

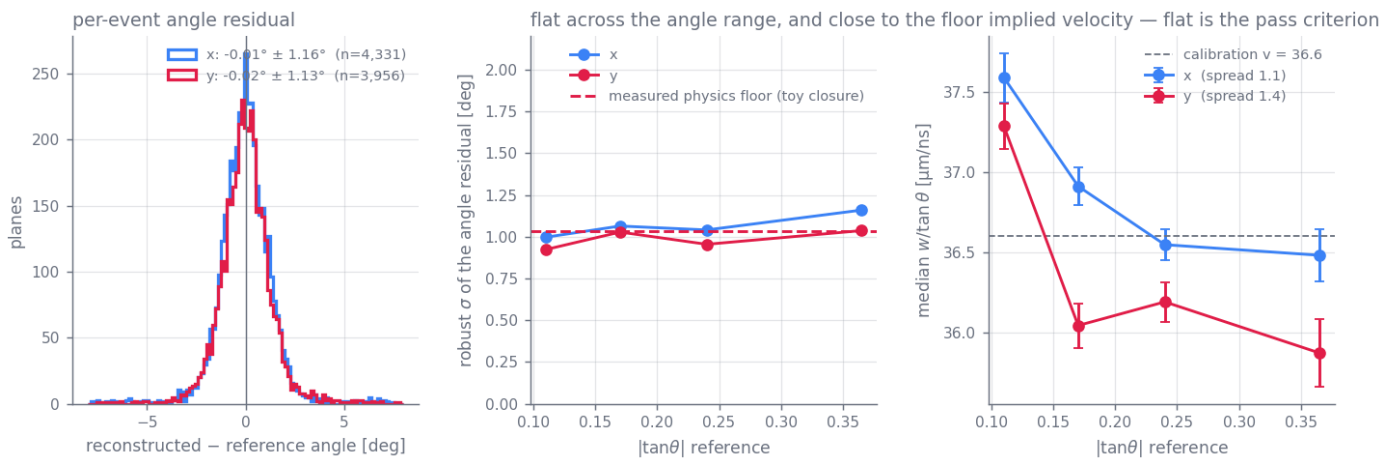
Backup

The setup to scale: top-down, and layer by layer



Same geometry as the 3-D build-up, drawn as a measured section rather than a render: distances 2026-07-15, per-arm gaps and vessel placements 2026-07-17/18. The intermediate build states (+SiPM, +plastics) are [setup_2_sipm.png](#) / [setup_3_plastic.png](#) in `assets/img/`.

The ~1° per-event scatter is physics, not fitting

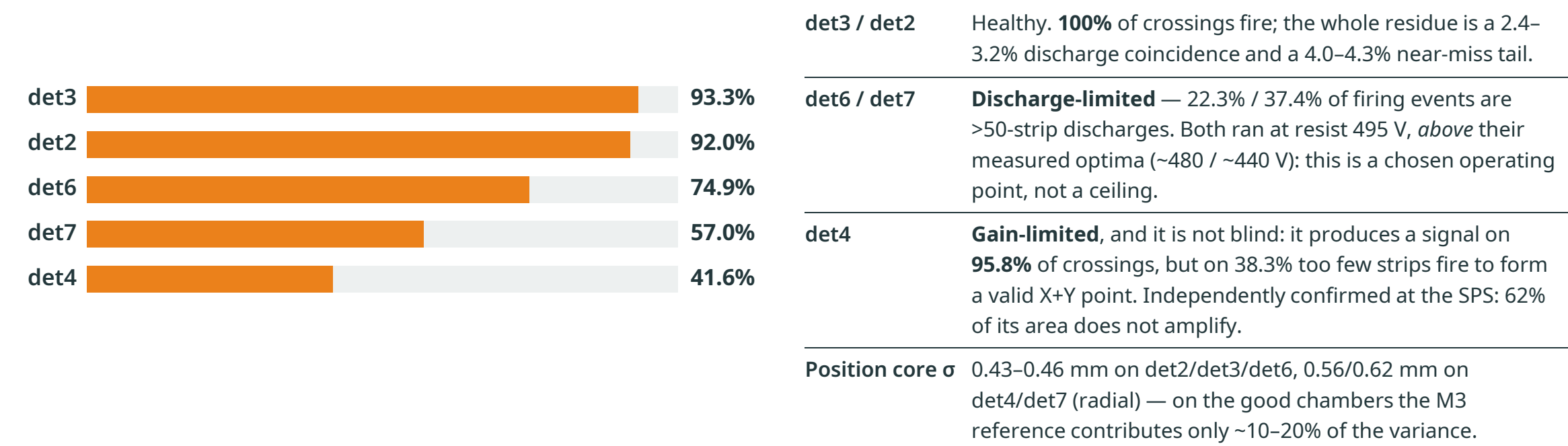


The closure test waveforms generated from the calibrated model with known tracks — real geometries, real charge profiles, real noise, real clipping — then fitted back. One knob: per-depth-bin transverse jitter of the charge centroid	
jitter 0	σ_θ 0.02° — electronics noise contributes essentially nothing
0.15 mm	0.55°
0.30 mm	1.05° — the scale diffusion + a few effective electrons per 2 mm of track + δ -rays actually give; reproduces the data exactly
0.50 mm	1.77°
errors honest?	pull width 1.16 (X) / 1.13 (Y) — ~15 % optimistic, and χ^2/dof genuinely ranks which fits to distrust

Full run, 6,852 / 6,850 reference-matched events, reference never in any fit: σ_θ = 1.16° / 1.13°, bias < 0.1°, **flat across the angle range and sitting on the measured floor** (middle, dashed); implied v flat to ~1.4 μm/ns (right)

So the per-event fit is extracting **essentially all the information the detector geometry physically has** — gains beyond ~1° come from ensemble averaging, not better fitters, which is why the reported errors carry a physics floor added in quadrature and the optimiser tolerances are deliberately loose. Ablations agree: dropping any single model refinement moves per-event σ by < 0.05°; what the refinements buy is *ensemble physicality* (implied-v flatness), the thing that cannot be tuned. Doc Part VII §41–43.

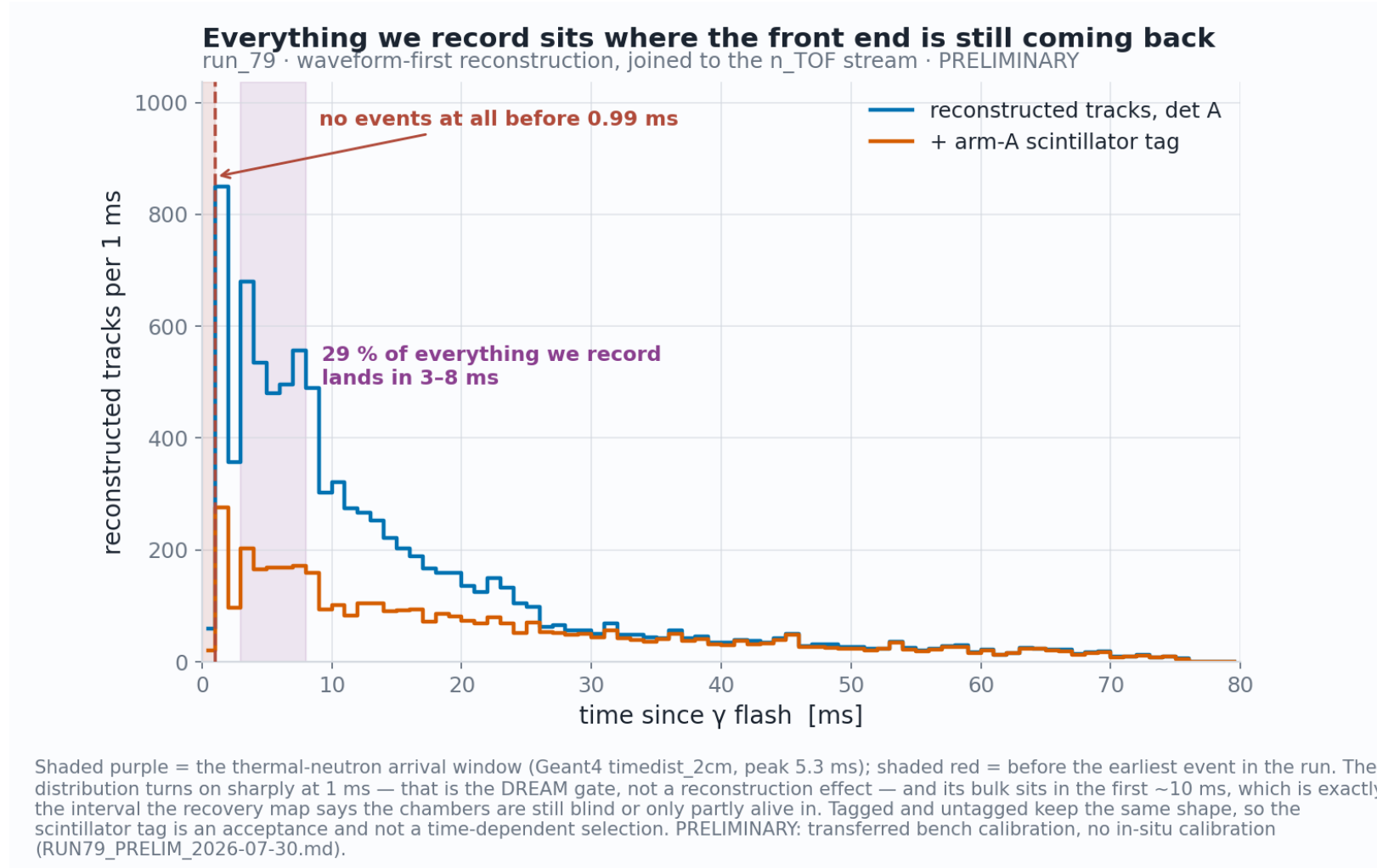
The fleet, and what limits each chamber



Reconstructed within 5 mm of the M3 reference track ($\chi^2 < 1.0$ & NClus=4). One high-statistics sub-run per chamber, 3.7k–12.3k reference muons, waveform-first reconstruction, 2026-08-13 reprocessing.

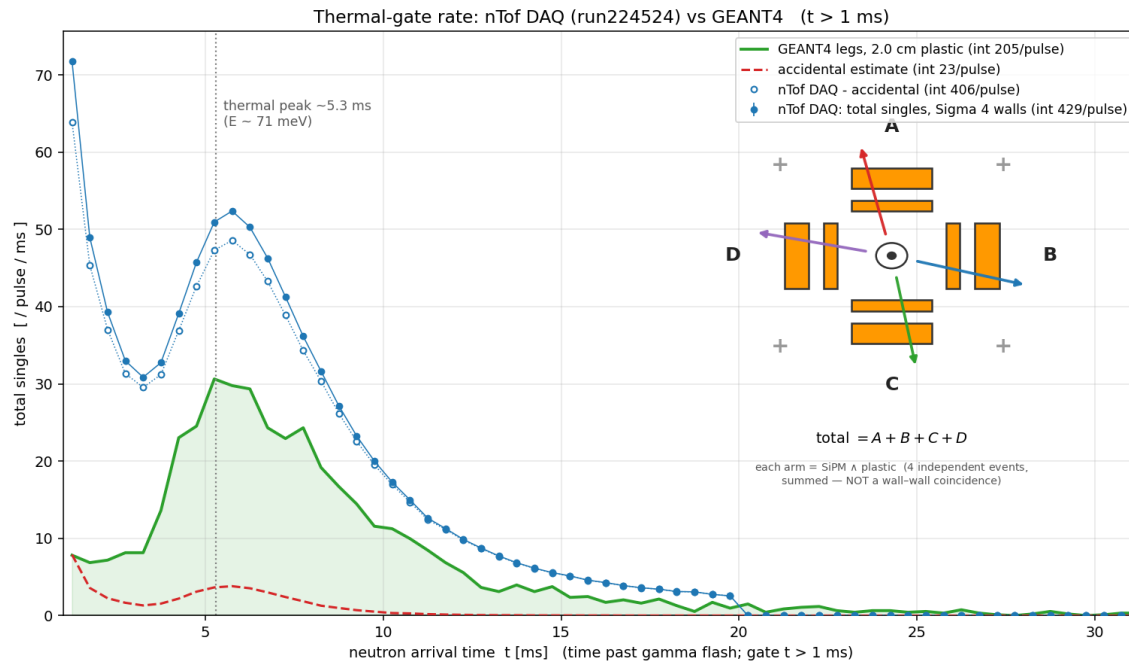
Superseded 2026-07-14 values were 92.9 / 91.3 / 57.8 / 43.1 / 20.7%. The last three chambers gain 14–21 points from the matched-filter reprocessing of the raw waveforms — low-amplitude hits the previous analyzer never admitted — plus a further 0.4–4 points from reconstructing position from the waveforms instead of the strip hit times.

What we record, and when



Reconstructed tracks on beam data, joined event-by-event to the n_TOF scintillator stream. The turn-on at 1 ms is the DREAM gate, not the reconstruction; the bulk of everything we record sits in the first ~10 ms — the same interval the recovery map says the chambers are only partly alive in. That overlap is the experiment's central operational problem, stated in data.

The trigger lives in the thermal window — by design



- ▶ **Capture loves slow neutrons, and slow neutrons are late.** Over ~20 m of flight path the thermal flux peaks at **5.3 ms** — recorded, and reproduced by Geant4 in shape.
- ▶ **The trigger:** per arm, the SiPM wall (any of 4 segments over $\frac{1}{2}$ MIP) **AND** a plastic bar over **0.90 MIP**, within 20 ns — a charged leg crossing the stack, not a γ .
- ▶ Simulation says **96 %** of those trigger legs are ^{27}Al -capture γ s converting in the capsule wall — the trigger is dominated by the background the analysis has to beat.

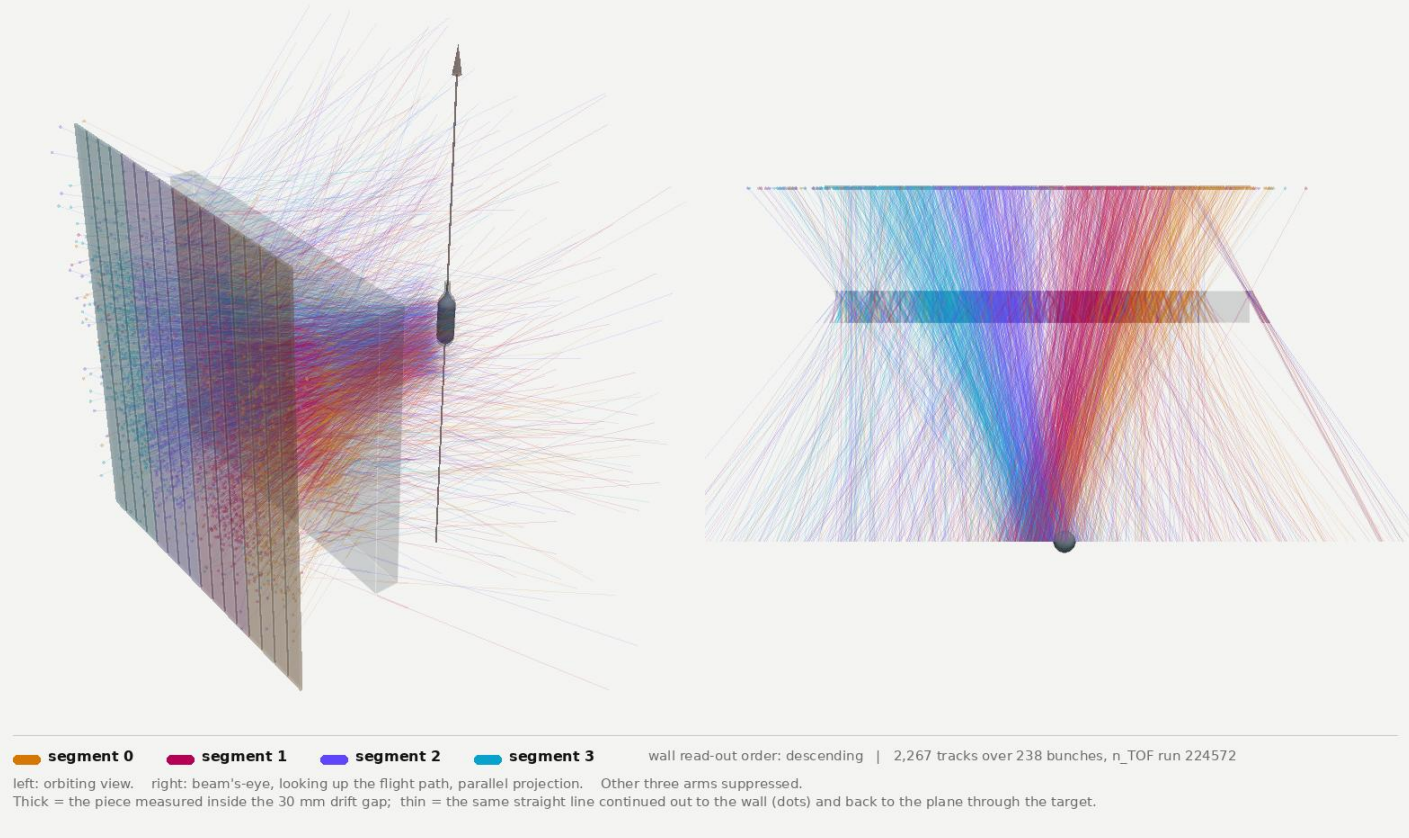
Recorded trigger rate vs neutron arrival time, against Geant4 — the 5.3 ms thermal peak reproduces; the normalisation runs **2.1×** high (per wall 1.1–2.6×), under investigation

The tracks point back at the capsule

run_79 / mx17_A -- tracks, by the SiPM wall segment that triggered them

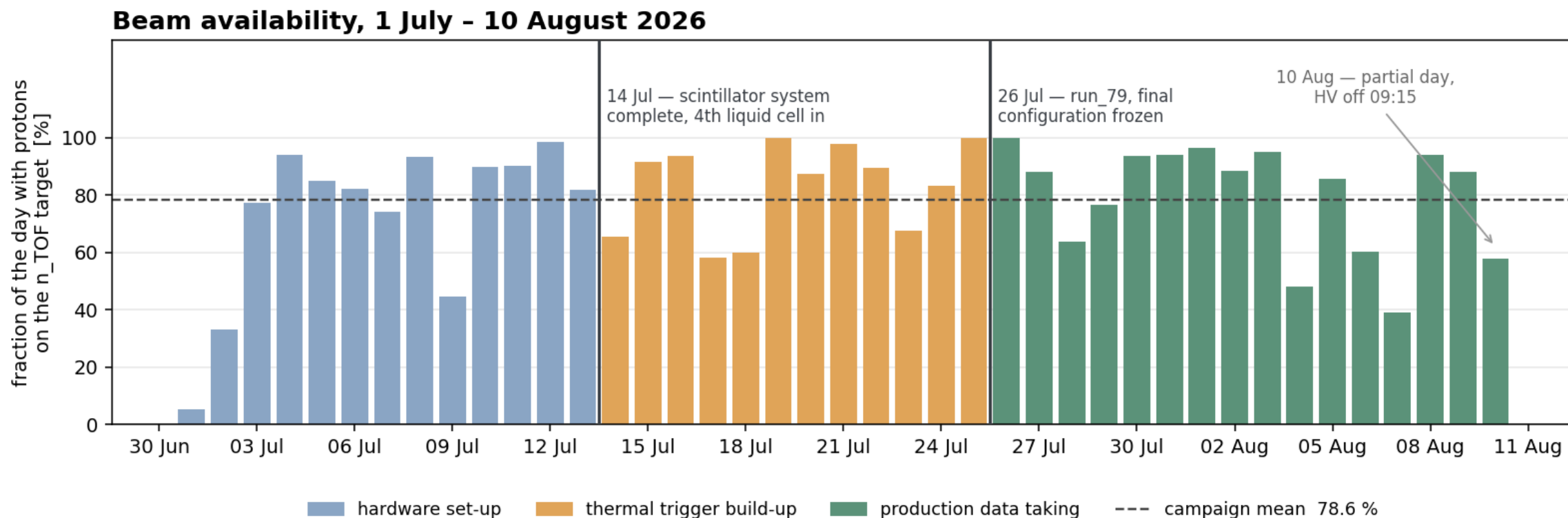
all 4 bundles -- 2,267 tracks: 251 mm apart at the wall, 9 mm apart at the target plane

PRELIMINARY -- transferred bench calibration, angles rescaled to $v = 36.0$ um/ns; see RUN79_PRELIM_2026-07-30.md



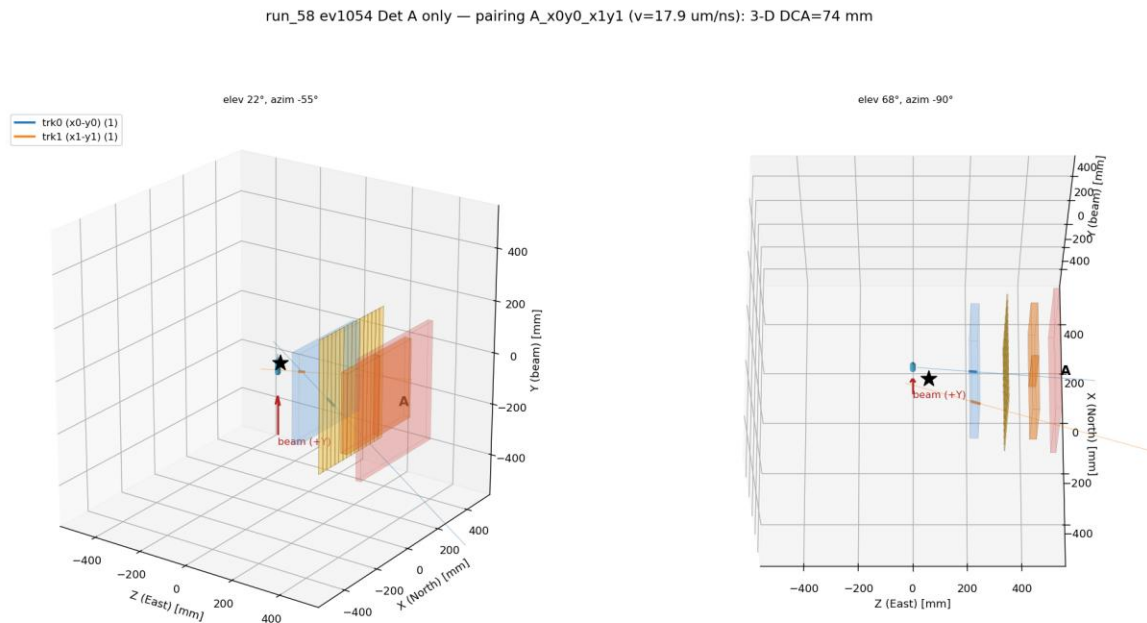
Waveform-first reconstruction on beam data, coloured by **which SiPM wall segment fired** — external position truth on a 100 mm granularity, on a lever arm the chamber knows nothing about. The four bundles are **251 mm apart at the wall and 9 mm apart at the target plane**; shuffling the segment labels collapses that to 28 mm and 3.2 mm. **Ensemble statement** — the waist is a convergence of medians, not a per-track pointing resolution.

Beam availability, day by day



Fraction of each day with protons on the n_TOF target, from the one-minute classification of the last beam-current transformer before the target (FTN.BCT477). Campaign mean **78.6 %**; the 21.4 % without beam is the usual mix of PS supercycle sharing and machine stops — the logger's attribution between the two is not reliable, so only the total is quoted. Colours are the three campaign phases; 10 August is a partial day (HV off 09:15).

Next: from single tracks to pairs



- ▶ **What a signal is:** two tracks in *different* arms, opening angle kinematically confined above **109°**, total pair energy 19.6 MeV (slide 4). Everything in this talk exists to make that measurable.
- ▶ **The reconstruction is ready for the easy half:** the forward fit now returns *every* candidate track, not just the winner — single-track output bit-identical, candidates shipped alongside. Two tracks merged closer than ~12 mm still fit as one bad- χ^2 compromise: the two-track forward model is the one designed-but-unbuilt piece.
- ▶ **The dataset is final and the search is offline:** 41.8 M events on EOS, with the trigger dominated by AI-capture pairs — an enormous calibration sample for exactly this topology.
- ▶ **The bar the tracker has to clear:** the AI background separates on *total pair energy*, which asks for $\sigma(p)/p \lesssim 30\%$ per lepton from scattering or range — the open MPGD question to leave the room with.

A real two-track event in one chamber (July scan run) — two track-grade lines in **both** planes, paired into 3-D. A demonstration, not a measurement: hits-basis, and the $X \leftrightarrow Y$ pairing is two-fold ambiguous